# Beyond Local Revenue: Commercial Property Taxes, Firm Behavior, and Fiscal Externalities

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**Abstract**

Property taxation is widely viewed as a promising source of revenue for developing countries, yet evidence on its economic effects remains limited. Using administrative data from corporate and employee tax returns in a staggered difference-in-differences design, this paper shows that a property tax increase in Kampala, Uganda significantly reduced rates of firm survival. Business rental expenses increased by 6.3 percent, and firms responded by reducing formal employment and wages, with wage informalization accounting for most of the decline in the formal wage bill. Higher-earning employees of ongoing firms bore disproportionate shares of the wage declines. Within government, there were vertical fiscal externalities: the tax increase reduced central government income tax revenues by 30 percent of the amount it contributed to local property tax collections. The estimates imply that property taxes have 53 percent marginal deadweight losses.

*JEL codes:* H22, H24, H32, H71.
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# I. INTRODUCTION

Property taxation is frequently highlighted as a promising revenue instrument in developing countries because immovable property constitutes a relatively inelastic and observable tax base, making it easier to identify, assess, and enforce than many alternative taxes. This has motivated a growing literature studying how administrative reforms and enforcement strategies can expand the effective property tax base and increase revenue collections (e.g., Manwaring and Regan 2025; Flores et al. 2025; Bergeron et al. 2024). Yet, this focus on revenue generation has largely overlooked a central question: what are the economic effects of property taxation? Much of the existing work abstracts from the downstream effects of property taxation on firms, workers, and production decisions, implicitly assuming that any resulting distortions are limited or negligible.

From a theoretical perspective, property taxes are often viewed as relatively efficient, with the burden falling primarily on property owners and limited implications for real economic activity. This view, however, assumes a separation between property ownership and production. In practice, many firms occupy their own premises, so that property taxes effectively tax a productive input, raising the cost of using commercial space and potentially affecting firms' production decisions. Even in rental markets, recent studies document substantial pass-through of property taxes into rents (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019), implying that firms operating in rented premises may also be affected by these taxes. These cost increases tighten liquidity constraints, which might induce firms to adjust along several margins, including survival, employment, wages, and the use of owned versus leased space. The extent to which such adjustments arise, and whether they meaningfully alter real economic activity, remains an open empirical question.

These firm-level responses may, in turn, have broader fiscal implications that extend beyond the local government imposing the tax. If higher property taxes induce firm exit, reduce profitability, or contract formal employment and wages, they can erode the bases of centrally administered taxes such as corporate income tax (CIT) and employee personal income tax. This creates a vertical fiscal externality that local governments do not internalize when setting tax policy. Yet direct evidence on the size of this externality, and thus the true fiscal cost of local property taxation, remains scarce.

This paper therefore provides a comprehensive analysis of the economic effects of property taxation, spanning firm activity, incidence, and distributional and welfare consequences. It also quantifies spillovers from local property taxes to central government revenues, highlighting the importance of vertical fiscal linkages for tax policy design.

I exploit a natural experiment in Kampala, Uganda, where property tax liabilities increased differentially across locations within the city through a revaluation and mass property enumeration exercise that began in 2016/17. Prior to this exercise, the last valuation had been conducted in 2006, meaning that taxable property values had remained unchanged for a decade. During this period, numerous new properties had been established that were not captured in the old tax roll. The 2016/17 exercise led to the creation of a new property tax roll, which included additional properties and assigned higher rateable values to existing ones. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, although the marginal tax rate remained unchanged in both cases. Following the exercise, the total number of properties on the tax roll increased by 76 percent, while projected property tax revenue more than tripled.

This exercise was implemented in a staggered manner, starting with one division and gradually extending to others over time.[1] I exploit this geographical variation using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with firms located outside the capital but within the Greater Kampala Metropolitan Area serving as never-treated units. This strategy allows me to credibly identify the causal effects of increases in property tax liabilities on firm-level outcomes while avoiding the well-known limitations of conventional two-way fixed effects estimators under staggered treatment adoption. The identifying assumption is that, in the absence of the reform, average outcomes for firms in treated and untreated locations would have evolved similarly, i.e., the parallel trends assumption. I provide evidence in support of this assumption using an event-study analysis. The main results are robust to an alternative stacked difference-in-differences approach following Cengiz et al. (2019). The empirical analysis utilizes administrative tax data from corporate income and employee tax returns. These novel firm-level panel datasets span nine years (2013/14 - 2021/22) and were recently compiled by the Uganda Revenue Authority with support from UNU-WIDER.

First, I analyze how the reform affected firm survival. The firm survival outcome is constructed following the approach of Saez et al. (2019), restricting the sample to firms present in 2015/16 (the year prior to the reform) that reported non-zero turnover, regardless of whether they operated in other years. Firms are assigned a turnover value of zero in years in which they do not operate. The outcome variable is then defined as binary, equal to one if the firm operated in that year and zero otherwise. Consistent with Saez et al. (2019), the sample is reweighted using the nonparametric reweighting method developed by DiNardo et al. (1996) (DFL reweighting). The results indicate that the reform

[1] For context, the capital city is divided into five administrative divisions.

reduced firms' survival probability by 3.2 percent. Scaling this reduced-form effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits citywide. Using cross-tax validation and alternative definitions of firm exit, I show that the estimated survival effects are not driven by evasion of corporate income tax. Moreover, the negative effects on survival are concentrated among loss-making firms with no CIT liability, for whom evasion provides no plausible motive to stop filing. At the county-industry level, I further show that the reform had negative effects on the stock of active firms.

Turning to revenues and profits, I find no statistically significant effects among active firms, although profits show a modest short-run increase. I also find a null effect on raw material purchases and a negative but imprecisely estimated effect on investment. The effects on revenues and profits differ sharply by firm size, however. Large firms see significant increases in both revenues and profits, while small firms experience an imprecisely estimated decline in revenues and no meaningful change in profitability. A potential explanation for this pattern is a reallocation channel operating via equilibrium adjustments in local input or output markets. To rationalize this channel, I develop a simple general-equilibrium model in which the reform disproportionately contracts space-intensive small firms and, through equilibrium adjustment of a local factor price, benefits large firms that use the factor more intensively.[2] The model, however, does not preclude a complementary market-power channel, under which larger firms pass a greater share of the tax burden onto consumers through higher output prices.

Firms preserved profitability by adjusting labor costs. Among active firms, the reform reduced formal employment by 3.8 percent and the formal wage bill by 3.7 percent in treated areas relative to untreated areas. To quantify the aggregate impact, I re-estimate the employment effects on an expanded sample that includes all pre-reform firms, assigning zero formal employment in any year in which a firm is no longer active.[3] The reduced-form estimate from this specification implies approximately 320 fewer formal jobs citywide per $1 million increase in aggregate property tax liabilities.[4] Robustness tests confirm that the employment effects are not driven by displaced workers relocating to firms in comparison areas. The results are also robust to restricting the sample to the pre-COVID period. The calendar-time-specific treatment effects also show that the

[2] A model that uses equilibrium adjustments in the output market would produce a similar pattern.

[3] The sample used here is similar to that used to analyze firm survival.

[4] Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, I can not calculate firm-level employment elasticities with respect to property tax liabilities.

employment declines predate the onset of the pandemic.

To reconcile the large employment effects with unchanged revenues, I examine the informalization margin, which is particularly relevant in settings with weak enforcement capacity. If higher property tax liabilities tighten firms' liquidity constraints, they raise the value of immediate cash-flow savings. Informalization is a potentially attractive margin of adjustment because it allows firms to reduce payroll-linked obligations while retaining workers' productive services through informal arrangements. I find that the reform increased wage informality by 2.6 percentage points, relative to a pre-reform mean of 10.3 percent. This accounts for 68 percent of the decline in the formal wage bill, suggesting that a substantial share of the observed employment decline reflects transitions to informal arrangements rather than outright job destruction. These results suggest that informality is an important margin of adjustment through which firms in developing countries respond to cost shocks.

I next examine the distributional consequences of the reform within firms by assessing which types of workers are most affected.[5] First, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. Second, I analyze how the firm-level wage distribution evolves around the reform in treated firms relative to untreated firms. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the share of workers in the top quartile declined by 2.8 percentage points relative to pre-reform total employment, while the share of workers in the $50^{\text{th}}$–$75^{\text{th}}$ percentile increased by 2.2 percentage points. The shares in other bins are unaffected. Consistent with these findings, I find that wages fall significantly at the top of the distribution, while there are no significant effects at the bottom. These results suggest that the reform reduced within-firm wage inequality and was therefore progressive at firm level.

Heterogeneity analysis reveals that pre-reform profitability and leverage are important moderators of the reform's effects on employment and survival. Effects are stronger among less profitable firms and monotonically decreasing in profitability. For leverage, I find a U-shaped pattern in the effects of the reform across the leverage distribution: high-leverage firms are vulnerable through a financial distress channel, whereby higher property tax obligations tighten liquidity constraints associated with existing debt service obligations; very low-leverage firms are vulnerable through a financial exclusion channel, whereby limited access to credit markets constrains their ability to smooth the shock, forcing adjustment on real margins. Firms with moderate leverage retain sufficient

[5]The sample is restricted to active firms with at least five workers in the pre-reform period to ensure adequate support across the wage distribution.

unused debt capacity to absorb the shock, explaining their relative insulation. I provide direct evidence in support of this mechanism: moderately leveraged firms experience a differential increase in debt following the reform, while firms at either tail of the leverage distribution do not. This asymmetry suggests that access to the borrowing margin is the key mechanism insulating moderately leveraged firms from the adverse consequences of the reform.

Estimating the effects separately by occupancy status, I find that employment and survival decline even among tenant firms, suggesting an important role for the rental cost channel in transmitting property tax burdens to firms. To quantify this channel, I estimate the effect of the reform on business rental expenses and calculate the implied pass-through rate. I consider an unbalanced panel of firms that reported paying rent in the year prior to the reform, estimating the effect of the reform on rent expenses scaled by their pre-reform values. The results show that the reform led to a 6.3 percent increase in business rental expenses. This reduced-form estimate implies that a UGX 1 increase in aggregate property tax revenue is associated with a UGX 0.52 increase in total rental expenses paid by CIT-registered tenant firms.[6] Interpreted as a pass-through rate to the formal sector, this suggests that approximately 52 percent of the aggregate property tax increase is transmitted to formal tenant firms through higher rents, which likely represents a lower bound, since the denominator includes tax revenue from all commercial properties while the numerator captures rent increases among CIT-registered firms only.

This result contrasts with the neoclassical view, which predicts that, under an inelastic supply of property, the burden of property taxation should fall almost entirely on landlords. To rationalize this finding, I propose a simple framework that embeds a key institutional feature of property tax enforcement in developing countries: targeted enforcement, whereby the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity than when the unit is vacant. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use. Under these conditions, property taxes raise the marginal cost of supplying space for rent, altering the landlord's pricing decisions.

Lastly, I quantify the vertical fiscal externality by examining how revenue from corporate and labor income taxation changes following the reform. I find that total central gov-

[6] Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, it is not possible to construct a firm-level measure of tax pass-through. This pass-through rate is therefore estimated at the aggregate level for the population of CIT-registered firms. It should therefore not be interpreted as a structural, firm-level pass-through rate.

ernment revenue collected from treated firms declines by an average of UGX 0.6 million per year relative to untreated firms, from a pre-reform mean of UGX 7.8 million, equivalent to a 7.9 percent decline. This result implies that the reform reduced central government income tax revenues by 30 percent of the amount it contributed to local property tax collections.

Embedding this fiscal externality within the Marginal Value of Public Funds framework proposed by Hendren and Sprung-Keyser (2020), I estimate that each shilling of net consolidated revenue raised through the reform costs the public 1.53 shillings, implying that the reform is welfare-improving only if the public goods it finances generate at least 1.53 shillings of benefits per shilling of spending.[7] These results underscore the need for coordination across levels of government when designing and evaluating subnational tax policy, as the true welfare cost of local tax instruments may be considerably higher than their face value suggests once vertical spillovers are taken into account. Even so, the implied marginal deadweight loss of 53 percent suggests that commercial property taxation imposes a lower cost on society than the corporate income tax, whose implied welfare cost is estimated at 68 percent (Global Tax Research Initiative 2026).[8]

Event-study analyses indicate that all the results are not driven by differential pre-trends. The event-study plots indicate that pre-period coefficients are statistically indistinguishable from zero, suggesting the absence of differential pre-trends and that outcomes would have evolved similarly in treated and untreated areas in the absence of the reform.

Relative to the existing literature, this paper provides a more comprehensive account of the economic effects of property taxation by examining real economic responses, fiscal spillovers across levels of government, and the resulting welfare implications. It contributes to four specific strands of the literature. First, it contributes to a growing literature on property taxation in developing countries. Recent work has studied optimal rate setting and enforcement (Brockmeyer et al. 2021; Bergeron et al. 2024), the role of local elites in tax collection (Balán et al. 2022), public disclosure and compliance (Manwaring and Regan 2025), and the social contract underpinning tax morale (Kresch et al. 2023). This literature has focused predominantly on residential properties and on the revenue and compliance dimensions of property taxation. Yet commercial properties often constitute

[7] Note that this conclusion abstracts from incidence.

[8] Global Tax Research Initiative (2026) estimate an MVPF of 1.68 for a corporate income tax cut in Uganda. Under the standard assumption of symmetric behavioral responses at the margin, this estimate identifies the underlying marginal welfare cost at the prevailing tax rate, from which the 68 percent is derived. The implied welfare cost of the property tax reform is also lower than the corresponding figures for corporate income taxation in neighboring East African economies, including Tanzania (71 percent) and Rwanda (73 percent).

a substantial share of the tax base and account for a disproportionate share of property tax revenues because of their higher compliance rates, making the economic effects of commercial property taxation a first-order policy question that remains largely unexplored.[9] This paper is among the first to study these effects using firm-level data.

Second, this paper contributes to the literature on how taxes affect business decisions. Existing studies have examined the effects of payroll taxes (Saez et al. 2019; Egebark and Kaunitz 2018), corporate taxes (Ljungqvist and Smolyansky 2014; Kennedy et al. 2026), business owner taxes (Risch 2024), and state and local tax policies in the United States (Rohlin et al. 2014; Thompson and Rohlin 2012; Fajgelbaum et al. 2019; Giroud and Rauh 2019). Far less attention, however, has been devoted to understanding how property taxes affect real economic activity and its distributional consequences. To my knowledge, this paper provides the first comprehensive analysis of the economic effects of property taxation, encompassing the real effects on firm activity, the incidence, and the distributional and welfare consequences.

Third, this paper contributes to the literature on spillover effects of tax enforcement. Existing work has focused primarily on horizontal spillovers, either within the same tax base through network effects, i.e., value added tax (e.g., Pomeranz 2015; Drago et al. 2020; Boning et al. 2020), or across taxes administered at the same level of government (Lopez-Luzuriaga and Scartascini 2019). I extend this literature along a vertical dimension, providing evidence that taxes administered at the sub-national level can erode the revenue base of the central government, a fiscal externality that operates through firms' behavioral responses rather than through direct administrative linkages.

Lastly, this paper contributes to the long-standing literature on property tax incidence (see England (2016) and Oates and Fischel (2016) for reviews). While earlier work found low pass-through of property taxes to rent (Carroll and Yinger 1994), more recent studies have documented substantial pass-through to rental costs (Watson and Ziv 2024; Baker 2024), particularly when the supply of housing is elastic (Löffler and Siegloch 2021). Prior research, however, has focused primarily on rental housing. This paper provides evidence on pass-through in the context of commercial properties.

The remainder of the paper is organized as follows. Section II describes the institutional context, including property taxation in Uganda and details of the revaluation and property enumeration reform. Section III describes the corporate income tax and employee tax return data used in the analysis and presents descriptive statistics. Section IV

[9] For instance, in Kampala, commercial properties constitute 33 percent of all properties, with compliance rates of 34 percent compared with only 11 percent for residential properties (Manwaring and Regan 2019). These properties account for 53 percent of property tax revenues (Badalov et al. 2025).

outlines the identification strategy. Section V presents the empirical results on the effects of the reform on various dimensions of business activity, including firm survival, revenues, profits, investment, and input use. Section VI examines the effects of the reform on labor market outcomes as well as the distributional consequences. Section VII presents heterogeneity along various dimensions. Section VIII presents empirical evidence on the effects of the reform on business rental expenses. Section IX quantifies the fiscal externality to central government revenues and the welfare implications using the Marginal Value of Public Funds framework. Section X concludes.

## II. CONTEXT

### *II.A. Property Taxation in Uganda*

Property taxation in Uganda is governed by the Local Government (Rating) Act, 2005, which allows local jurisdictions to levy rates based on the rateable value of any property within their jurisdiction. The tax applies to all immovable property, including buildings and structures used for commercial purposes (e.g., rented residential houses, rented shops, factories, hotels). It also applies to owner-occupied properties used for business purposes but excludes vacant sites and owner-occupied residential properties.[10] Local governments have the discretion to levy rates ranging from a minimum of UGX 2,000 (approximately $0.60) up to 12 percent of the rateable value, where the rateable value is defined as the assessed property value net of a 22 percent conservancy allowance. In Kampala Capital City Authority (KCCA), the property tax rate is set at 6 percent of the rateable value.

Currently, most cities and municipalities in Uganda, including Kampala, use a market-value approach in which property rates are applied to the estimated annual rental value or business income value (Bakibinga and Ngabirano 2019). The law requires each local government to prepare an initial valuation list and update it at least once every five years. In practice, however, these valuation lists are often not updated on schedule, largely due to a shortage of qualified valuation surveyors in both the private and public sectors (Bakibinga and Ngabirano 2019) and inadequate resources.

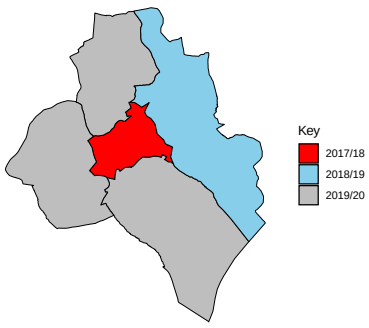
[10]The full list of exempted properties includes: official residence of the president; official residences of cultural/traditional leaders; public worship places; cemeteries and crematoriums; charitable and educational institutions; outdoor sports and recreational facilities; local council-owned properties; property owned by privileged organizations under Diplomatic Privileges Act; International Treaty Tax-exempt organizations; and property belonging to any institution with which Government has contractual obligation not to levy fees and tax.

### *II.B. Property Revaluation, Enumeration, and Tax Collection in Kampala*

To increase local revenue mobilization, the Kampala Capital City Authority (KCCA) initiated a citywide property revaluation and enumeration exercise beginning in the financial year 2016/17. Prior to this, the last valuation had been conducted in 2006, implying that the taxable values of properties remained unchanged for a decade. Over the same period, numerous newly developed properties were not captured in the existing tax roll. The number of properties on the tax roll increased by 76 percent, from 127,021 to 223,329, while the average property tax liability increased by 90 percent after the exercise. The revaluation and enumeration exercise was therefore expected to substantially expand tax revenue for KCCA, where property taxes had accounted for 35 percent of own-source revenue prior to the exercise. During the mass enumeration and survey exercise, KCCA collected detailed information on ownership, location (including GIS coordinates), and property attributes for valuation purposes. Property valuation was conducted primarily by professional valuers, although in some cases it relied on regression model predictions.

The exercise was implemented in a staggered manner, starting with one division and gradually extending to others over time. Figure 1 shows the rollout of the new tax roll across the five divisions of the capital city, following the revaluation and enumeration exercise. The different colors indicate the rollout phases of the exercise, as indicated in the legend.

Panel A: Kampala Capital City




Panel B: Greater Kampala Metropolitan Area




Figure 1: Kampala Capital City and Surrounding Areas

*Notes:* Panel A shows the five divisions of Kampala Capital City and the different phases of the rollout of the new property tax roll, as indicated in the legend. Panel B shows the Greater Kampala Metropolitan Area (GKMA), which includes counties in the neighborhood of the capital city. Firms in these locations serve as the never-treated units in the identification strategy, as these areas did not experience similar changes in property tax enforcement during the study period.

The exercise resulted in a new property tax roll that both expanded coverage and increased the rateable values assigned to existing properties. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive-margin increase in tax liability, i.e., from zero to a positive tax liability, while the statutory tax rate remained unchanged in both cases. Figure 2 reports the number of properties and potential property tax revenue across the five divisions under both the old and new tax rolls. All divisions experienced an increase in the number of properties, with the number more than doubling in some divisions (Figure 2, Panel A). Similarly, potential revenue per property—a proxy for tax liability—increased across all divisions, with some divisions experiencing increases exceeding 180 percent (Figure 2, Panel B).

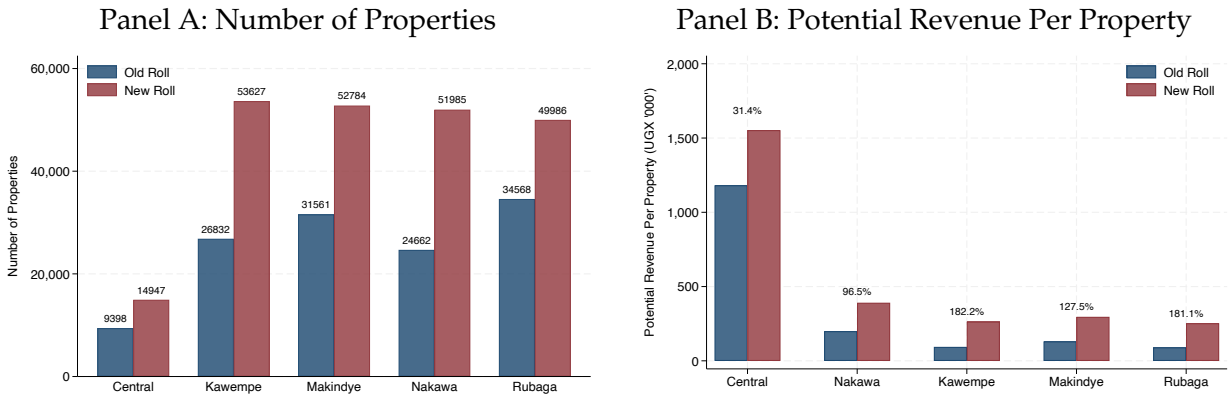



Figure 2: Number of Properties and Potential Property Tax Revenue

*Notes:* Panel A shows that number of properties in the old and new roll across the five divisions. Panel B shows the potential revenue per property before and after the reform. The data is extracted from Ahabwe et al. (2020) and Bakibinga and Ngabirano (2019).

*Property Tax Collection:*— Following the valuation exercise, preliminary valuation lists are displayed in hard copy at the respective division offices, with additional dissemination through the press, radio announcements, and local leaders. Property owners are then granted a one-month period to file objections, after which the valuation court convenes to hear all appeals. Appeal decisions may result in either upward or downward revisions to tax liabilities. Final valuation rolls are then generated, and bills and demand notes are printed and delivered to property owners. Although the City Authority is not able to physically deliver demand notes to every property owner, the digital platform, eCitie, sends SMS and email notifications to all billed property owners.[11] The City Authority

[11]Most of the information on the valuation and billing process was obtained from (Manwaring and Regan

is also legally authorized to issue a demand notice to a tenant if the property owner is unavailable, and the tenant is legally permitted to deduct the tax payment from rental payments.

Billing for all properties included on the final valuation roll occurs at the start of the financial year (July 1). Taxpayers have until December 31 to remit property tax payments without penalty. They may request payment plans, which are officially limited to a maximum of four equal installments. If payment is not made in time, KCCA may issue a demand notice requiring the taxpayer to settle the outstanding rate, along with any accrued interest, within two months of the notice being served. Interest begins accruing 30 days after the end of the financial year, at a rate of 2 percent per month, for as long as the liability remains unpaid. Payments can be made through commercial banks or via mobile money platforms.

## III. DATA AND DESCRIPTIVE STATISTICS

### *III.A. Data Sources and Sample Construction*

The paper uses administrative tax data from corporate income and employee tax returns as its primary datasets. The corporate income tax (CIT) return data comprise a firm-level panel covering the universe of firms registered for CIT in Uganda, spanning the period 2013/14 – 2021/22. The employee tax return data cover the universe of individuals in formal employment over the same period. These are newly compiled datasets launched in 2022 by the Uganda Revenue Authority with support from UNU-WIDER.

The CIT data contain annual balance sheets and profit and loss accounts, which provide detailed measures of firm activity, including revenues, profits, and assets. All monetary values are recorded in Uganda shillings. To ensure comparability across years, I deflate them to constant 2016/17 values using the consumer price index (CPI). The data also include information on firms' sectors of operation and locations. Location information is central to treatment assignment, as discussed later in the paper. Because the CIT data do not contain information on employment, I link them to the employee tax return data to construct a measure of formal employment by counting the number of workers filing under each firm. Since employee tax returns are filed monthly while corporate income tax returns are filed annually, I compute average employment over the twelve months of each year. A notable limitation of the employee-level data is that employees can only be uniquely identified at firm level and therefore cannot be individually tracked either across employers or over time (Agaba et al. 2023). Consequently, all analyses using

2019).

employee-level data are conducted at the firm level.

*Sample Construction*:— The full CIT dataset comprises 404,527 firm–year observations. However, some firms report zero turnover throughout the entire sample period while others report positive revenues in some years but not others. For analyses other than firm survival, I restrict the sample to firms that report positive revenues whenever present in the sample. This yields 133,494 firm-year observations. The sample is further restricted to firms located in the Greater Kampala Metropolitan Area (GKMA) to ensure geographic comparability between treated and never-treated firms. Because the reform affects only firms in the capital city, firms in surrounding locations in the rest of the GKMA (Figure 1, Panel B) serve as the comparison group. This restriction yields 94,527 firm–year observations, representing 71 percent of the observations from firms that reported positive revenues whenever present in the data. Of these, 33,021 observations form a balanced panel comprising 3,669 unique firms.

Additional sample restrictions vary by section or analysis. For the baseline analyses of firm survival effects, the sample is restricted to firms that were active and reported positive revenues in the pre-reform period. For the analysis of the effect on business rental expenses, the sample is restricted to firms that reported paying rent in the pre-reform period, as this outcome is normalized by its pre-reform value.

### *III.B. Descriptive Statistics*

Figure 3 plots the industry composition and size distribution across treated and comparison locations for firms that were active in the pre-reform year. The figure reveals substantial overlap along these dimensions. The firm size distribution is highly skewed to left: the majority of firms have annual revenues below UGX 150 million and fewer than five formal employees. However, the distribution is not systematically different for firms inside versus outside the city. Appendix Figure A.1 shows a similar overlap using density plots for revenues and assets. There are, however, some differences in industry composition across locations. For example, in retail and wholesale trade, the share of firms inside the city exceeds that outside the city by a considerable margin, while the reverse holds for the education sector. In the analysis, I account for these differences by conditioning the treatment effects on firm size and industry.[12] There is sufficient overlap in both dimensions to allow for comparisons of firms within the same industry and size bins across treated and comparison locations. Nonetheless, my identification strategy does not require industry composition or the firm-size distribution to be balanced across locations,

[12] I use industry and sector interchangeably to mean the same thing.

as it instead relies on the parallel trends assumption, as discussed in the next section.




Figure 3: Firm Size Distributions and Industry Composition

*Notes:* This figure plots the industry and size mix of pre-reform active firms located inside (treated) and outside (comparison) the city. Each graph's bars sum to one within each location. Panels A and B show the distribution of revenues and employment, respectively. Panel C shows the industry composition of firms in the sample following the International Standard Industrial Classification of all Economic Activities (ISIC, Rev.4). Utilities combines electricity, gas, steam, and air conditioning supply with water supply, sewerage, waste management, and remediation activities. Administration combines administrative and support service activities with public administration & defense and compulsory social security. I exclude ISIC categories for activities of households as employers; undifferentiated goods- and services-producing activities of households for own use; and activities of extraterritorial organizations and bodies, as these account for negligible shares in the sample.

Table 1 reports pre-reform means and standard deviations for selected variables across

different groups. The average firm in the universe of corporate income tax (CIT) filers has annual revenues of UGX 556 million and total assets of UGX 267 million, of which 70 percent are fixed assets. Average revenues for treated and comparison firms exceed those in the CIT universe, but are broadly comparable across the two groups. In contrast, firms in the comparison group hold more assets on average than treated firms, suggesting that firms in the city generate higher revenues per unit of assets. Similarly, average formal employment is higher among comparison firms than among treated firms. Again, these differences are inconsequential for the identification strategy as it relies on the parallel trend assumption.

The average firm in the CIT universe pays annual rent of UGX 6.2 million and owns land and buildings worth UGX 72 million. Comparison firms have lower average rent expenses and are less likely to be renters (i.e., report positive rent expenses) compared to treated firms. In contrast, comparison firms have higher average values of land and buildings and are more likely to own property (report positive values for land and buildings). To assess the implications of these differences, I disaggregate the main results by whether firms are tenants or owner-occupiers.

Lastly, I present statistics on variables capturing firms' financial positions. Firms with better financial situations may respond differently to cost increases associated with property tax liabilities. Table 1 shows that the share of loss-making firms is balanced at 23 percent across comparison and treated firms and slightly lower in the CIT universe. Similarly, average profit margins (EBIT/revenue) are similar across comparison and treated firms. Treated firms have higher liquidity (cash-to-assets) relative to comparison firms. While leverage is slightly higher in the CIT universe at 41 percent, it is balanced across treated and comparison firms. Overall, treated and comparison firms exhibit similar pre-reform financial characteristics.

## IV. IDENTIFICATION STRATEGY

I use a difference-in-differences design to estimate the causal effects of the reform on various measures of business activity, distributional consequences, and the fiscal externalities. Identification comes from comparing outcomes for firms located in Kampala with those for firms located in surrounding areas that were not exposed to changes in property tax liabilities during the sample period. The identifying assumption is that, absent the reform, average outcomes for firms in treated and untreated areas would have evolved similarly. This parallel trends assumption does not require that the two groups of firms are identical in levels—treated and comparison firms may differ in size, sector compo-

Table 1: Summary Statistics

| | (1) Universe | (2) Ever-Treated | (3) Comparison Group |
|---|---|---|---|
| Revenues | 556.09 | 614.96 | 629.81 |
| | (1,232.30) | (1,294.14) | (1,328.00) |
| Total Assets | 267.43 | 284.87 | 361.35 |
| | (595.70) | (620.41) | (696.65) |
| Fixed Assets | 188.27 | 190.70 | 294.15 |
| | (461.14) | (468.84) | (568.66) |
| Formal Employees | 5.16 | 5.15 | 7.45 |
| | (10.69) | (10.43) | (13.00) |
| Rent Expenses | 6.15 | 8.39 | 4.43 |
| | (12.84) | (14.83) | (11.27) |
| Landing and Buildings | 72.33 | 63.74 | 130.18 |
| | (209.02) | (202.33) | (270.43) |
| $\mathbb{I}\{\text{Rent Expense} > 0\}$ | 0.44 | 0.50 | 0.34 |
| | (0.50) | (0.50) | (0.47) |
| $\mathbb{I}\{\text{Landing and Buildings} > 0\}$ | 0.20 | 0.15 | 0.31 |
| | (0.40) | (0.36) | (0.46) |
| $\mathbb{I}\{\text{Loss Making}\}$ | 0.20 | 0.23 | 0.23 |
| | (0.40) | (0.42) | (0.42) |
| Profit Margins | 0.15 | 0.13 | 0.14 |
| | (0.31) | (0.31) | (0.33) |
| Cash-to-Assets | 0.17 | 0.17 | 0.13 |
| | (0.22) | (0.23) | (0.20) |
| Debt-to-Assets | 0.41 | 0.35 | 0.36 |
| | (0.36) | (0.35) | (0.35) |

Note: The table reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. Standard deviations are reported in parentheses. Column 1 presents statistics for the CIT universe, while the remaining columns present statistics for different groups according to their treatment status. The ever-treated group consists of firms located in Kampala City, whereas the comparison group consists of firms from the rest of the GKMA. All monetary values are denominated in Uganda Shillings (UGX), constant 2016/17.

sition, or baseline performance—but rather that average outcomes would have evolved along similar trajectories had the reform not occurred. I provide evidence in support of this assumption using event-study analysis.

A conventional approach would estimate the average and dynamic effects of the re-

form using Two-Way Fixed Effects (TWFE) difference-in-differences regressions, as in equations 1 and 2, respectively.

$$y_{it} = \alpha_i + \alpha_t + \beta D_{it} + X'_{i0}\gamma_t + \epsilon_{it}, \tag{1}$$

$$y_{it} = \alpha_i + \alpha_t + \sum_{\substack{e=-5 \\ e\neq -1}}^{4} \beta_t D^e_{it} + X'_{i0}\gamma_t + \nu_{it}, \tag{2}$$

where $y_{it}$ is the outcome of firm $i$ in year $t$. $D^e_{it} = \mathbf{1}\{t - G_i = e\}$ is an indicator for unit $i$ being $e$ periods away from initial treatment at time $t$, and $G_i$ denotes the period when unit first becomes treated. $\alpha_i$ is firm fixed effect to absorb time-invariant differences across firms, while $\alpha_t$ is time fixed effect. $X_{i0}$ is a vector of pre-reform firm characteristics, including sector of operation, firm size, and accounting month. The term $X'_{i0}\gamma_t$ allows these characteristics to have year-specific effects, thereby allowing for differential trends. $\epsilon_{it}$ and $\nu_{it}$ are error terms.

However, recent literature shows that under staggered adoption, TWFE estimators are consistent only under homogeneous treatment effects. In the presence of heterogeneous effects across cohorts or over time, TWFE coefficients are generally biased and difficult to interpret (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Callaway and Sant'Anna 2021; Borusyak et al. 2024). Bias in TWFE estimators arises from negative weighting induced by "forbidden comparisons", whereby already-treated units are used as controls for later-treated units (Borusyak et al. 2024). Heterogeneous effects are plausible in this setting for two reasons. First, firms' responses to increases in property tax liabilities may evolve over time as firms adjust along different margins. Second, changes in property tax liability vary across cohorts (Figure 2), implying that treatment effects may differ across cohorts.

Appendix Table A.14 presents the De Chaisemartin and d'Haultfoeuille (2020) decomposition of treatment effects according to whether they receive positive or negative weights under conventional TWFE estimation. Across the firm survival, firm revenue, formal employment, and total tax revenue outcomes, between 27 and 35 percent of the treatment effects receive negative weights, although the aggregate weight assigned to these negatively weighted effects is relatively small for each outcome.

I therefore estimate treatment effects using the doubly robust estimator proposed by Callaway and Sant'Anna (2021), with never-treated units serving as the comparison group. This estimator remains consistent in the presence of cohort-specific treatment effects and accommodates conditional parallel trends, requiring only that treated and comparison units follow similar trajectories after conditioning on observed covariates. All

results are robust to using not-yet-treated firms as the comparison group instead.

Rather than estimating dynamic effects directly from the event-study regression in equation 2, the approach recovers group-time average treatment effects and aggregates them by exposure length, as in equation 3, thereby producing event-study estimates that are free from the bias induced by forbidden comparisons under TWFE.

$$\beta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{I}\{g + e \leq \mathcal{T}\}\, \mathbb{P}(G = g \mid G + e \leq \mathcal{T})\, ATT(g, g + e), \tag{3}$$

where, $e$ denotes event-time (the number of periods since initial treatment, i.e., $e = t - g$), G denotes the first period a unit is treated, and $ATT(g, t)$ is the average treatment effect for units who are members of a particular group $g$ at a particular time period $t$, i.e., $ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid X, G_g = 1]$. $\beta_{es}(e)$ is therefore the average effect of participating in the treatment $e$ time periods after the treatment was adopted across all groups that are ever observed to have participated in the treatment for exactly $e$ time periods. Note that I condition the group-time estimates on observed pre-treatment covariates (e.g., sector of operation, firm size) to allow for covariate-specific trends. The identification therefore relies on conditional parallel trends assumption. Appendix Section III shows that the estimates for the main outcomes are robust to a stacked difference-in-differences approach following Cengiz et al. (2019), which likewise avoids relying on already-treated units as controls. The regressions are weighted by the logarithm of average revenues over two pre-reform years, yielding policy-relevant estimates of the average treatment effects. I also report unweighted regressions in the Appendix.

The timing of the reform was not randomly assigned across divisions. The City Authority initially prioritized areas with relatively high-value properties, beginning with Central Division and subsequently Nakawa (Figure 2, Panel B). This targeted rollout does not in itself threaten the difference-in-differences design, but it raises the possibility that early-treated areas may have followed different outcome trajectories even in the absence of the reform. I address this concern by conditioning the group-time treatment effects on predetermined firm characteristics (such as size, sector of operation, and account month) and exploiting not-yet-treated firms as the primary comparison group. The event-study estimates further allow me to assess whether treated and comparison firms exhibited differential trends prior to the reform.

A potential threat to the identification strategy is that other shocks, such as changes in local regulations, economic or political conditions, or other city-level policies, may have coincided with the reform and differentially affected economic activity within Kampala relative to the surrounding metropolitan area. For such shocks to explain the results, how-

ever, they would need to coincide not only with the overall reform period but also with the staggered timing of implementation across Kampala divisions. Otherwise, specifications using not-yet-treated firms as the comparison group would minimize the potential influence of such shocks. The COVID-19 pandemic presents a particularly important potential confounder, as it generated substantial disruptions to economic activity and may have affected firms in the urban core differently from those in surrounding areas. I directly examine this possibility in Section VI.C and find no evidence that differential exposure to the pandemic drives the estimated effects.

## V. EFFECTS ON BUSINESS ACTIVITY

### *V.A. Effects on Firm Survival*

A natural starting point for assessing the effects of property taxation is the extensive margin: whether firms continue operating. A firm's decision to remain in business depends on whether the expected value of continued operation exceeds the opportunity cost of exit, a margin that a permanent increase in property tax liabilities can directly erode. Unlike variable costs, which scale with output, property taxes are largely fixed obligations that do not adjust with firm performance, making them particularly burdensome during periods of low revenue and disproportionately consequential for marginal firms. The survival margin is therefore an important channel through which property taxation can affect both the level and composition of local economic activity. This section investigates whether the reform affected the probability of firm survival.

*Constructing the Survival Outcome*:— Analyzing firm survival in firm-level panel data is challenging because firm exits inherently render the sample endogenous to the reform. To address this, I restrict the cohort to firms present in 2015/16 (the year prior to the reform), with positive turnover, regardless of whether they operated in other years.[13] I then construct a balanced panel by expanding each firm to all sample years and assigning zero values for turnover in the years the firm did not operate. The survival outcome is then defined as an indicator equal to one if a firm has positive turnover in year $t$ and zero otherwise. A similar approach was used by Saez et al. (2019) when analyzing whether payroll taxes affect firm survival in Sweden. By construction, the mean of the outcome variable is one in the reference year; it can be below one before 2015/16 because some firms had not yet entered, and below one after 2015/16 because some firms exit following the reform.

[13] As a robustness check, I repeat this procedure using the cohort of firms present in the first year of the sample (with positive turnover).

However, as highlighted by Saez et al. (2019), using this approach to estimate the effects on firm survival requires reweighting firms so that pre-reform distributions are aligned along key dimensions across different groups, since unweighted estimates can easily reflect compositional differences rather than the causal effects. For example, if younger firms, which tend to have lower baseline survival rates, are unevenly distributed across treated and untreated locations, unweighted estimates may simply capture age-related compositional differences rather than the effect of the tax reform itself. To address this, I reweight firms using the nonparametric reweighting method of DiNardo et al. (1996), with the weights constructed following Yagan (2015). Observations are first weighted by average pre-reform turnover, so that estimates reflect firms' economic scale and larger firms contribute proportionally more to the parameter estimates. I then reweight the sample across industry-size bins to ensure that treated and control groups are balanced along these dimensions in the pre-reform period. Appendix Figure A.2 shows the raw trends in weighted survival rates for eventually treated and never-treated firms over time, without adjusting for differences in treatment timing or covariates.

*Effects on Firm Survival*.— Figure 4 shows the effects of the reform on firm survival using the reweighted sample constructed following the procedure described above. In the pre-reform period, there is no evidence of differential trends in survival probabilities between firms located in areas that experienced increases in property tax liabilities and those in areas that did not, as all pre-reform coefficients are statistically indistinguishable from zero. This finding supports the underlying identifying assumption that, absent the reform, survival probabilities for treated and untreated firms would have evolved similarly, supporting a causal interpretation of the post-reform estimates.

Following the reform, there is a differential decline in survival probabilities for firms located in areas that experienced an increase in property tax liabilities relative to those in locations that did not experience the reform, implying that the reform increased firm exit. The negative effects start to manifest after one year of the reform, grow in the second year before it stabilize at approximately 5 percentage points and persists up to the end of the sample. The corresponding average effects are presented in Table 2. Column 4 shows the preferred specification which includes the full set of controls. Comparing firms in locations exposed to increases in property tax liabilities with firms in other locations, the reform led to a 3.2 percent decline in survival probabilities with a standard error of 1.1 percent. Scaling this effect by the aggregate increase in potential property tax liabilities implies that a $1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits (see bottom row of Table 2).

A key concern with this approach to measuring firm survival is that it may be con-

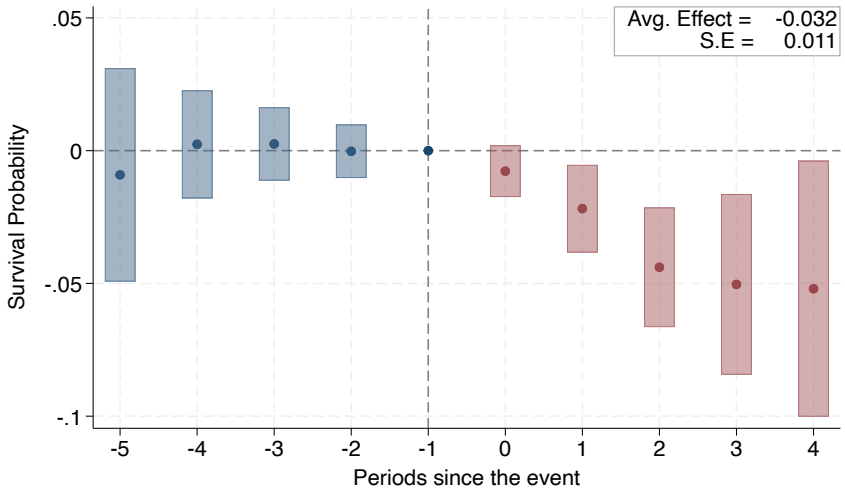



Figure 4: Effects of the Property Tax Reform on Firm Survival

*Notes:* The figure presents the effects of the reform on firm survival probability. The outcome variable is an indicator equal to one if the firm is operating at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The regressions are estimated on a balanced panel comprised of 123,864 firm-year observations. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at −1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The group-time treatment-effect estimates condition on firms' baseline sector of operation, pre-reform size, and accounting month.

founded by evasion responses. Firms may respond to the higher costs associated with increased property tax liabilities by ceasing to file income tax returns, in which case this approach would misclassify them as exits. I implement several strategies to address this concern. First, I use a cross-tax validation strategy to identify firms' operational footprints. Using employee-level tax return data in combination with CIT data, I classify evasion as cases where the firm disappears from the CIT records but still has employees associated with its tax identification. Accordingly, I classify a case as exit if the firm disappears from the CIT records and has no footprints in the employee tax returns, i.e., both tax records go silent. Appendix Figure A.4 shows that my estimates are unlikely to be driven by an evasion response. A likely criticism of this validation approach is that firms may collude with workers to evade both taxes. However, the timing mismatch between the two tax returns makes such collusion practically difficult. Employee tax returns are filed monthly, whereas corporate income tax (CIT) is filed annually. Since I classify a case as evasion even if the firm's identifier appears in only one month, for collusion to occur, the firm would have to decide to evade at the start of the income year, before the realization of revenues. Moreover, workers would need to be compensated for agreeing to

Table 2: Effects of the Property Tax Reform on Firm Survival

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| ATE | -0.036*** | -0.029*** | -0.029** | -0.032*** |
| | (0.011) | (0.011) | (0.011) | (0.011) |
| Sector | No | Yes | Yes | Yes |
| Pre-reform Size | No | No | Yes | Yes |
| Other Controls | No | No | No | Yes |
| Observations | 130864 | 123864 | 123864 | 123864 |
| Exits per $1 Million | 28.35 | 22.90 | 22.71 | 25.34 |

Note: The table presents average effects of the reform on firm survival probabilities. The outcome variable is an indicator equal to one if the firm is operating at time $t$, zero otherwise. Because the mean of the dependent variable is one in the pre-reform year by construction, the coefficient estimates can be interpreted as a percent change in survival probability. In the preferred specification (column 4), the estimation also conditions on pre-reform accounting month and occupancy status in addition to the covariates indicated in table. The bottom row reports the citywide reform-induced firm exits per additional $1 million in aggregate property tax liability, calculated as $\frac{\hat{\beta} \cdot N}{\Delta T^{\text{City}}}$, where $N$ denotes the total number of active formal firms in the city prior to the reform and $\Delta T^{\text{City}}$ is the reform-induced change in citywide property tax liability. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with never-treated units used as the comparison group. Standard errors are clustered at the firm level.

collude, as doing so would entail forfeiting other benefits such as social security. This creates an additional friction limiting firms' ability to engage in coordinated evasion. To my knowledge, this paper is the first to cleanly separate survival effects from evasion response using cross-tax referencing.

*Other Robustness Checks*.— I conduct several robustness checks to assess the sensitivity of the survival effects, with results reported in the Appendix. First, I redefine exit as the absence of a corporate income tax (CIT) return for at least two consecutive years, thereby excluding firms that temporarily disappear for one year before subsequently re-entering. Second, I impose a stricter definition of exit that requires no CIT return in year $t$ and no record in employee tax returns in both years $t$ and $t+1$. As shown in Figure A.5, the estimated effects remain stable under both definitions. Third, I reconstruct the survival outcome following the same procedure as before but restrict the sample to the cohort of firms that were present and reported positive turnover in the first year of the sample. Figure A.6 shows that the results are consistent with those obtained using the baseline sample. Fourth, I exclude Central Division and restrict the treated sample to firms in divisions that border the comparison areas, thereby improving the geographic comparability of treated and control firms. Figure A.7 shows that the negative survival effects persist in this narrower sample. As a final robustness check, I assess whether the findings

are sensitive to the choice of comparison group in the Callaway and Sant'Anna (2021) difference-in-differences design. Figure A.8 shows that the results remain robust when using not-yet-treated firms as the comparison group rather than never-treated firms.

Another concern is that general equilibrium adjustments to the reform may violate the stable unit treatment value assumption (SUTVA), specifically the no-interference condition, by transmitting effects to firms in comparison locations. Two channels are plausible. First, if the reform depresses entry in treated locations, firms that would otherwise have entered treated areas may relocate to comparison areas, intensifying competition faced by comparison-location incumbents and raising their exit rate. This would attenuate the estimated treatment effect. Second, if the reform pushes economic activity from treated to comparison locations more broadly, comparison-location incumbents may benefit from increased demand and lower their exit rate, inflating the estimated treatment effect. The net direction of any resulting bias is therefore ambiguous a priori. To assess whether these spillovers materially affect the estimates, Appendix Figure A.9 restricts the never-treated group to firms geographically distant from treated areas, where spillover exposure should be smallest. Survival effects are similar to the baseline, suggesting that if spillovers are present, they do not drive the results.

*Net Effects*.— The above analysis focuses only on firm exits and therefore does not capture net effects operating through both entry and exit. A reform that raises operating costs may simultaneously accelerate exit among incumbents and deter entry by potential entrants, both of which would reduce the stock of active firms. To examine these net effects, I construct a balanced panel at the county–industry level, using four-digit industry classifications, and analyze how the number of firms evolves before and after the reform across locations that experienced an increase in property tax liabilities relative to those that did not. I define the outcome variable as the log number of firms in each county-industry cell, which captures both incumbent exit and reduced entry. The results, presented in Appendix Figure A.10, show a differential decline in the number of firms operating in treated locations relative to untreated areas following the reform. This finding complements the survival analysis by confirming that the reform reduced the total stock of active firms, not merely the survival rates of incumbents. The decline in firm counts therefore reflects a combination of accelerated exit and reduced entry, both of which are consistent with higher operating costs reducing the returns to formal business activity in treated locations.

### *V.B. Revenues, Profits, Investment, and Raw Material Purchases*

Higher property tax liabilities raise the cost of occupying commercial space, directly for owner-occupiers through higher tax bills, and indirectly for tenants to the extent that landlords pass the tax through to rents. These cost increases tighten liquidity constraints, which may affect multiple dimensions of firm performance. To examine these effects, I estimate the reform's impact on revenues and profits using a balanced panel of active firms.[14] Revenues are expressed in logarithms, while profits are scaled by pre-reform revenues to facilitate economically meaningful interpretation. All regressions are weighted by the logarithm of pre-reform revenues. Unweighted results on the same sample are qualitatively similar (Appendix Figure A.12).

Figure 5 presents the results. The absence of differential pre-reform trends across both outcomes among firms in treated areas relative to those in untreated areas supports the parallel trends assumption underlying the identification strategy. Following the reform, I find no statistically significant effects on revenues, suggesting that the reform did not substantially alter firms' scale of operations. Profits show a modest short-run positive effect that dissipates quickly, which, combined with the null revenue result, suggests that firms adjusted on cost margins to preserve profitability in the face of higher property taxes. I explore this possibility in the next section by analyzing employment responses. Appendix Figure A.11 shows a null effect on raw material purchases and negative but imprecisely estimated effect on investment, providing suggestive evidence that the reform may have discouraged capital accumulation.

These estimates are conditional on firm survival, as the sample is restricted to firms that reported positive revenues throughout the panel. To the extent that the reform induced firm exit, as shown in the preceding section, these results capture the experience of firms that successfully weathered the shock. Conditioning on continued operation may, however, generate survivorship selection. To assess the quantitative importance of this selection, Appendix Figure A.14 expands the sample to include all firms reporting positive revenues in the pre-reform period, follows this fixed cohort in all subsequent years, and assigns zero revenue in years in which a firm no longer operates. This specification therefore incorporates the extensive-margin response through firm exit. The estimated average effect on revenues is $-0.033$ (S.E. $= 0.028$) and remains statistically imprecise. The downward shift in the point estimate is consistent with some positive selection among surviving firms, but accounting directly for the exit margin does not materially alter the conclusion that the reform did not generate a statistically detectable average decline in

[14] Active firms are defined as those reporting positive turnover throughout the sample period. Results from the unbalanced panel are reported in Appendix Figure A.13 and are qualitatively similar.

firm revenues.

As a complementary sensitivity exercise, Appendix Table A.1 reports Lee-style bounds that allow for differential post-reform selection into the positive-revenue sample. For log revenues, the lower and upper bounds are $-0.070$ and $0.024$, respectively, while the corresponding bounds for profits are $0.002$ and $0.042$. The revenue bounds therefore indicate that survivorship selection could mask a negative effect on revenues, although the identified range still includes effects close to zero. For profits, both Lee-style point bounds are positive, providing no evidence that selection masks adverse effect on profitability. Taken together, these exercises suggest that positive selection among surviving firms may attenuate the estimated revenue response, but offers little indication that survivorship selection conceals declines in profits.

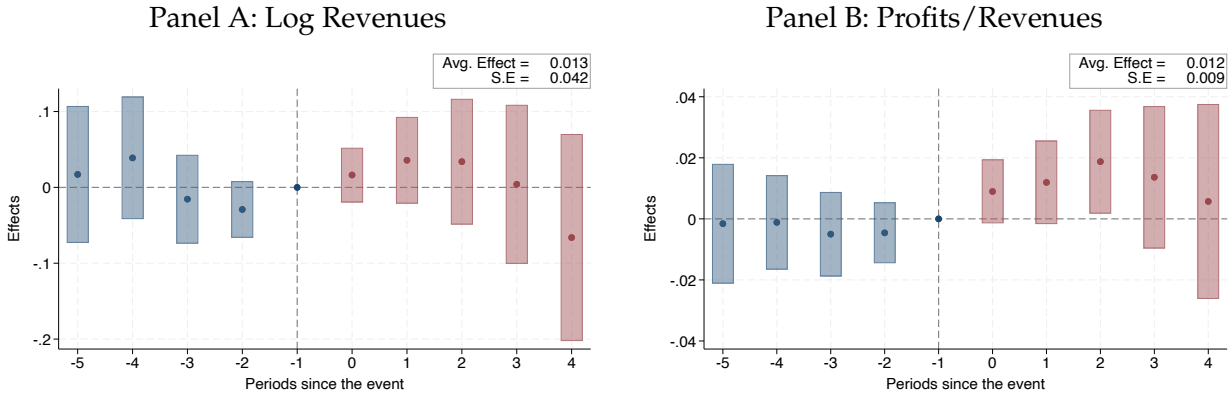



Figure 5: Effects of the Reform on Revenues and Profits

*Notes:* The figure presents the effects of the reform on firm revenues and profits. Revenues are measured in logarithms, while profits are measured as earnings before interest and taxes (EBIT) and scaled by the firm's average revenue over the two pre-reform years. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 33,012 firm-year observations. The horizontal axis indicates the number of years since the reform. Each dot represents the estimated treatment effect for a given event year, with event time $-1$ omitted as the reference period. The vertical bars indicate 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

The aggregate effects reported in Figure 5 mask substantial heterogeneity by firm size. Figure 6 decomposes the treatment effects on log revenues (Panel A) and profit margins (Panel B) by size, with average effects summarized in Panel C. The two groups respond in opposite directions. Large firms experience a statistically significant increase in revenues and profit margins. Small firms experience a decline in revenues following the

reform, although the point estimate is imprecisely estimated, while profit margins show no meaningful change.

Two mechanisms could explain this pattern. One possibility is that large firms possess greater market power and pass a share of the tax burden onto consumers through higher output prices, while small firms in more competitive segments cannot (Weyl and Fabinger 2013; Ganapati et al. 2020). An alternative is a reallocation channel operating via equilibrium adjustments in local input or output markets. For instance, the reform might have disproportionately induced exit or downsizing among small firms, freeing up productive inputs whose local prices then fall and benefit surviving large firms.[15] This second mechanism can be formalized within a simple general-equilibrium model, fully developed in Appendix IV.

The model features two inputs: commercial space and a locally supplied input $X$ whose price is unobservable in the data.[16] It rests on two ingredients. First, firms differ in the intensity with which they use commercial space, with small firms more space-intensive than large firms, a pattern documented in Figure A.15. Second, firms are linked through the local market for $X$, whose price adjusts in general equilibrium. The reform raises the marginal cost of space, disproportionately contracting space-intensive small firms, and thereby reducing aggregate demand for $X$. Because $X$ is locally supplied, its equilibrium price falls, and the decline disproportionately benefits large firms, which use $X$ more intensively. Appendix Figure A.16 provides direct evidence consistent with this reallocation. Following the reform, small firms reduce their local input purchases, while large firms modestly expand theirs, consistent with the shift in local input demand from space-intensive small firms toward less space-intensive large firms that the model predicts.

## VI. LABOR MARKET OUTCOMES AND DISTRIBUTIONAL CONSEQUENCES

### *VI.A. Effects on Formal Employment and Wage Bill*

The preceding results broadly suggest that firms preserved revenues and profitability by adjusting on cost margins. A natural candidate is labor, both employment levels and wages. I examine these outcomes on the balanced panel of active firms used in the preceding analysis, isolating workforce and wage adjustments among continuing firms independently of the exit margin.

[15] A model that uses equilibrium adjustments in the output market would produce a similar predictions.

[16] Candidates for $X$ include local labor, particularly for low-to-medium-skill workers with limited mobility across jurisdictions; local intermediate inputs supplied by other firms in the same geographic market; or local services such as logistics whose prices reflect local supply and demand conditions.

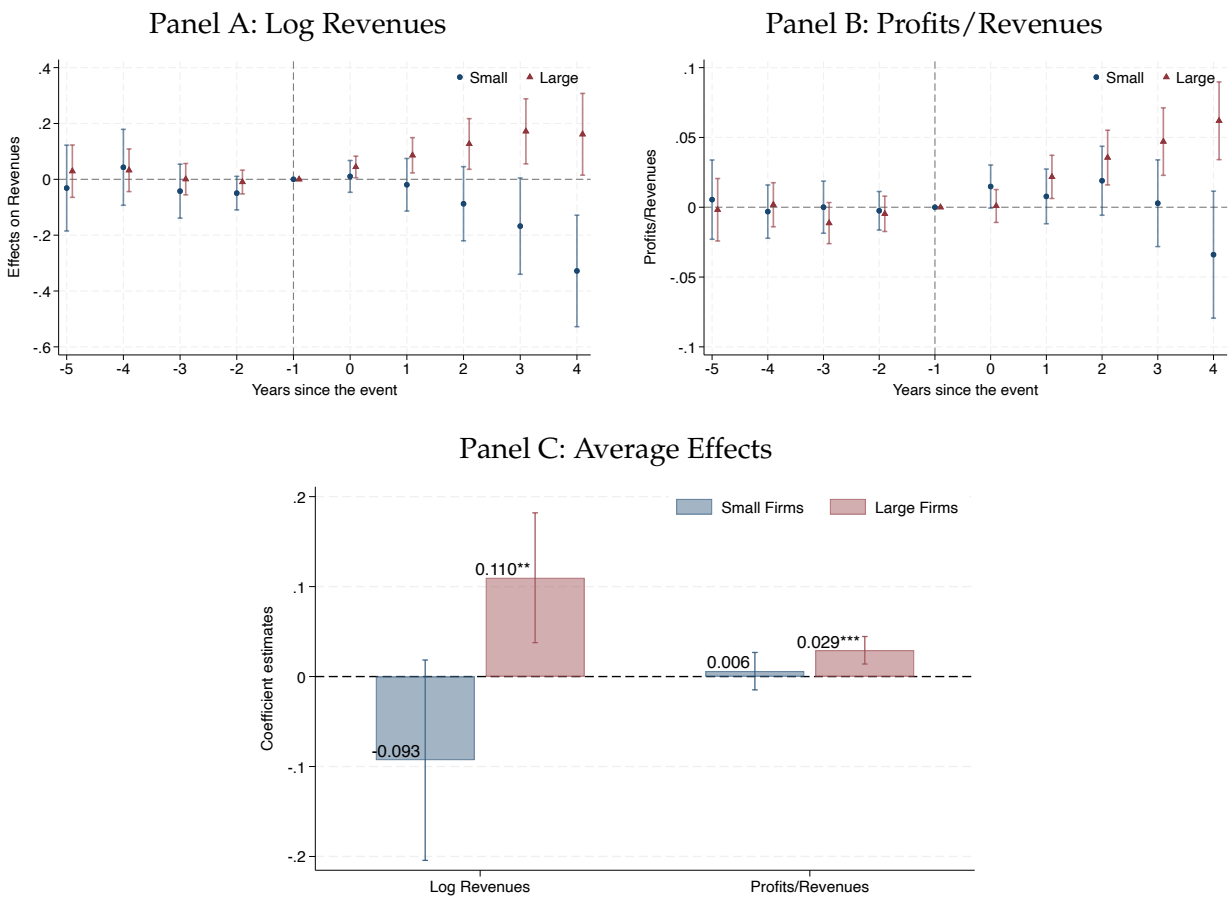



Figure 6: Effects of on Revenues and Profits, by Firm Size

*Notes:* The figure presents the effects of the reform on firm revenues and profits, by firm size. Firms are classified as small and large using the median level of pre-reform employment as the cutoff. Revenues are measured in logarithms, while profits are measured as earnings before interest and taxes (EBIT) and scaled by the firm's average revenue over the two pre-reform years. In Panel C, the outcome variables are labeled on the horizontal axis. All regressions are estimated on a balanced panel of firms, with observations weighted by the logarithm of pre-reform revenues. The sample for large firms comprises 16,470 firm-year observations while that for small comprises 16,542 firm-year observations. The horizontal axis in Panels A and B indicates the number of years since the reform. Each dot represents the estimated treatment effect for a given event year, with event time –1 omitted as the reference period. The vertical bars indicate 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

Because I observe only formal employment, which can equal zero even for active firms, I do not log-transform the outcome variables in the main specification. As discussed by Chen and Roth (2024), log transformations of outcomes with zeros complicate the interpretation of coefficients as percentage effects. Instead, I estimate effects in levels, winsorizing at the 5$^{th}$ and 95$^{th}$ percentile by year, and convert the estimates into

percentage terms using pre-reform averages. In the Appendix, I report results from an alternative specification that normalizes outcome variables by their two-year pre-reform averages prior to estimation, again winsorized at the same level.[17] This approach is common in the literature (e.g., Saez et al. 2019; Harju et al. 2022) and yields coefficients directly interpretable as percent changes. Appendix Figure A.19 shows that the results are qualitatively similar.

Figure 7 presents event-study estimates and 95 percent confidence intervals for the effects of the reform on formal employment and the total wage bill, estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). In both panels, pre-reform coefficients are close to zero and statistically insignificant, supporting the parallel trends assumption. Following the reform, formal employment declines differentially among firms in treated areas. The negative effect emerges in the reform year, grows through the third post-reform year, and persists through the end of the sample. The total wage bill follows a similar pattern, with treated firms experiencing differential declines of comparable timing. The growing magnitude of the employment effect over the first three post-reform years is consistent with gradual workforce adjustment: firms facing persistently higher operating costs shed workers incrementally rather than through immediate large-scale layoffs, a pattern consistent with the presence of adjustment costs in labor demand.

On average, formal employment among firms in treated areas declined by 3.8 percent relative to those in untreated areas, while the formal wage bill declined by 3.7 percent (Figure 7 and Appendix Table A.2). These findings offer a plausible explanation for why profitability remained largely unaffected, as documented in Section V.B: firms absorbed higher property tax costs in part by reducing their formal workforce. The fact that formal employment and the formal wage bill decline while revenues remain unchanged could be explained by firms substituting toward informal labor arrangements to maintain productive capacity at lower cost. In settings where enforcement is imperfect, formally registered firms can adjust on the intensive margin of informality by shifting workers off the books, reducing the formal payroll without proportionate losses in output (Ulyssea 2018). I examine this informalization channel directly in Section VI.B.

The results on employment and wages are robust to alternative sample restrictions and weighting choices. Appendix Figure A.17 shows that estimates from an unweighted specification, which gives equal weight to all firms regardless of pre-reform revenue, yield qualitatively similar patterns, indicating that the results are not driven by a small number of large firms. Appendix Figure A.18 further shows that estimates from an unbalanced

[17] This normalization restricts the sample to firms with positive employment in the pre-reform period.

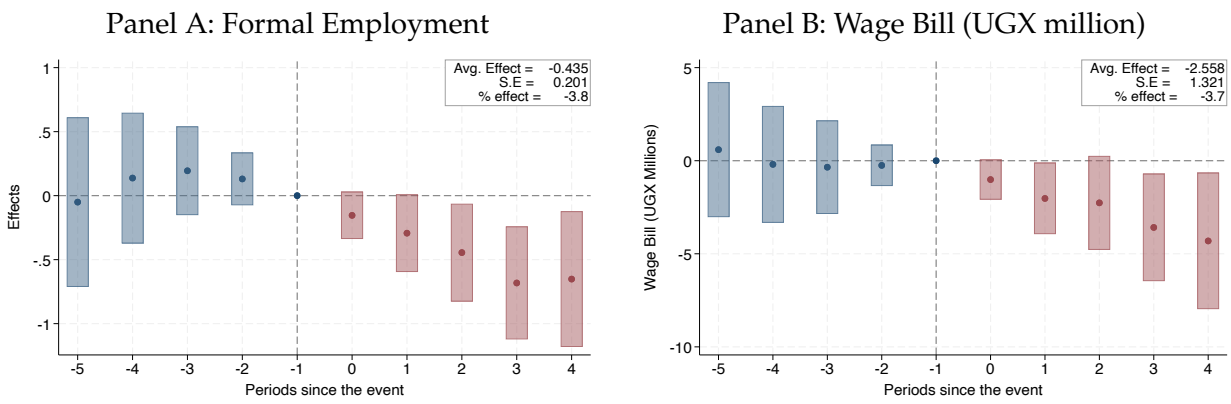



Figure 7: Effects of the Reform on Formal Employment and Wage Bill

*Notes:* The figure presents the effects of the reform on the number of formal workers and the total formal wage bill. The percent effects are computed by dividing the coefficient estimates by the pre-reform average of the outcome variable. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 33,012 firm-year observations. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Vertical bars indicate 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

panel yield similar patterns but somewhat larger magnitudes.

To assess the aggregate magnitude of the reform's effects on formal employment, I re-estimate the employment specification on a broader sample that includes all firms with positive revenues in the pre-reform period, expanded to all years of the sample and assigning zero formal employment in any year in which the firm is not active.[18] This sample differs from both the balanced and unbalanced panels used above: whereas the balanced panel restricts attention to firms active in every year of the sample and the unbalanced panel includes firms only in years in which they are active, this specification retains firms in the data after exit, with their employment set to zero. The estimated effect from this sample therefore captures the full scale of the reform's impact on formal employment, encompassing both workforce reductions among surviving firms and the complete destruction of formal jobs associated with firm exit. Appendix Figure A.20 shows that formal employment declined by 5.7 percent in this expanded sample.

I translate the estimated effect from this specification into an aggregate citywide measure. Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. More-

[18]This sample is similar to that used to analyze firm survival.

over, multiple firms may operate within the same property. I therefore cannot estimate firm-level employment elasticities with respect to property tax liabilities. Instead, I aggregate the reduced form employment effects across firms and scale them by the aggregate increase in potential property tax liabilities induced by the reform. The bottom row of Appendix Table A.3 shows that a $1 million increase in aggregate property tax liabilities is associated with approximately 320 fewer formal jobs citywide.

While not directly comparable owing to differences in the tax instruments studied, the populations affected, and the institutional settings, the negative employment effects documented here are broadly consistent with estimates from related tax literature, including studies of payroll taxes in Sweden (Saez et al. 2019; Egebark and Kaunitz 2018) and corporate taxes in the United States (Ljungqvist and Smolyansky 2014; Giroud and Rauh 2019).

A potential concern with the employment analysis is that workers displaced from treated firms may be absorbed by comparison firms, inflating the estimated treatment effect through labor reallocation rather than genuine job destruction. This concern is a violation of the stable unit treatment value assumption (SUTVA), specifically the no-interference condition, which requires that the outcomes of comparison firms are unaffected by the treatment of other firms in the sample. If displaced workers move to nearby untreated firms, comparison firms experience a mechanical increase in employment, inflating the estimated employment loss at treated firms and yielding a treatment effect that combines genuine contraction with cross-firm reallocation. I address this concern using two complementary approaches.

First, I construct an alternative comparison group restricted to firms outside the Greater Kampala Metropolitan Area (GKMA).[19] If reallocation is driving the results, displaced workers are most likely to be absorbed by nearby firms, precisely those in the original comparison group within the GKMA. Excluding these firms removes the channel through which reallocation could bias the estimates, so one would expect the employment effects to attenuate or disappear if reallocation were the primary driver. Appendix Figure A.21 shows that the negative employment effects persist under this restricted comparison group. Second, I conduct a placebo test that directly asks whether nearby firms gained employment. I restrict the sample to firms located outside Kampala city and compare those within the GKMA, which are geographically proximate to treated firms and therefore most likely to absorb displaced workers, against firms located outside the GKMA en-

[19]Specifically, this comparison group comprises firms in the 15 districts closest to the GKMA that are not themselves within the GKMA.

tirely.[20] If reallocation were occurring, employment should increase among GKMA firms relative to more distant firms following the reform. Appendix Figure A.22 shows no such differential increase, which is inconsistent with systematic absorption of displaced workers. The raw trends in Appendix Figure A.3 also show no evidence that displaced workers were absorbed in comparison locations. Taken together, these results indicate that the documented employment decline reflects genuine destruction of formal jobs rather than compositional reallocation across firms.

### *VI.B. Wage Informality*

Firms facing higher costs associated with increases in property tax liabilities may have an incentive to shift workers off formal payrolls, reducing their social security contributions and other payroll-linked obligations, while retaining their productive services through informal contracts. To the extent that the property tax shock tightens firms' liquidity constraints, it raises the value of actions that generate immediate cash-flow savings, making informalization particularly attractive even if the associated regulatory risk and organizational cost remain unchanged. In other words, the reform increases the firm's shadow value of cash without changing the informalization costs. To analyze this margin, I construct a proxy for informal wages, as the difference between the wage bill reported on firms' corporate income tax returns and the total wage bill derived from employee-level returns.[21] I normalize the outcome variable by the CIT wage bill, so that coefficients are interpretable as percentage-point changes.[22]

Figure 8 shows that the share of informal wages increased by 2.6 percentage points against a pre-reform mean of 10.3 percent. Appendix Table A.4 presents the detailed results. Appendix Table A.5 shows that wage informalization accounts for 62 to 68 percent of the decline in the formal wage bill.[23] This suggests that a substantial share of the observed decline in formal employment reflects a reallocation of labor toward informal arrangements rather than outright job destruction, implying that the aggregate employment estimates above overstate true job losses to the extent that some workers transition into informal roles rather than unemployment. Beyond this measurement implication, the informalization margin points to a broader distributional consequence of property taxation in contexts where informal labor markets are prevalent: the tax burden is partly shifted onto workers in the form of reduced labor protections and benefits associated with

[20] The comparison group and specifications mirror those used in the first approach.

[21] Informal wage $= \frac{\text{CIT wage bill - Total wage bill from employee returns}}{\text{CIT wage bill}}$

[22] The underlying assumption is that firms report truthfully on their wage bill on the corporate income tax returns.

[23] The event-study analysis corresponding to Table A.5 is presented in Figure A.23

formal employment.

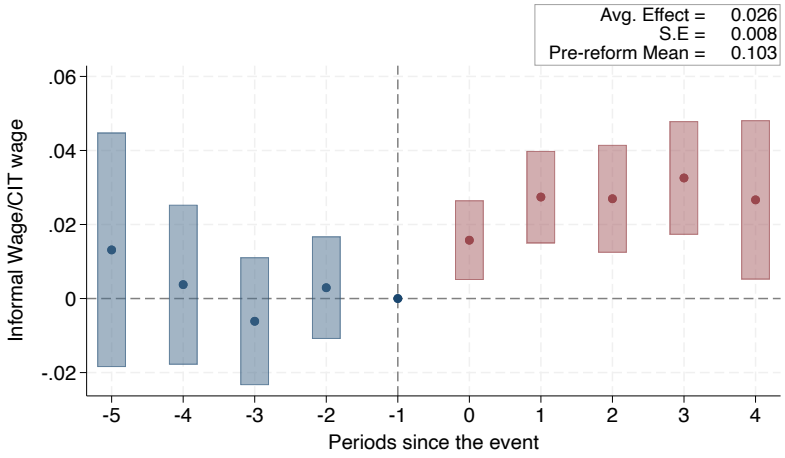



Figure 8: Effects on Wage Informality

*Notes:* The figure presents the effects of the reform on wage informality, measured as the fraction of the wage bill reported in corporate income tax returns that does not appear in individual employee income tax filings. This means coefficients can be interpreted as a percentage point changes. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 24,579 firm-year observations. Results from unweighted regressions are reported in Appendix Figure A.27 and are similar to the weighted estimates presented here. The horizontal axis reports the number of years relative to the reform implementation. Each dot represents the estimated treatment effect for a given year, with year −1 as the reference category. The vertical bars denote the corresponding 95 percent confidence intervals. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

A potential concern with this approach of analyzing wage informality is that the reform may incentivize firms to inflate their reported wage bill on corporate income tax returns in order to reduce taxable income, which would generate an artificial wedge between CIT-reported and employee-level wages. I provide three pieces of evidence against this interpretation. First, Appendix Figure A.23 shows that the reported CIT wage bill remains unchanged while the reported PAYE wage bill declines following the reform. If firms were inflating the CIT-reported wage bill, one would expect it to increase after the reform, not remain stable.

Second, I examine firms with no employees registered for PAYE in the pre-reform period, i.e., those for which any labor compensation reported on CIT returns must, by construction, correspond entirely to informal labor payments. This subgroup provides a particularly stringent test of the over-reporting hypothesis. If the increase in the informal wage share were primarily driven by strategic inflation of deductible labor expenses on CIT returns, then firms with no formally registered workers should face at least as strong an incentive to overstate wage bills following the reform. Indeed, because these firms

have no formal payroll against which reported labor expenses can be validated using PAYE filings, they are arguably the least constrained in their ability to inflate reported wages for corporate tax purposes. Appendix Figure A.24 provides no evidence of such behavior. The CIT-reported wage bill among these firms remains stable following the reform and, if anything, appears to decline. This pattern is difficult to reconcile with the hypothesis that the main results are driven by strategic over-reporting of deductible wage expenses on CIT returns.

Lastly, if firms were systematically over-reporting labor costs in response to the reform, one would expect a broad-based inflation of other deductible expense categories as well. Appendix Table A.6 provides evidence against this interpretation: expenses on machine repair decline while expenses on training, travel, telephone, and legal remain unchanged following the reform. Rather than reflecting strategic over-reporting, this pattern is consistent with firms cutting specific operating costs in response to higher property tax liabilities, supporting the interpretation that the wedge between CIT-reported and employee-level wages reflects genuine labor informalization rather than tax-motivated accounting manipulation.

*Survey Evidence on Workers' Social Security Coverage.*— As an out-of-sample validation of the wage-record results, I turn to the labour force module of the Uganda National Household Survey and test whether workers in the areas exposed to the reform experienced a decline in access to formal employment benefits relative to workers in unaffected areas. I focus on a single question that asks respondents whether they receive social security contributions from their current employer, and use it as a direct measure of formal-sector coverage. Appendix Table A.7 reports the results. Following the reform, the probability that a worker in the treated areas reports receiving social security contributions falls by roughly 3.6 percentage points relative to workers in the comparison areas. These results, however, should be interpreted as descriptive rather than causal: they rely on two independent cross-sections rather than a worker-level panel, so I cannot rule out compositional changes at the individual level or test the parallel-trends assumption directly. Even so, the survey-based evidence points in the same direction as the wage-record results, reinforcing the interpretation that the reform shifted employment toward less formal arrangements.

### *VI.C. Addressing the COVID-19 Threat*

A potential threat to identification is that the later years of the sample overlap with the COVID-19 pandemic, which disrupted firm operations and labor markets in ways

that may confound the estimated effects of the reform. The identifying assumption required for my estimates to be uncontaminated by COVID-19 is that the pandemic did not differentially affect treated and comparison firms. This assumption is plausible for three reasons. First, I restrict never-treated comparison units to firms located just outside the city boundary, which are likely to have faced similar pandemic exposure as treated firms within the city. Second, the results are similar when using not-yet-treated firms rather than never-treated firms as the comparison group. Because not-yet-treated group includes firms located within the city and thus subject to the same local economic conditions, this comparison further limits the scope for differential COVID-19 exposure between treated and comparison groups.

Third, as I show later, the effects on employment are concentrated among tenant businesses and that the reform led to a significant increase in rental expenses. If COVID-19 was driving the results, one would not expect COVID-19 to operate differentially through the rental cost channel, since the pandemic would not systematically raise business rents. The fact that treatment effect heterogeneity aligns closely with exposure to higher rental costs is more naturally explained by the property tax mechanism than by pandemic disruptions.

Nevertheless, I conduct three additional exercises to assess whether the results are driven by COVID-19. First, I restrict the sample to the pre-COVID period by dropping all observations after 2018/19. Panel A of Appendix Figure A.25 shows that differential decline in employment persists in the restricted sample. Although the average effects are smaller in magnitude, the event-study plots reveal that the effects were still evolving by the end of the truncated sample, suggesting that the full-sample estimates reflect the continuation of a trend that predates the pandemic rather than a COVID-induced shock, thereby confirming that the results are not an artifact of the pandemic period.

Second, I exploit the calendar-time-specific effects estimator embedded in Callaway and Sant'Anna (2021)'s staggered difference-in-differences framework, which recovers the average treatment effect for each calendar year separately, using only firms treated by that date. This allows me to assess whether the employment effects emerge prior to the COVID-19 period or materialize only during the pandemic. Panel B of Appendix Figure A.25 shows that employment effects were already building before the onset of the pandemic, with estimates growing in magnitude through 2018/19 and persisting into the pandemic period. The fact that the estimated effects predate the pandemic makes it unlikely that COVID-19 is the primary driver of the estimated impacts.

Finally, I conduct a robustness check excluding the central division, which is entirely surrounded by other divisions (see Figure 1). This ensures that treated locations are only

those bordering never-treated locations, and therefore more likely to have experienced similar pandemic exposure, strengthening their comparability. Appendix Figure A.26 shows that the employment effects are robust to this sample restriction.

### *VI.D. Distributional Consequences*

The aggregate employment effects documented above mask substantial heterogeneity in how the burden of the reform is distributed across workers. In particular, if firms respond by adjusting employment at the bottom of the wage distribution, disproportionately reducing employment among low-skill, low-wage workers, the reform may disproportionately harm the most economically vulnerable workers, even when its aggregate employment effect appears moderate. As with other business taxes, the distributional consequences of property taxation thus depend critically on which workers bear the burden of firms' adjustment responses (e.g., Risch 2024; Fuest et al. 2018).

To identify which types of workers are most affected by the reform, I implement two complementary firm-level analyses.[24] First, I examine changes in the composition of employment by tracking how the share of workers in each pre-reform wage quartile, relative to pre-reform total employment, evolves following the reform. Formally, the outcome variable is $\frac{E_{q,t}}{E_0}$, where $E_{q,t}$ denotes the number of workers in quartile $q$ at time $t$ and $E_0$ denotes pre-reform total employment.[25] Second, I analyze how the within-firm wage distribution evolves in treated relative to untreated firms following the reform. In both analyses, I restrict the sample to firms that employed at least five workers in the pre-reform period and remained active throughout the sample period, to ensure sufficient support across the wage distribution. All estimates follow the staggered difference-in-differences approach of Callaway and Sant'Anna (2021).

Figure 9 presents the effects of the reform on employment composition across the pre-reform wage distribution. The reform led to a differential decline in the share of workers in the top quartile among treated relative to untreated firms, accompanied by a corresponding increase in the share of workers in the $50^{th}$–$75^{th}$ percentile. The absence of pre-reform differential trends in Panel A supports a causal interpretation of these patterns. On average, the share of workers in the top quartile declined by 2.8 percentage points, while the share in the $50^{th}$–$75^{th}$ percentile increased by 2.2 percentage points (Panel B). Effects on the bottom quartile and the $25^{th}$–$50^{th}$ percentile are not statistically significant, although the point estimates are positive.

[24] I conduct the analysis at the firm level rather than the worker level, as workers are uniquely identified only at the firm level in the data.

[25] The focus on the quartiles reflects sample size limitations.

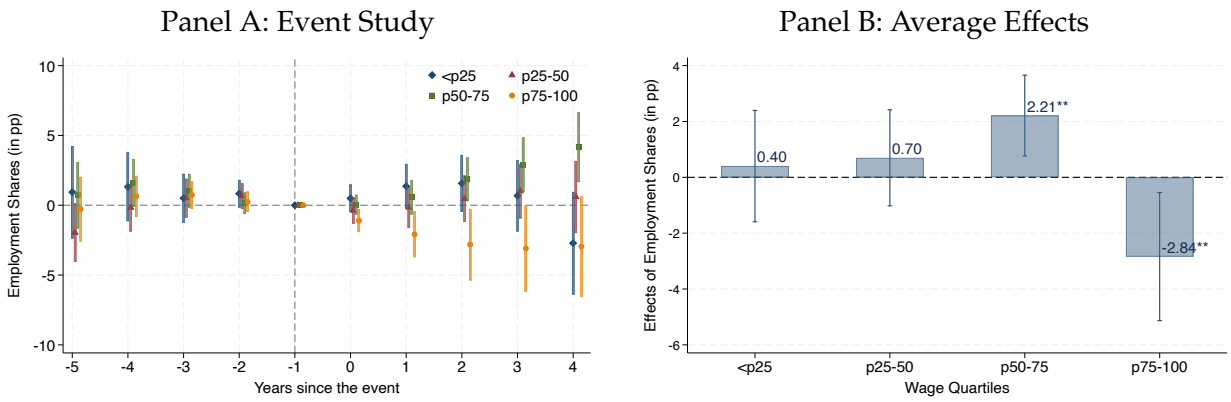



Figure 9: Effects of the Reform on Employment Composition

*Notes:* The figure presents the effects of the reform on within-firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,q,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm $i$'s number of workers in quartile $q$ at time $t$ and $E_{i,0}$ denotes firm $i$'s total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to pre-reform total employment. Panel A presents the event-study estimates, with vertical bars indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

Two mechanisms could account for this pattern. First, firms may substitute high-earning workers with lower-wage workers of relatively similar productivity, reducing their wage bill without proportionally contracting their workforce. Second, wages at the top of the distribution may have fallen more sharply than those at the bottom, mechanically shifting workers from the top quartile into the third quartile without any change in employment composition itself. Because of data limitations, I am not able to separate these mechanisms.

In the second analysis, I examine the effects of the reform on within-firm wage dispersion. Specifically, I estimate the effects of the reform on the logarithm of the ratio between the 80$^{th}$ percentile of the within-firm formal earnings distribution and progressively lower percentiles. A decline in these ratios indicates that wages at the upper end of the distribution fall relative to those lower down, implying compression of the within-firm wage distribution. Figure 10 shows the results of this analysis. Panel A presents the event-study estimates, showing no evidence of differential pre-trends. Following the reform, however, the upper-to-lower wage ratios decline steadily, with the magnitude of the effect increasing as the comparison moves further down the distribution. Panel B summarizes the average treatment effects, showing economically meaningful declines in the $p_{80}/p_{40}$ and $p_{80}/p_{20}$ ratios of 5.3 and 7.4 percent, respectively. These results provide

direct evidence that the reform compressed the formal wage distribution within firms.

This compression could, in principle, arise either because wages at the top declined or because wages at the bottom increased. To distinguish between these possibilities, I next examine the effects of the reform on individual wage percentiles. Appendix Figure A.28 shows that while the estimated effects are negative throughout the distribution, they are concentrated at the upper end. The 60$^{th}$ percentile declines by 0.1 percentage points relative to the pre-reform formal wage bill, while the 80$^{th}$ percentile declines by 0.2 percentage points, and these are the only statistically significant estimates. Appendix Figure A.29, which instead scales each percentile by its own pre-reform value, yields qualitatively similar results.[26] The evidence therefore indicates that the observed compression is driven by declines in formal earnings at the top of the within-firm wage distribution rather than gains at the bottom. Evidence in Appendix Figure A.31 suggests that the decline at the top is not driven by wage informalization, pointing instead to a genuine reduction in wages at the top of the distribution.

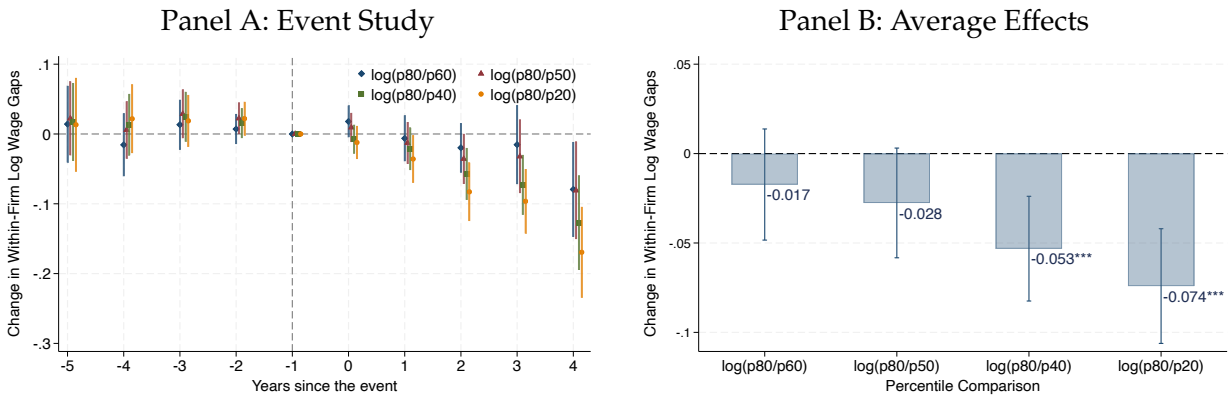



Figure 10: Effect of the Property Tax Reform on Within-Firm Earnings Dispersion

*Notes:* The figure examines the effects of the property tax reform on within-firm wage dispersion. The outcomes are the logarithm of the ratio of the 80$^{th}$ percentile of the within-firm formal earnings distribution to the indicated lower percentile, i.e., $\log(p_{80}/p_q)$, where $q \in \{20, 40, 50, 60\}$. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B reports the average treatment effects. A negative coefficient indicates a decline in the ratio between upper-tail and lower-tail formal earnings, implying a compression of the within-firm formal earnings distribution. All regressions are weighted by the logarithm of pre-reform formal wage bill. The coefficients are estimated using the staggered difference-in-differences design following Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

Taken together, these results indicate that the burden of higher property tax liabilities

[26] Appendix Figure A.30 reports analogous estimates for the 25$^{th}$, 50$^{th}$, and 75$^{th}$ percentiles and reaches the same conclusion.

falls disproportionately on higher-earning workers within firms. The reform thus compressed the within-firm wage distribution, reducing inequality between low- and high-wage workers, a pattern that is progressive in this narrow sense. These findings relate closely to Risch (2024), who shows that a disproportionate share of the tax burden on business owners is ultimately borne by relatively high-earning employees. While property taxes differ from business owner taxes in their statutory incidence, both share the feature of being levied outside of direct business income, suggesting a common transmission channel through which firm-level cost shocks are absorbed along the upper tail of the within-firm wage distribution.

## VII. HETEROGENEITY ANALYSIS ON EMPLOYMENT AND FIRM SURVIVAL

### *VII.A. Heterogeneity by Profitability and Leverage*

To better understand the underlying mechanisms, I examine heterogeneity by baseline profitability and leverage. Property taxes increase fixed operating costs and thereby reduce profitability; firms with thin operating margins may be pushed into losses, increasing the likelihood of firm exit or employment adjustment. To assess this channel, I first examine effects among loss-making firms. I then partition the sample into terciles and examine how the effects vary across the distribution of pre-reform profitability.[27] Figure 11 shows that the magnitude of both employment and survival effects declines monotonically with profitability. The event-study plots in Appendix Figures A.32 and A.34 show that this pattern is not driven by pre-trends. The finding that survival effects are strongest among loss-making firms further rules out the possibility that the overall survival effects are driven by evasion responses to corporate income tax, since such firms have zero corporate income tax liability.

Turning to leverage, Figure 11 reveals a U-shaped pattern in employment and survival effects across the leverage distribution. The effects are statistically significant only at the two tails, among firms with very low and very high leverage, while moderately leveraged firms are largely insulated. Appendix Figures A.33 and A.35 confirm that these patterns are not driven by differential pre-trends, supporting the causal interpretation of these patterns. This U-shape points to two distinct vulnerability channels operating at opposite ends of the distribution. High-leverage firms are exposed through a financial distress channel: higher property tax obligations tighten liquidity constraints associated with existing debt service commitments, compressing already thin repayment margins and amplifying the probability of distress. At the other end, very low-leverage firms are

[27] Profit margins are defined as the ratio of profits (EBIT) to revenue.

exposed through a financial exclusion channel: absent meaningful access to credit markets, these firms cannot borrow to smooth the shock and are instead forced to adjust on real margins, cutting employment or exiting altogether. Firms with moderate leverage, by contrast, retain sufficient unused debt capacity to absorb the shock through additional borrowing, explaining their relative insulation. This U-shaped pattern of incidence suggests that it is financial flexibility, rather than financial strength per se, that shields firms from cost shocks.

Figure 12 provides direct evidence consistent with this mechanism. Using a kernel-weighted local difference-in-differences approach that permits treatment effects to vary smoothly across the leverage distribution, I find an inverted-U pattern in the borrowing response: moderately leveraged firms in treated areas exhibit a relative increase in debt following the reform, while firms at either tail of the leverage distribution exhibit little or no borrowing response. Appendix Figure A.37 confirms this pattern using tercile-based estimates, showing that the borrowing response is concentrated among moderately leveraged firms, and is not driven by differential pre-trends. The combination of a U-shaped pattern in employment and survival effects with an inverted-U pattern in borrowing provides a coherent account of how financial structure mediates the effects of cost shocks: moderately leveraged firms draw on available debt capacity to absorb the shock, while firms at either extreme—constrained by excessive leverage or excluded from credit markets—cannot access this financing margin and instead adjust through real margins. Taken together, the heterogeneity patterns by profitability and leverage point to financial constraints and financial distress as key mechanisms underlying the reform's effects.

## VIII. RENTAL COST CHANNEL

The heterogeneity analysis by occupancy status reveals that firm survival probability and employment decline even among tenant firms (Appendix Figure A.36). This suggests that the rental cost channel plays an important role in transmitting property tax burdens to firms. To quantify the importance of this mechanism, I estimate the effect of the reform on business rental expenses and calculate the implied pass-through rate. The sample is restricted to firms that reported positive rental expenses in the pre-reform year and the outcome variable is normalized by its pre-reform value.

The results are presented in Figure 13. Panel A reports estimates from a balanced panel, while Panel B reports estimates from an unbalanced panel. Prior to the reform, rent expenses for firms in locations exposed to increases in property tax liabilities and those in unaffected areas followed similar trends, supporting the parallel trends assump-

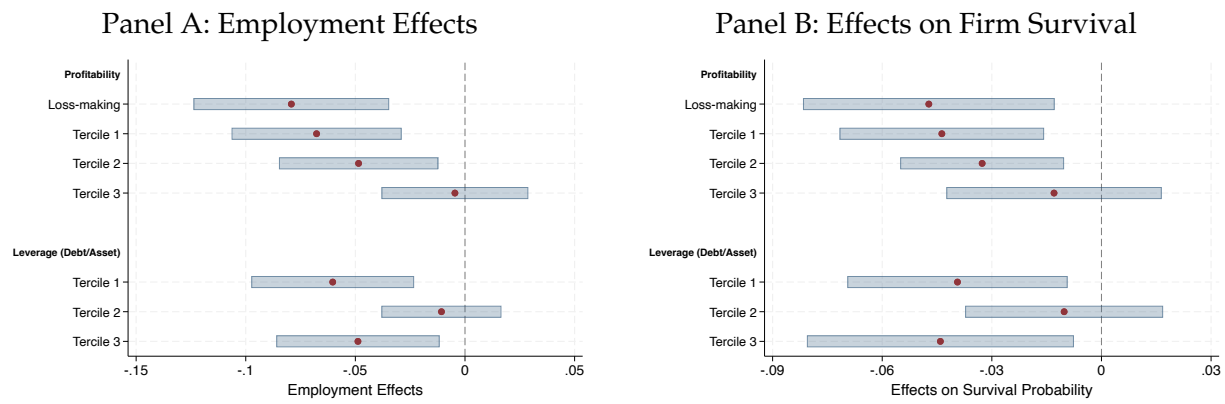



Figure 11: Employment and Survival Effects: Heterogeneity by Profitability and Leverage

*Notes:* The figure shows the heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient from a separate regression. Firms are distinguished by pre-reform profitability (EBIT/revenues) and leverage (debt-to-asset ratio). For comparability across categories, employment is normalized by pre-reform values. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

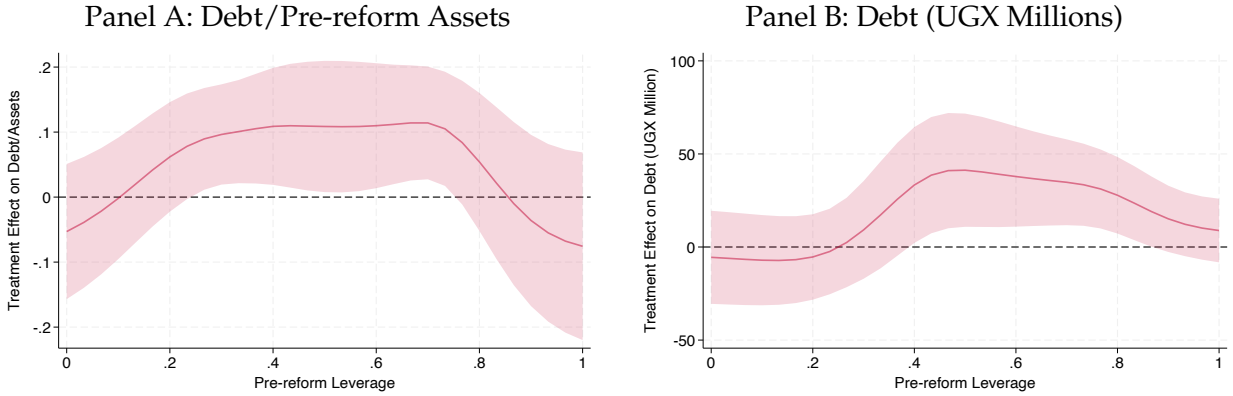



Figure 12: Effects of the Reform on Firms' Debt by Initial Leverage

*Notes:* The figure presents estimates of the treatment effect on debt over the distribution of pre-reform leverage, using a kernel-weighted local difference-in-differences approach. At each point in the pre-reform leverage distribution, I estimate the treatment effect following Callaway and Sant'Anna (2021), using the full sample, with Gaussian kernel weights that assign greater weight to firms whose pre-reform leverage is closer to the evaluation point. This procedure allows treatment effects to vary smoothly across the leverage distribution. Panel A shows the effects on debt scaled by pre-reform assets, while Panel B shows the effects on total debt, measured in UGX millions. Both outcomes are winsorized at the $5^{th}$ and $95^{th}$ percentiles. Both panels are estimated using balanced panels, comprising 29,106 firm-year observations in Panel A and 33,012 firm-year observations in Panel B. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

tion. Following the reform, firms in treated locations experienced a differential increase in rent expenses relative to those in untreated areas. The average effects presented in Table 3 show that the reform led to a 9.1 percent increase in business rental expenses in

the balanced panel (column 3) and 6.3 percent in the unbalanced panel (column 6). Appendix Figure A.38 shows that small tenant firms experience large increase in business rental expenses compared to large tenant firms, consistent with larger firms possessing higher bargaining power in the commercial rental market.

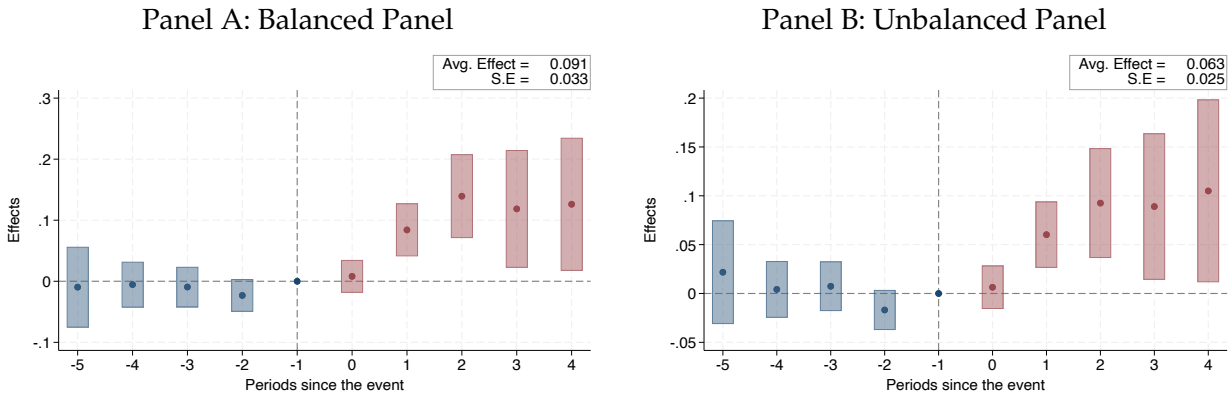



Figure 13: Effects of the Reform on Business Rental Expenses

*Notes:* The figure shows the effects of the reform on business rental expenses. Panel A presents the results estimated on a balanced panel of firms that reported paying rent in the pre-reform period while Panel B presents the results from an unbalanced panel. The sample in Panel A comprises 30,798 firm-year observations while that in Panel B comprises 56,761 firm-year observations. The dependent variable is rent costs scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Appendix Figure A.39 shows the effects using never treated units as the comparison group. Standard errors are clustered at the firm level. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals.

A potential concern with these estimates is that the reform may indirectly affect rents paid by firms in comparison locations through a general equilibrium mechanism. If the reform reduces business entry in treated areas and redirects it toward comparison areas, the resulting increase in demand for commercial space would raise rents faced by incumbent tenants in comparison locations for reasons unrelated to the direct effect of the reform. This would inflate rents in the comparison group and attenuate the estimated treatment effect. Appendix Figure A.40 addresses this concern by restricting the comparison group to firms located outside the Greater Kampala Metropolitan Area, where such cross-location spillovers are less plausible. The estimated coefficients under this restriction are, if anything, smaller than in the baseline specification, providing no evidence that spillovers to comparison-area rents are attenuating the estimates.

*Implied Pass-Through Rate*:— To interpret the economic magnitude of the reform's effects on rental expenses, I compute an implied pass-through rate using the reduced-form estimate from the unbalanced panel (Figure 13 Panel B and Table 3 Column 6). As noted in Section VI, I do not observe firm-specific property tax liabilities, which makes it impossible to estimate pass-through at the individual firm level.

Given these constraints, I estimate the pass-through rate for the population of formal firms, i.e., those registered for corporate income tax, operating in Kampala.[28] Conceptually, this approach treats the reform as an aggregate shock to the cost of occupying commercial property and asks what share of the resulting increase in property tax liabilities is transmitted to formal tenant firms through higher rents. I operationalize this by scaling the total increase in rental expenses among CIT-registered tenant firms by the aggregate increase in property tax liability induced by the reform. Formally, I define the pass-through rate as: $\frac{\hat{\beta}_{rent} * \bar{R} * N_{renters}}{\Delta T^{\text{City}}}$, where $\hat{\beta}_{rent}$ is the average treatment effect on rental expenses (in percents) estimated on an unbalanced panel of tenant firms, $\bar{R}$ is the average pre-reform rent expense, $N_{renters}$ is the number of CIT-registered firms that reported positive rental expenses in the pre-reform period, and $\Delta T^{\text{City}}$ is the reform-induced change in citywide commercial property tax liabilities.[29]

Results in Table 3 (second row from the bottom) indicate that a UGX 1 increase in aggregate property tax revenue is associated with a UGX 0.52 increase in total rental expenses paid by CIT-registered tenant firms (column 6). This measure captures the share of the aggregate increase in property tax liabilities borne by formal tenant firms through higher rents. It should be interpreted as a pass-through rate to the formal sector specifically, rather than a citywide rate encompassing all commercial tenants, since the CIT-filing population does not cover the universe of businesses operating in Kampala. Even within this population, the estimate is likely a lower bound on the true pass-through to formal tenant firms, since the denominator reflects the total property tax increase across all commercial properties citywide, including owner-occupied properties. To the extent that the denominator overstates the tax increase attributable to properties occupied by formal tenant firms, the resulting pass-through rate is biased downward. Moreover, the estimate is an aggregate measure that pools across firms facing different levels of tax exposure and occupying properties of varying types and values, reflecting the average pass-through across a heterogeneous population rather than the pass-through on any individual property. It should therefore not be interpreted as a structural, firm-level pass-through rate.

[28] Here, formal refers specifically to firms registered for corporate income tax with the Uganda Revenue Authority, which constitutes a subset of all businesses operating in Kampala.

[29] Note that $\hat{\beta}_{rent}$ is estimated on an unbalanced panel that includes all pre-reform tenant firms such that the numerator captures the change arising from the population of CIT-registered tenant firms.

Table 3: Effects of the Reform on Business Rental Expenses

| | Balanced Panel | | | Unbalanced Panel | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Average Effect | 0.102*** | 0.102*** | 0.091*** | 0.067*** | 0.066*** | 0.063** |
| | (0.032) | (0.032) | (0.033) | (0.025) | (0.025) | (0.025) |
| Sector of Operation | Y | Y | Y | Y | Y | Y |
| Pre-reform Size | N | Y | Y | N | Y | Y |
| Accounting Month | N | N | Y | N | N | Y |
| Pre-reform Mean | 39.72 | 39.72 | 39.72 | 28.68 | 28.68 | 28.68 |
| Implied Pass-Through Rate | | | | 0.545 | 0.538 | 0.517 |
| Observations | 30798 | 30798 | 30798 | 56761 | 56761 | 56761 |

Note: The table presents average effects of the reform on business rental expenses. Columns 1 to 3 present the results estimated on a balanced panel of firms that reported paying rent in the pre-reform period, while columns 4 to 6 present results estimated on an unbalanced panel. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change. The second row from the bottom presents the pass-through rate to formal firms calculated from the expression: $\frac{\hat{\beta}_{rent} * \bar{R} * N_{renters}}{\Delta T^{\text{City}}}$, where $\hat{\beta}_{rent}$ is the estimated average treatment effect on rental expenses in percent, $\bar{R}$ is the average pre-reform rent expense, $N_{renters}$ is the number of CIT-registered firms that reported positive rental expenses in the pre-reform period, and $\Delta T^{\text{City}}$ is the reform-induced change in citywide commercial property tax liabilities. The coefficients are estimated using a staggered Difference-in-Differences approach following Callaway and Sant'Anna (2021), with standard errors are clustered at the firm level. Pre-reform mean is measured in millions of Uganda Shillings.

Although not directly comparable to property-level pass-through estimates in the housing literature, the pass-through rate to formal firms estimated here is broadly consistent in magnitude with recent studies of property tax incidence in rental markets. These studies typically estimate pass-through at the property or unit level and, in some cases, embed the analysis in general equilibrium frameworks that allow for endogenous adjustment in rents, occupancy, and supply, finding pass-through rates ranging from \$0.50 to \$1.39 per \$1 of tax (e.g., Baker 2024; Watson and Ziv 2024; Löffler and Siegloch 2021; Rolheiser 2019). By contrast, my estimate is a reduced-form aggregate incidence measure for CIT-registered tenant firms, pooling across heterogeneous properties, landlords, and tenant firms, and abstracting from equilibrium reallocation across locations. Despite these differences in the level of aggregation, empirical design, and institutional context, the similarity in magnitude suggests that a substantial share of the commercial property tax burden is transmitted to firms through higher rents.

My results are consistent with the "old view" of property tax incidence, which posits that the portion of the tax falling on land is borne entirely by owners, since the supply of land is perfectly inelastic, while the portion falling on improvements, i.e., structures, can be shifted to final consumers and occupants, since the supply of capital to the construction

industry is relatively elastic (Oates and Fischel 2016). In the setting that I study, it is plausible that the tax falls primarily on improvements rather than on land itself, since idle land is exempt.

The finding that higher property tax liabilities increase business rental expenses contrasts with neoclassical predictions, which suggest little to no pass-through of property taxes (Mieszkowski 1972; Zodrow 2001). In Appendix V, I propose a framework that rationalizes this result. The model incorporates a key institutional feature of targeted enforcement: the probability of enforcement, and thus compliance, is higher when a property is occupied and generating observable economic activity, and lower when it is vacant. This feature is likely prevalent in developing country contexts, where limited administrative capacity leads tax authorities to target visibly active properties. As a result, the tax shifts from being a cost of owning property to a cost of supplying space to the rental market. In this setting, property taxes act as a marginal cost of renting out space, and standard incidence logic predicts positive pass-through to tenants.

*Occupancy Status: Own or Lease?*— Since the incidence of property taxes depends on occupancy status, higher tax liabilities may prompt firms to reconsider whether to own or lease their operating space. I examine this margin by classifying firms into pure owners, pure renters, and mixed firms based on pre-reform records, and tracking transitions across these categories following the reform. Appendix Figure A.41 shows that the reform did not induce complete switches between full ownership and full tenancy, unsurprising given the lumpiness and illiquidity of commercial property transactions. The more informative margin operates at the intensive margin: the reform differentially induced pure owners to transition into mixed arrangements, supplementing owned premises with leased space, while pure renters were less likely to acquire property and mixed firms were more likely to unwind their ownership stake. These findings indicate that property tax reform reshapes occupancy decisions primarily through gradual adjustments along the intensive margin rather than complete tenure switching, with owner-occupiers incorporating leased space into their operations rather than divesting owned property outright.

## IX. VERTICAL FISCAL EXTERNALITY AND WELFARE IMPLICATIONS

### *IX.A. Implications for Central Government Tax Revenue*

The firm-level effects documented in Sections V and VI have direct implications for central government tax revenues. To the extent that the reform induces firm exit, reduces firm profitability, or contracts formal employment and wages, it erodes the tax bases of

both the corporate income tax and the employee income tax (Pay-As-You-Earn, PAYE), potentially offsetting part of the additional property tax revenue raised by the local government. This fiscal externality represents a form of vertical tax competition: when a sub-national government raises property taxes, it may inadvertently compress the tax bases of taxes administered at the national level, generating a negative fiscal spillover that is not internalized by the local authority setting the tax rate but is consequential for consolidated public revenues.[30] Whether and to what extent such spillovers materialize is ultimately an empirical question. To examine this question directly, I study how central government tax revenues evolve around the reform.

Figure 14 presents the event-study estimates and average effects on central government tax revenues, measured in millions of Uganda shillings. I examine three outcomes in turn: corporate income tax payments (Panel A), employee income tax (PAYE) payments aggregated at the firm level (Panel B), and the combined revenue from both taxes (Panel C).[31] Despite inducing firm exit, the reform had no statistically significant effect on corporate income tax revenues. This result is consistent with the earlier finding that the firms most likely to exit were precisely those with little or no corporate income tax liability—loss-making and least profitable firms that contributed little to the corporate tax base even before the reform.

The picture is different for employee income tax revenues. Panel B reveals a differential decline in PAYE receipts from treated relative to untreated firms following the reform, driven by declines in formal employment and wages, as well as increases in wage informality documented in Section VI. These labor market responses translate directly into a smaller formal wage bill from which PAYE is withheld, reducing central government receipts. Combining both taxes, Panel C shows that total central government revenue collected from treated firms declined by an average of UGX 0.6 million annually relative to untreated firms, against a pre-reform mean of UGX 7.8 million, a decline of 7.9 percent. As reported in Appendix Table A.11, this is equivalent to UGX 0.3 of lost central government revenue for every UGX 1 increase in property tax revenues, suggesting that the fiscal gains at the local level are substantially offset by revenue losses at the central level.

### *IX.B. The Marginal Value of Public Funds*

The preceding results document that the property tax reform generated significant real effects on firms, reducing central government income tax revenue. In this section, I

[30] This vertical fiscal externality is distinct from the horizontal tax competition literature, which focuses on competition between jurisdictions at the same level of government (see, e.g., Wilson 1999).

[31] I aggregate employee taxes at firm level because the data does not allow me to follow workers over time, as workers are only uniquely identified at firm level.

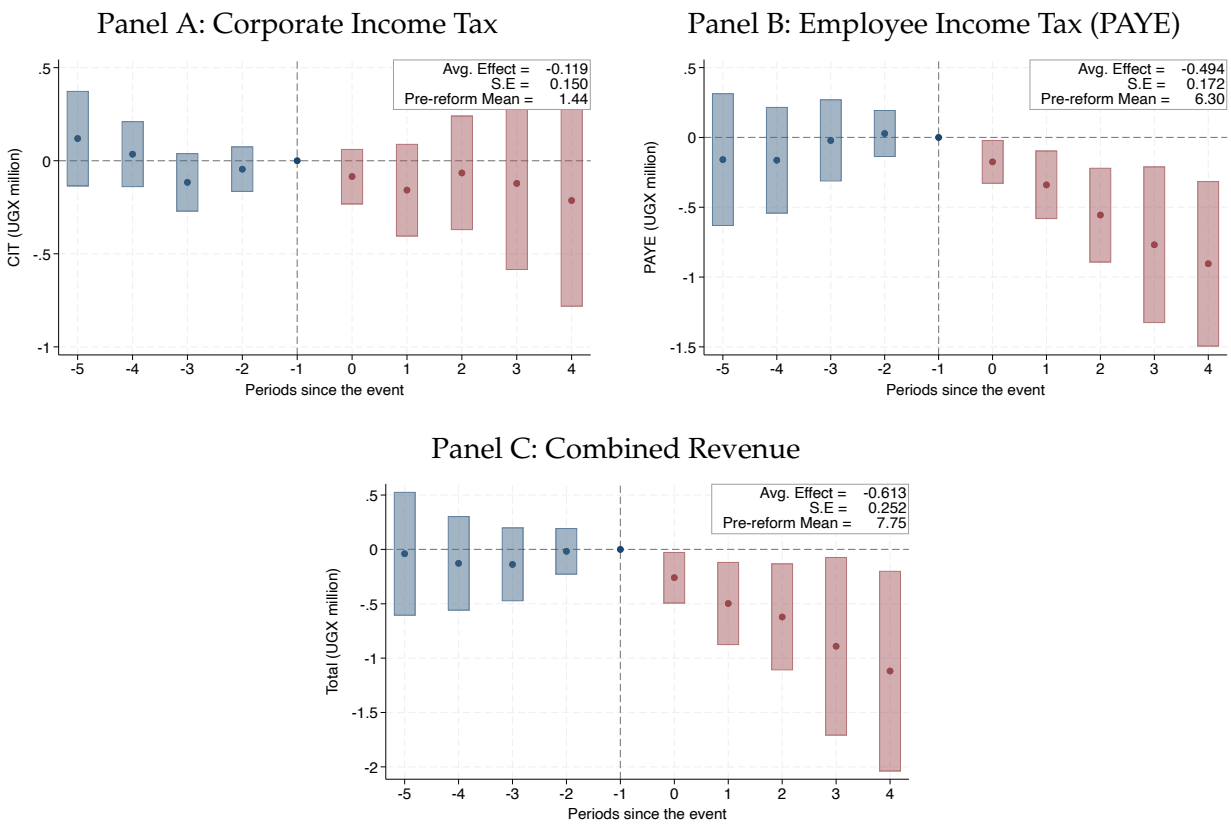



Figure 14: Implications for Central Government Tax Revenue

*Notes:* The figure presents the effects of the reform on central government tax revenue. All analysis is conducted at the firm level, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 142,677 firm-year observations. The outcome variable is in millions of Uganda shillings (UGX), constant 2016/17. Panel A shows the effects on corporate income tax, Panel B shows the effects on employee tax revenue (PAYE), while Panel C shows the effects on the combined revenue. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

bring these estimates together within a unified welfare framework to assess the overall cost of raising revenue through the property taxation. I follow the Marginal Value of Public Funds (MVPF) approach developed by Hendren and Sprung-Keyser (2020), which provides a transparent method for evaluating the welfare consequences of government policies. The MVPF is defined as the ratio of beneficiaries' willingness to pay for a policy to its net cost to the government:

$$MVPF = \frac{WTP}{\text{Net Cost to Government}} \tag{4}$$

For a given policy with $MVPF = A$, the policy delivers $A$ dollars of benefits per dollar of net government spending.

As noted by Hendren and Sprung-Keyser (2020), the theoretical and empirical literature on optimal income taxation observes that a tax cut providing \$1 in additional after-tax income is valued at exactly \$1 by mechanical beneficiaries, that is, the tax cut is valued at cost by those who would receive it in the absence of any behavioral response. The willingness to pay for a \$1 tax cut is therefore equal to 1. The cost to the government, however, depends on how taxpayers respond. A tax cut that provides \$1 of mechanical benefits costs the government $1 + FE$, where $FE$ is the impact of the behavioral response on government revenue. The MVPF for a tax cut is thus given as $\frac{1}{1+FE}$.

The reform I study operates in the opposite direction: it *increases* property tax liabilities. The MVPF framework applies symmetrically, but the interpretation of the numerator and denominator changes. The numerator now captures the taxpayer's willingness to pay to *avoid* the tax increase rather than to receive a tax cut. By the envelope theorem, this is again exactly \$1 per dollar of mechanical tax revenue: the marginal dollar of additional tax liability is a transfer from taxpayer to government, and the welfare cost of behavioral adjustments is second order. The denominator now captures the net revenue *gained* by the government rather than the net revenue lost from a tax cut. For each \$1 of mechanical property tax revenue raised, the government retains $1 + FE$, where $FE$ reflects the revenue consequences of firms' behavioral responses to higher tax liabilities. If these responses shrink other tax bases, by reducing profitability, shedding formal employment, or increasing wage informality, then $FE < 0$ and the government retains less than the full dollar. Note that this represents net revenue for the consolidated government since the reform generates revenue only for the local government, while the associated fiscal externality operates through the central government's tax base.

In principle, the fiscal externality in the setting I study comprises two components: a local component, capturing the effect of behavioral responses on the property tax base itself, and a vertical component, capturing the erosion of centrally administered tax bases through firms' responses to higher costs. The estimates in this section capture only the vertical component. Although data limitations prevent me from quantifying the local component, I expect it to be relatively limited in the short run because immovable property constitutes an inelastic tax base.[32]

The discussion thus far treats the tax increase as a costless policy action. In practice, the reform required a substantial upfront investment in property revaluation and mass

[32]The presence of a meaningful local fiscal externality would further increase the MVPF. Abstracting from this component therefore makes my estimates conservative.

enumeration before any additional revenue could be collected. This administrative cost must be recovered from the revenue stream and therefore reduces the net fiscal gain to the government. Let $C$ denote the total cost of the revaluation and enumeration exercise and $PV(\Delta R^{PT})$ the present value of the additional property tax revenue generated over the horizon for which the updated valuation roll remains valid.[33] The administrative cost per dollar of property tax revenue is $a = \frac{C}{PV(\Delta R^{PT})}$. Incorporating both the vertical fiscal externality and the administrative cost, the MVPF from the perspective of the consolidated government becomes:

$$MVPF^c = \frac{1}{1 + FE^v - a} \tag{5}$$

Values above 1 indicate that each dollar of net consolidated revenue costs taxpayers more than a dollar, implying that the reform is welfare-improving only if the public goods it finances generate benefits exceeding the MVPF per dollar of spending.

The local government, by contrast, does not internalize the vertical fiscal externality. From its perspective, the MVPF is $MVPF^l = \frac{1}{1-a}$, which is close to 1 when administrative costs are small. The wedge between $MVPF^c$ and $MVPF^l$ provides a concrete measure of the welfare cost of decentralized tax setting, i.e., the efficiency loss that arises because the local government does not bear the consequences of the vertical fiscal externality its tax generates.

Table 4 presents the estimated MVPFs under alternative assumptions about the administrative cost per shilling of revenue. In the most conservative scenario with zero administrative costs, the reform generates a consolidated MVPF of 1.43 with a 95% confidence interval ranging from 0.94 to 1.92. After accounting for the cost of the reform, the MVPF rises to 1.53 (95% CI of [0.97, 2.09]).[34] Throughout, the local government's perceived MVPF remains close to 1, and wedge between the local and consolidated MVPF ranges from 0.43 to 0.48. The estimated MVPF implies that each UGX 1 of net consolidated revenue raised through the reform cost the public UGX 1.53, implying that property taxes generate 53 percent marginal deadweight losses. Abstracting from incidence, the property tax reform is welfare-improving only if the public goods financed by the net revenues generate benefits of at least the MVPF per shilling spent.

Existing MVPF estimates for public spending programmes, while concentrated in high-income countries, suggest that the threshold implied by my estimates is attainable for a broad range of public investments. As illustrated in Appendix Figure A.43, many

[33] If the reform generates $\Delta R^{PT}$ per year in additional revenue, the valuation roll is valid for $T$ years before requiring an update, and the discount rate is $r$, then $PV(\Delta R^{PT}) = \Delta R^{PT} \times \frac{1-(1+r)^{-T}}{r}$

[34] It's important to note that the MVPF estimated in this paper represents a lower bound because it assumes away local fiscal externalities.

programs have MVPFs exceeding 1.53, including investments in early childhood, education, job training, and selected infrastructure and environmental policies, while several program are effectively self-financing. By contrast, a number of social insurance programs and cash transfer interventions have MVPFs below this threshold. These comparisons suggest that financing high-return public investments through commercial property taxation could generate positive net social value. Whether the specific expenditures financed by property tax revenues in Kampala satisfy this criterion, however, remains an empirical question beyond the scope of this paper.

The estimated MVPF implies 53 percent marginal deadweight loss, suggesting that the property tax, despite being levied on an immobile base, is not free of distortions. Placed in comparative perspective, however, the property taxation is a relatively efficient instrument in Uganda compared to the corporate income tax, whose implied marginal deadweight loss is estimated at 68 percent (Global Tax Research Initiative 2026).[35] Appendix Table A.12 shows that the implied welfare cost of the property tax reform is also lower than the corresponding figures for corporate income taxation in neighboring East African economies, including Tanzania and Rwanda, as their MVPFs are higher.

The property tax reform also compares favorably to income and capital gains tax reforms studied in high-income settings, despite substantial differences in economic context. Hendren and Sprung-Keyser (2020) estimate an MVPF of 1.85 for the 1993 US top marginal income tax rate increase, while Agersnap and Zidar (2021) estimate an MVPF of 2.65 for a US capital gains tax reform, both well above the 1.53 estimated here. Property taxation appears less efficient, however, than tax enforcement: Boning et al. (2025) report MVPFs of 1.15 and 1.30 for audits of high- and low-income taxpayers, respectively. This gap is intuitive, since enforcement raises revenue by improving compliance with existing taxes rather than by introducing new margins of distortion, but it does highlight audit-based instruments as a particularly efficient revenue-raising tool. Table A.12 presents a detailed comparison of the property tax MVPF against a range of other tax instruments studied in the literature, notwithstanding differences in context.

## X. CONCLUSION

This paper studies the economic effects of commercial property taxation, providing new evidence on its effects on real activity, incidence, and vertical fiscal externalities, using firm-level data. I exploit a natural experiment in Kampala, the capital city of Uganda, where property tax liabilities increased following a revaluation and mass property enu-

[35]The 68 percent marginal deadweight loss is derived from the Global Tax Research Initiative (2026)'s MVPF estimate of 1.675 for a tax cut, assuming symmetric behavioral responses at the margin.

Table 4: Marginal Value of Public Funds for the Property Tax Reform

| | Administrative cost per UGX of revenue ($a$) | |
|---|---|---|
| | $a = 0$ | $a = 0.045$ |
| *Local Government's Perspective* | | |
| Net revenue per UGX 1 | 1.00 | 0.95 |
| $MVPF^l$ | 1.00 | 1.05 |
| *Consolidated Government* | | |
| Net revenue per UGX 1 | 0.70 | 0.66 |
| $MVPF^c$ | 1.43 | 1.53 |
| 95% CI | [0.94, 1.92] | [0.97, 2.09] |
| *Wedge* ($MVPF^c - MVPF^l$) | 0.43 | 0.48 |
| 95% CI | [-0.06, 0.92] | [-0.08, 1.04] |

Note: The table reports the Marginal Value of Public Funds (MVPF) for the property tax reform. The MVPF is computed as $1/(1 + FE^v - a)$ for the consolidated government and $1/(1 - a)$ for the local government, using the estimated vertical fiscal externality $FE^v = -0.3$. The administrative cost $a$ is computed as $C/PV(\Delta R^{PT})$ where $C$ is the total cost of the revaluation and enumeration exercise, obtained from KCCA's administrative reports. The present value is computed over a five-year horizon, consistent with the statutory requirement to update the valuation roll at five-year intervals. Following the Second Kampala Institutional and Infrastructure Development Project (KIIP II), a World Bank-funded project that supported the reform studied in this paper, I use a discount rate of 12 percent. 95 percent confidence intervals are reported in brackets and computed using the delta method.

meration exercise, while surrounding areas were unaffected. This setting allows me to estimate causal effects using a difference-in-differences identification strategy, using novel administrative data from corporate income tax and employee tax returns.

First, I find that the reform reduced firm survival probability. Focusing on active firms, I find no significantly effects on firms' revenues, profits, investment, and non-labor inputs. The effects are, however, heterogeneous by firm size with large firms experiencing significant increases in both revenues and profits, while small firms experience an imprecisely estimated decline in revenues and no meaningful change in profitability. This evidence is consistent with a neoclassical reallocation channel in which the reform contracts space-intensive small firms and, through equilibrium price adjustment in the local input or output markets, benefits large firms. A complementary market-power channel where larger firms pass a greater share of the tax burden onto consumers via output prices may also contribute.

I provide evidence that firms preserve their profitability by adjusting labor costs, cutting formal employment and wages, with wage informalization being a major component of these adjustment. The reform also had progressive distributional consequences within firms. Workers at the top of the earnings distribution were disproportionately negatively

affected. In treated firms, the share of workers in the top quartile declined relative to untreated firms, with a corresponding increase in the share in the $50^{th}$–$75^{th}$ percentile range. There is no evidence of a significant change at the bottom of the distribution. Consistent with these compositional shifts, wages declined significantly at the top, while remaining unchanged at the bottom. Taken together, these findings indicate a reduction in within-firm wage inequality.

Heterogeneity analysis reveals that pre-reform profitability and leverage are important moderators of the employment and survival effects: effects are strongest among less profitable firms and decline monotonically with profitability. Moreover, access to borrowing plays a critical role in enabling firms to absorb cost shocks, insulating moderately leveraged firms from the most adverse real effects.

Further, I provide evidence that the rental cost channel plays an important role in transmitting property tax burdens to firms. Both formal employment and survival probabilities decline among tenant firms following the reform. These effects are consistent with a significant increase in business rental expenses of 6.3 percent. This result implies that 52 percent of the burden falls on formal tenant firms, contrary to the neoclassical prediction that landlords bear nearly the full burden.

Finally, I document evidence of a vertical fiscal externality and the associated welfare implications. The tax increase reduced central government income tax revenues by 30 percent of the amount it contributed to local property tax collections, suggesting that local fiscal gains are substantially offset by losses at the central level. Using the Marginal Value of Public Funds framework of Hendren and Sprung-Keyser (2020), I estimate that each UGX 1 of net consolidated revenue raised through the reform cost the public UGX 1.53, implying that the property taxes generate 53 percent marginal deadweight losses. The reform is therefore welfare-improving only if the public goods it finances generate benefits of at least the MVPF per spending spent.

Taken together, these findings reveal that commercial property taxation in low-income urban settings generates consequences that extend well beyond the property market itself, reshaping firm behavior, altering the distribution of wages within firms, and eroding the tax base of other levels of government. The tax is therefore not free of distortions, generating a non-trivial welfare cost. Even so, the welfare cost per shilling of revenue is smaller than that of corporate income taxation in Uganda and its regional neighbors, and smaller than the marginal welfare cost of income taxation in high-income countries. Property taxation therefore emerges as a relatively efficient revenue instrument, but one whose full welfare cost is only visible when the analysis extends beyond the taxing jurisdiction.

REFERENCES

Agaba, G., Tieu, Q., Jouste, M. and Arinaitwe, B. (2023), The uganda revenue authority pay-as-you-earn (paye) data, Technical report, WIDER Technical Note 2023/3 Helsinki: UNU-WIDER. https://doi. org/10.35188 ....

Agersnap, O. and Zidar, O. (2021), 'The tax elasticity of capital gains and revenue-maximizing rates', *American Economic Review: Insights* **3**(4), 399–416.

Ahabwe, J., Kikonyogo, H., Manwaring, P., Raikes Mugangaizi, R., Mutebi, J., Muhumuza Nuweabine, D. and Regan, T. (2020), 'Property tax compliance in kampala: preliminary evidence from taxpayer feedback and administrative data', *London: International Growth Centre* .

Badalov, N., Mutebi, J., Ravindranath, K., Regan, T. and Ssebuwufu, E. (2025), 'Property tax in kampala'.

Baker, S. (2024), 'Property tax pass-through to renters: A quasi-experimental approach'.

Bakibinga, D. and Ngabirano, D. (2019), 'Enhancing property rates administration, collection and enforcement in uganda: The case of kampala capital city authority (kcca) and four other municipalities'.

Balán, P., Bergeron, A., Tourek, G. and Weigel, J. L. (2022), 'Local elites as state capacity: How city chiefs use local information to increase tax compliance in the democratic republic of the congo', *American Economic Review* **112**(3), 762–797.

Bergeron, A., Tourek, G. and Weigel, J. L. (2024), 'The state capacity ceiling on tax rates: Evidence from randomized tax abatements in the drc', *Econometrica* **92**(4), 1163–1193.

Boning, W. C., Guyton, J., Hodge, R. and Slemrod, J. (2020), 'Heard it through the grapevine: The direct and network effects of a tax enforcement field experiment on firms', *Journal of Public Economics* **190**, 104261.

Boning, W. C., Hendren, N., Sprung-Keyser, B. and Stuart, E. (2025), 'A welfare analysis of tax audits across the income distribution', *The Quarterly Journal of Economics* **140**(1), 63–112.

Borusyak, K., Jaravel, X. and Spiess, J. (2024), 'Revisiting event-study designs: robust and efficient estimation', *Review of Economic Studies* **91**(6), 3253–3285.

Brockmeyer, A., Estefan, A., Arras, K. R. and Serrato, J. C. S. (2021), Taxing property in developing countries: Theory and evidence from mexico, Technical report, National Bureau of Economic Research.

Callaway, B. and Sant'Anna, P. H. (2021), 'Difference-in-differences with multiple time periods', *Journal of econometrics* **225**(2), 200–230.

Carroll, R. J. and Yinger, J. (1994), 'Is the property tax a benefit tax? the case of rental housing', *National Tax Journal* **47**(2), 295–316.

Cengiz, D., Dube, A., Lindner, A. and Zipperer, B. (2019), 'The effect of minimum wages on low-wage jobs', *The Quarterly Journal of Economics* **134**(3), 1405–1454.

Chen, J. and Roth, J. (2024), 'Logs with zeros? some problems and solutions', *The Quarterly Journal of Economics* **139**(2), 891–936.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020), 'Two-way fixed effects estimators with heterogeneous treatment effects', *American Economic Review* **110**(9), 2964–2996.

DiNardo, J., Fortin, N. M. and Lemieux, T. (1996), 'Labor market institutions and the distribution of wages, 1973-1992: A semiparametric approach', *Econometrica* **64**(5), 1001–1044.

Drago, F., Mengel, F. and Traxler, C. (2020), 'Compliance behavior in networks: Evidence from a field experiment', *American Economic Journal: Applied Economics* **12**(2), 96–133.

Egebark, J. and Kaunitz, N. (2018), 'Payroll taxes and youth labor demand', *Labour economics* **55**, 163–177.

England, R. W. (2016), 'Tax incidence and rental housing: a survey and critique of research', *National Tax Journal* **69**(2), 435–460.

Fajgelbaum, P. D., Morales, E., Suárez Serrato, J. C. and Zidar, O. (2019), 'State taxes and spatial misallocation', *The Review of Economic Studies* **86**(1), 333–376.

Flores, T., Bermúdez, J. C., Cruces, G., Schiavoni, J. L., Scot, T. and Tortarolo, D. (2025), *Exploring the Gender Divide in Real Estate Ownership and Property Tax Compliance*, World Bank.

Franzsen, R. and McCluskey, W. (2017), 'Property tax in africa', *Cambridge, MA: Lincoln Institute of Land Policy* .

Fuest, C., Peichl, A. and Siegloch, S. (2018), 'Do higher corporate taxes reduce wages? Micro evidence from Germany', *American Economic Review* **108**(2), 393–418.

Ganapati, S., Shapiro, J. S. and Walker, R. (2020), 'Energy cost pass-through in U.S. manufacturing: Estimates and implications for carbon taxes', *American Economic Journal: Applied Economics* **12**(2), 303–342.

Giroud, X. and Rauh, J. (2019), 'State taxation and the reallocation of business activity: Evidence from establishment-level data', *Journal of Political Economy* **127**(3), 1262–1316.

Global Tax Research Initiative (2026), 'Cost benefit analysis', Dataset. Accessed: July 2026. **URL:** *https://globaltaxresearchinitiative.org/cost-benefit-analysis*

Goodman-Bacon, A. (2021), 'Difference-in-differences with variation in treatment timing', *Journal of econometrics* **225**(2), 254–277.

Harju, J., Koivisto, A. and Matikka, T. (2022), 'The effects of corporate taxes on small firms', *Journal of Public Economics* **212**, 104704.

Hendren, N. and Sprung-Keyser, B. (2020), 'A unified welfare analysis of government policies', *The Quarterly journal of economics* **135**(3), 1209–1318.

Kennedy, P. J., Dobridge, C. L., Landefeld, P. and Mortenson, J. (2026), 'Corporate tax cuts, firm growth, and workers' earnings', *American Economic Review, forthcoming* .

Kresch, E. P., Walker, M., Best, M. C., Gerard, F. and Naritomi, J. (2023), 'Sanitation and property tax compliance: Analyzing the social contract in brazil', *Journal of Development Economics* **160**, 102954.

Lee, D. S. (2009), 'Training, wages, and sample selection: Estimating sharp bounds on treatment effects', *Review of Economic Studies* **76**(3), 1071–1102.

Ljungqvist, A. and Smolyansky, M. (2014), To cut or not to cut? on the impact of corporate taxes on employment and income, Technical report, National Bureau of Economic Research.

Löffler, M. and Siegloch, S. (2021), Welfare effects of property taxation, Technical report, CESifo Working Paper.

Lopez-Luzuriaga, A. and Scartascini, C. (2019), 'Compliance spillovers across taxes: The role of penalties and detection', *Journal of Economic Behavior & Organization* **164**, 518–534.

Lucas, Robert E., J. (1978), 'On the size distribution of business firms', *Bell Journal of Economics* **9**(2), 508–523.

Manwaring, P. and Regan, T. (2019), 'Property taxes in kampala', *ICG policy note* p. 43448.

Manwaring, P. and Regan, T. (2025), No shame in my name: Public disclosure and tax compliance in a low-capacity state, Technical report.

Mieszkowski, P. (1972), 'The property tax: An excise tax or a profits tax?', *Journal of Public Economics* **1**(1), 73–96.

Oates, W. E. and Fischel, W. A. (2016), 'Are local property taxes regressive, progressive, or what?', *National Tax Journal* **69**(2), 415–433.

Pomeranz, D. (2015), 'No taxation without information: Deterrence and self-enforcement in the value added tax', *American Economic Review* **105**(8), 2539–2569.

Risch, M. (2024), 'Does taxing business owners affect employees? evidence from a change in the top marginal tax rate', *The Quarterly Journal of Economics* **139**(1), 637–692.

Rohlin, S., Rosenthal, S. S. and Ross, A. (2014), 'Tax avoidance and business location in a state border model', *Journal of Urban Economics* **83**, 34–49.

Rolheiser, L. (2019), 'Commercial property tax incidence: Evidence from urban and suburban office rental markets', *Available at SSRN 2993371* .

Saez, E., Schoefer, B. and Seim, D. (2019), 'Payroll taxes, firm behavior, and rent sharing: Evidence from a young workers' tax cut in sweden', *American Economic Review* **109**(5), 1717–1763.

Suárez Serrato, J. C. and Zidar, O. (2016), 'Who benefits from state corporate tax cuts? A local labor markets approach with heterogeneous firms', *American Economic Review* **106**(9), 2582–2624.

Sun, L. and Abraham, S. (2021), 'Estimating dynamic treatment effects in event studies with heterogeneous treatment effects', *Journal of econometrics* **225**(2), 175–199.

Thompson, J. P. and Rohlin, S. M. (2012), 'The effect of sales taxes on employment: New evidence from cross-border panel data analysis', *National Tax Journal* **65**(4), 1023–1041.

Ulyssea, G. (2018), 'Firms, informality, and development: Theory and evidence from brazil', *American Economic Review* **108**(8), 2015–2047.

Watson, C. L. and Ziv, O. (2024), 'A test for pricing power in urban housing markets', *Unpublished Manuscript* .

Weyl, E. G. and Fabinger, M. (2013), 'Pass-through as an economic tool: Principles of incidence under imperfect competition', *Journal of political economy* **121**(3), 528–583.

Wilson, J. D. (1999), 'Theories of tax competition', *National tax journal* **52**(2), 269–304.

Yagan, D. (2015), 'Capital tax reform and the real economy: The effects of the 2003 dividend tax cut', *American Economic Review* **105**(12), 3531–3563.

Zodrow, G. R. (2001), 'The property tax as a capital tax: A room with three views', *National Tax Journal* **54**(1), 139–156.

APPENDIX

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# I. ADDITIONAL FIGURES

## *I.A. Additional Descriptive Statistics*

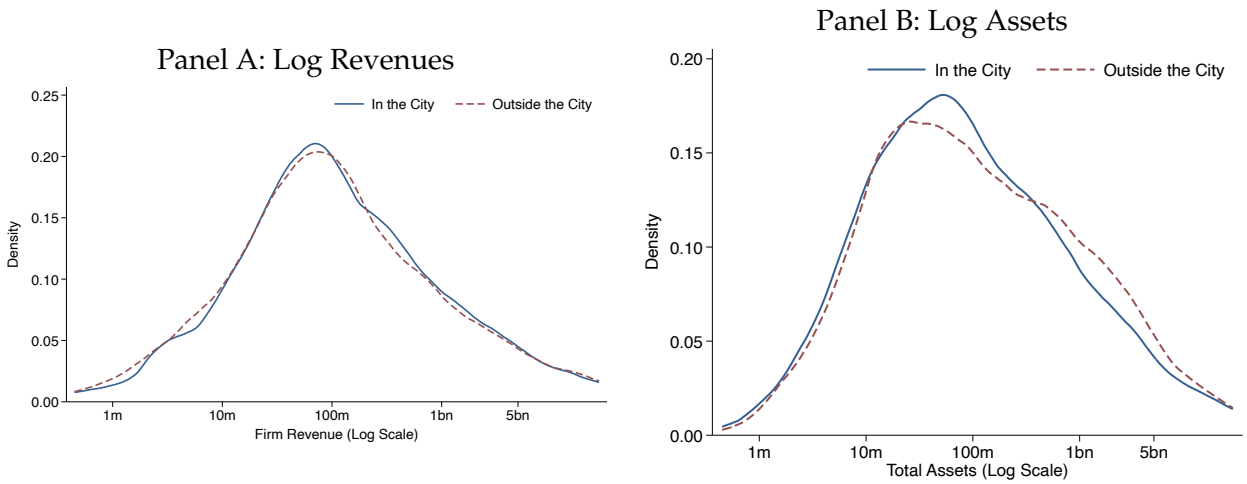



Figure A.1: Firm Size Distributions by Locations

*Notes:* This figure plots the size distributions of active pre-reform firms on the log scale for firms located inside the city (treated) and outside the city (comparison). Panel A shows the distribution of firm revenues, while Panel B shows the distribution of assets.

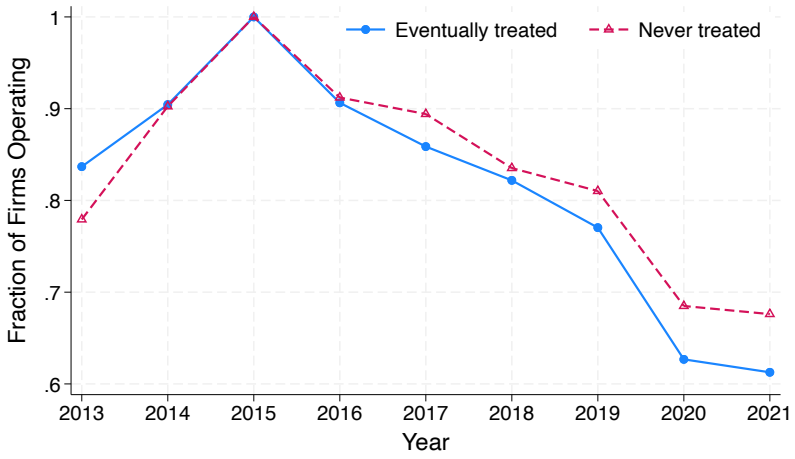



Figure A.2: Fraction of Firms Operating: Raw Trends

*Notes:* The figure plots weighted raw survival rates for firms that are eventually treated and firms that are never treated over time. Firm survival is measured by an indicator for whether the firm is operating in a given year. All treated cohorts are combined into a single eventually treated group, while firms that are never exposed to the reform form the comparison group. Means are calculated using the same DFL weights as in the main survival analysis. The figure is descriptive and does not adjust for covariates or differences in treatment timing across cohorts.

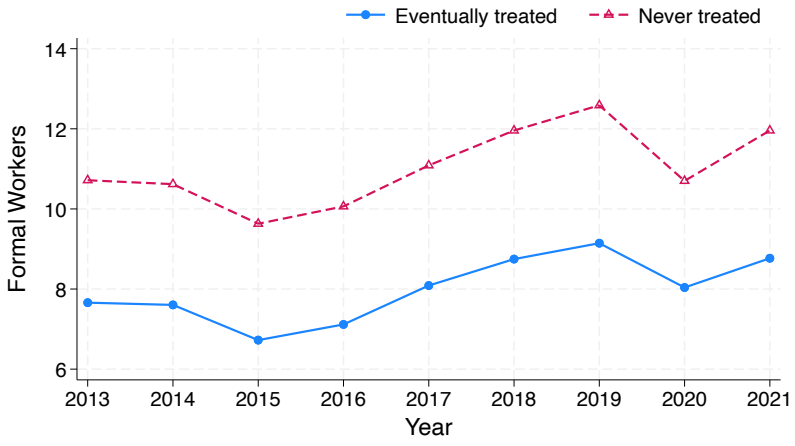



Figure A.3: Raw Trends in Formal Employment

*Notes:* The figure plots raw means for formal employment for eventually treated and never-treated firms over time. All treatment cohorts are combined into a single eventually treated group, while firms that are never exposed to the reform form the comparison group. Means are calculated on an unbalanced panel of firms reporting positive revenues whenever present in the data. The figure is descriptive and does not adjust for differences in treatment timing or covariates.

*I.B. Business Activity*

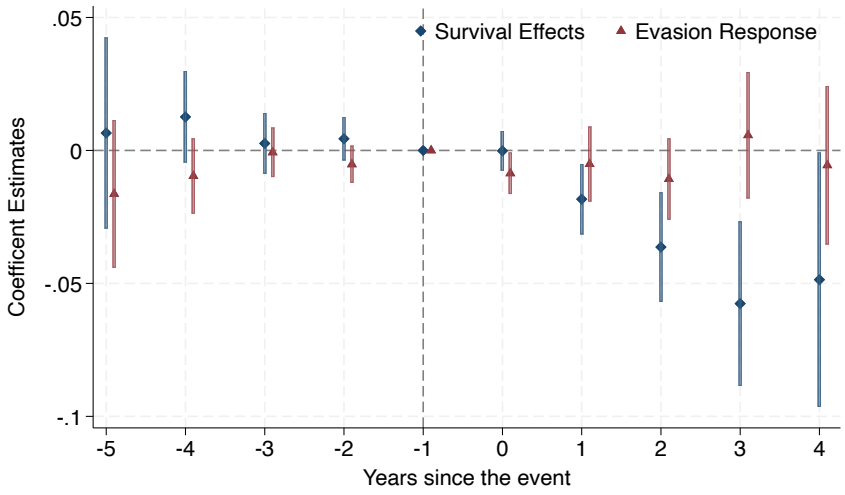



Figure A.4: Effects on Firm Survival versus Evasion Responses

*Notes:* The figure presents the dynamic effects of the reform on firm survival, distinguishing firm exit from evasion responses. The outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

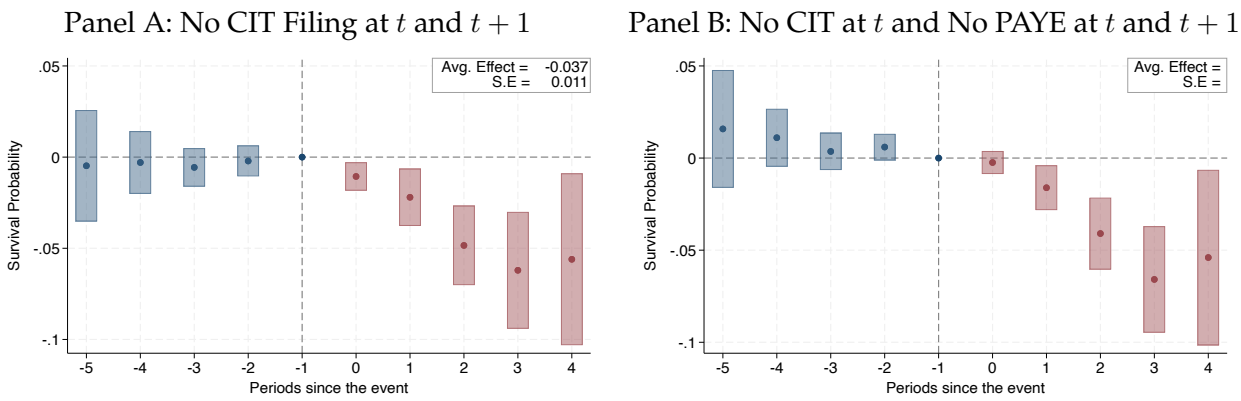



Figure A.5: Property Taxation and Firm Survival: Robustness to Alternative Exit Definitions

*Notes:* The figure presents robustness estimates for the effects of the reform on firm survival. In Panel A, I define exit as the absence of a corporate income tax (CIT) return for at least two consecutive years. In Panel B, I restrict exit to firms that have no CIT return in year $t$ and no records in the employee tax returns in both years $t$ and $t+1$. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

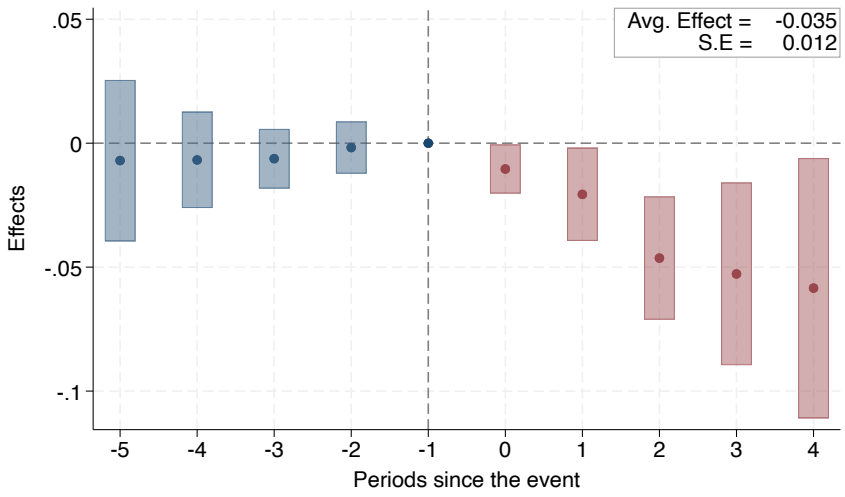



Figure A.6: Property Taxation and Firm Survival: Firms Present in the First Sample Year

*Notes:* The figure presents the dynamic effects of the reform on firm survival, restricting the sample to firms active in the first year of the sample rather than firms active in the year immediately preceding the reform. The outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

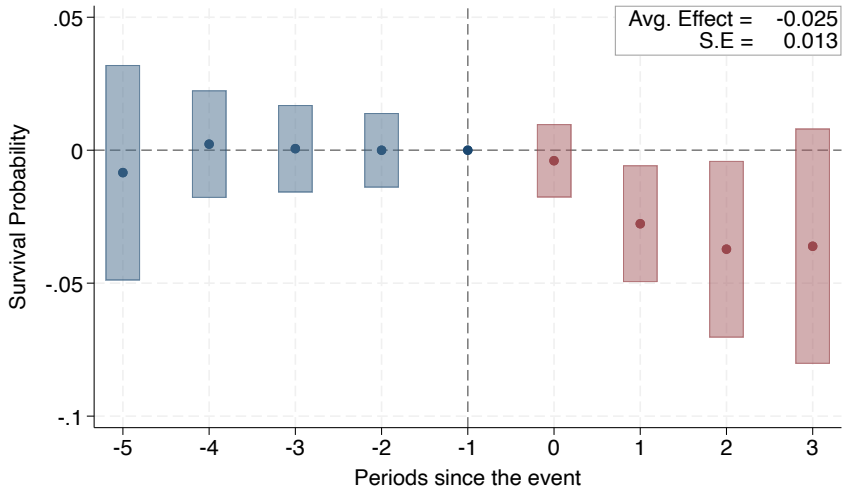



Figure A.7: Property Taxation and Firm Survival: Excluding the Central Division

*Notes:* The figure presents event-study estimates of the effects of the reform on firm survival, excluding the central division from the sample. The outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform, where negative values correspond to pre-treatment periods. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

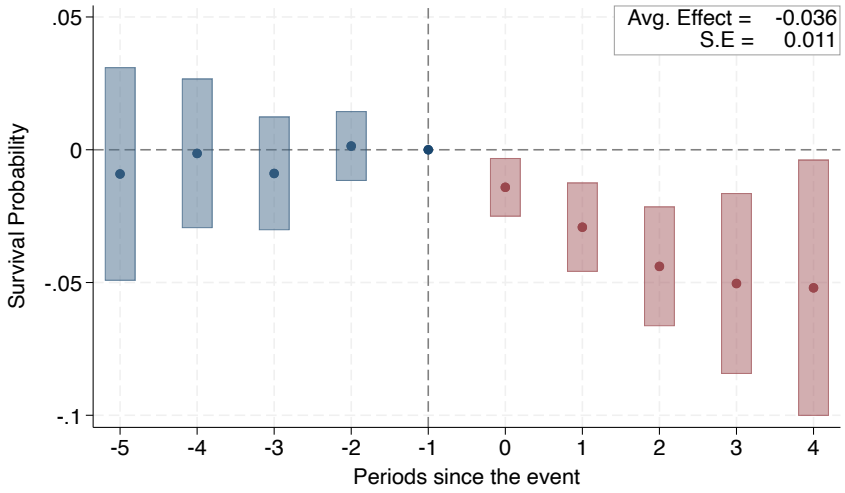



Figure A.8: Property Taxation and Firm Survival: Not-Yet-Treated Firms as the Comparison Group

*Notes:* The figure presents event-study estimates of the effects of the reform on firm survival, using not-yet-treated units as the comparison group rather than never-treated units. The outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform, where negative values correspond to pre-treatment periods. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

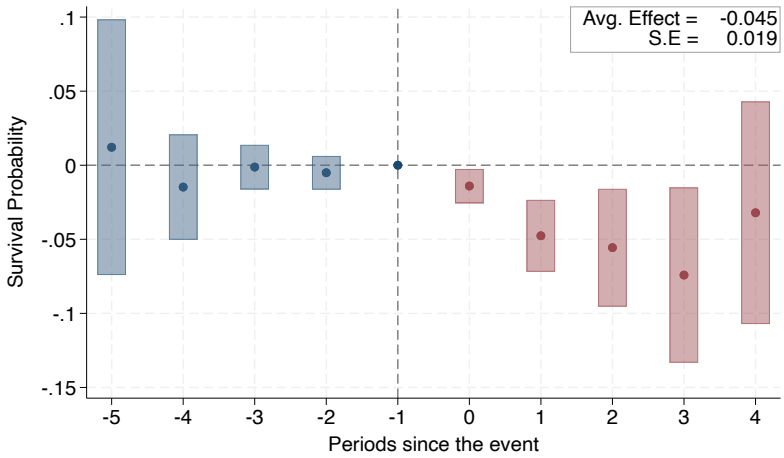



Figure A.9: Survival Effects Using Firms Outside GKMA as the Never-Treated Units

*Notes:* This figure presents the effects of the reform on firm survival with the never-treated group restricted to firms located outside the Greater Kampala Metropolitan Area (GKMA), thereby excluding nearby firms that are most likely to experience spillovers arising from the general equilibrium adjustments. Specifically, the never-treated group comprises firms in the 15 districts closest to the GKMA that are not themselves within the GKMA. This group is exactly similar to that used in SUTVA validation in Section VI.A. The regressions are estimated on an balanced panel comprised of 110,392 firm-year observations. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

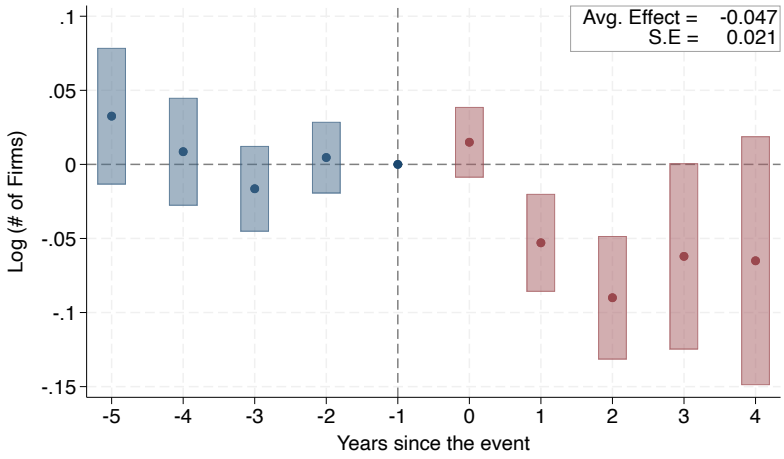



Figure A.10: Effects of the Reform on the Number of Operating Firms

*Notes:* The figure presents the effects of the reform on the number of operating firms in each county-industry cell. The data are aggregated to the county-industry level using four-digit industry classifications, and the outcome variable is the log number of firms in each cell. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with not-yet-treated units serving as the comparison group. Each point estimate represents the average treatment effect for a given event-time period relative to the reform, with period −1 as the reference category. The vertical bars denote the corresponding 95 percent confidence intervals. Standard errors are clustered at the county-industry level.

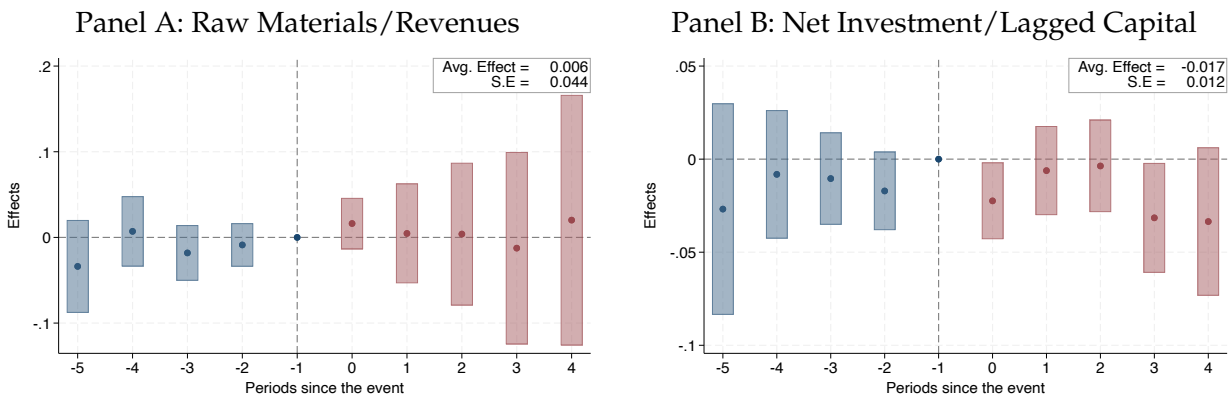



Figure A.11: Effects of the Reform on Investment and Raw Materials

*Notes:* The figure presents the effects of the reform on firms' investment rate and raw material purchases. The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. Raw material purchases are scaled by pre-reform revenues. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. For investment, the sample comprises 25,785 firm-year observations, while the sample for raw material purchases comprises 33,012 firm-year observations. The horizontal axis indicates the number of years since the reform. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. The vertical bars indicate 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

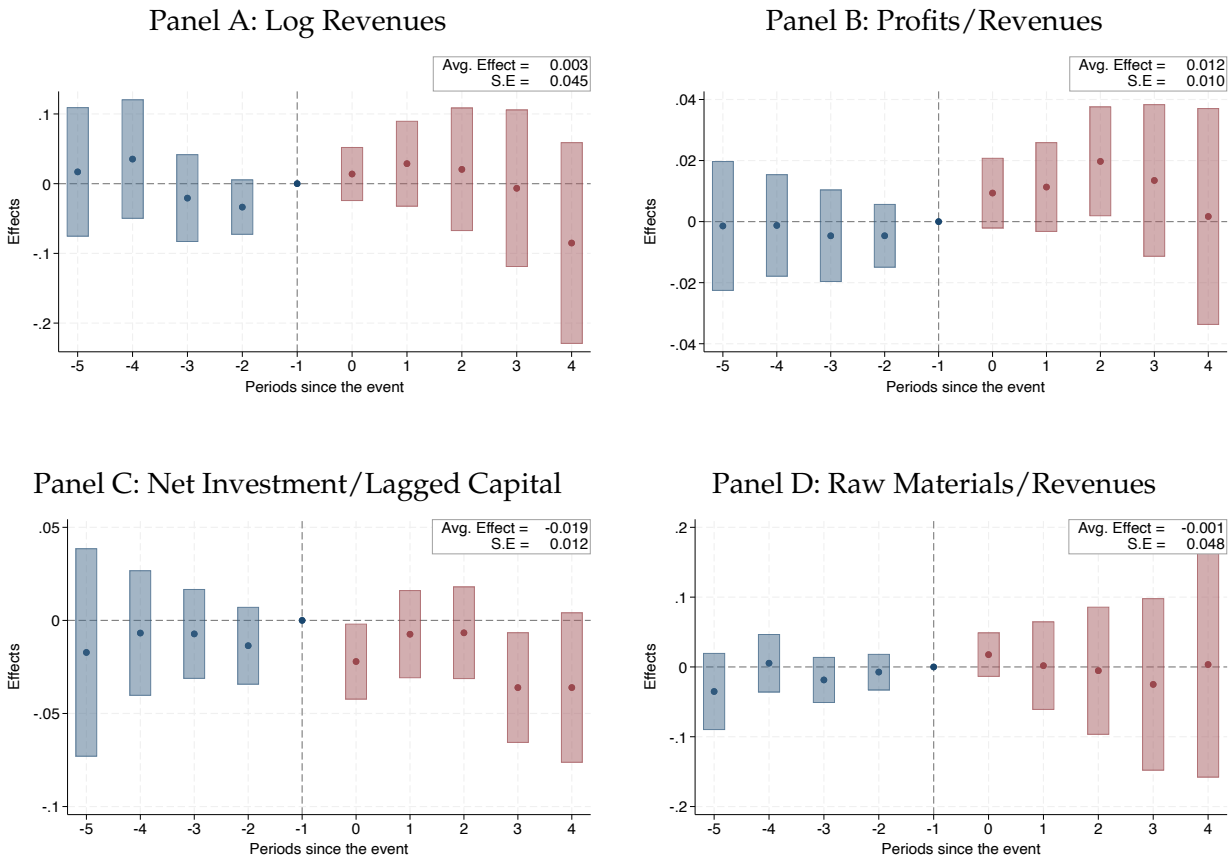



Figure A.12: Effects of the Reform on Revenues, Profits, Investment, and Raw Materials: *Unweighted Regressions*

*Notes:* The figure presents the effects of the reform on firm revenues, profits, investment, and raw material purchases. All regressions are estimated on an unweighted sample of firms in a balanced panel. Revenues are expressed in logarithms. Profits are measured as earnings before interest and taxes (EBIT), scaled by pre-reform revenues. The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. Raw material purchases are scaled by pre-reform revenues. For outcomes scaled by pre-reform revenues, the denominator is the average revenue over the two pre-reform years. For revenues, profits, and raw material purchases, the sample comprises 33,012 firm-year observations. For investment, the sample comprises 25,785 firm-year observations. The horizontal axis indicates the number of years since the reform. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. The vertical bars indicate 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

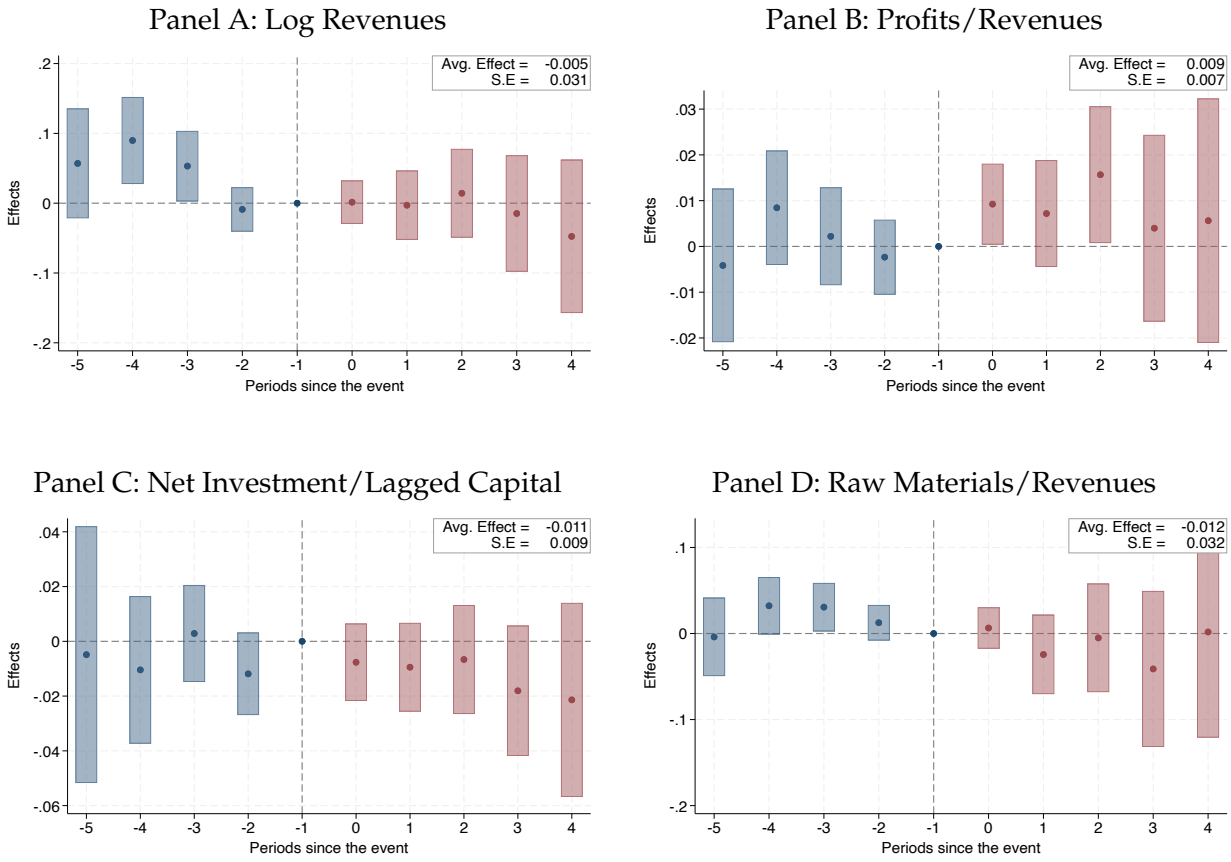



Figure A.13: Effects of the Reform on Revenues, Profits, Investment, and Raw Materials: *Unbalanced Panel*

*Notes:* The figure presents the effects of the reform on firm revenues, profits, investment, and raw material purchases. All regressions are estimated using an unbalanced panel and are weighted by the logarithm of pre-reform revenues. Revenues are expressed in logarithms. Profits are measured as earnings before interest and taxes (EBIT), scaled by pre-reform revenues. The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. Raw material purchases are scaled by pre-reform revenues. For outcomes scaled by pre-reform revenues, the denominator is the average revenue over the two pre-reform years. For revenues, profits, and raw material purchases, the sample comprises 96,744 firm-year observations. For investment, the sample comprises 49,348 firm-year observations. The horizontal axis indicates the number of years since the reform. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. The vertical bars indicate 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

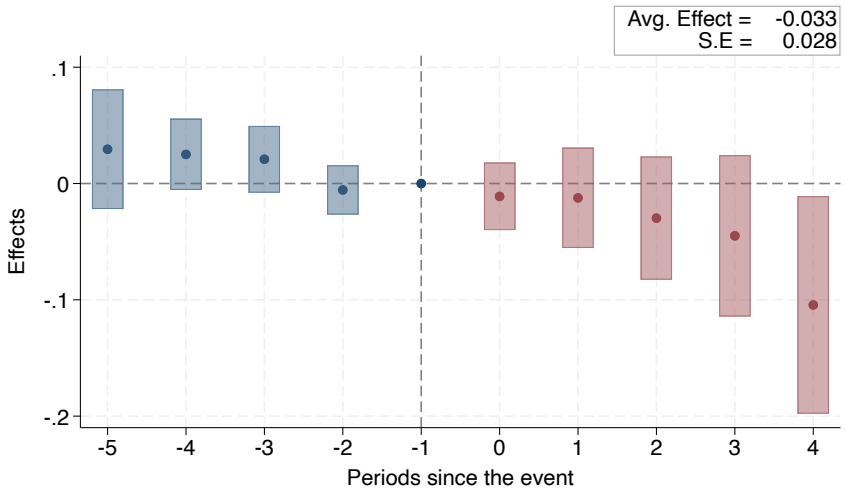
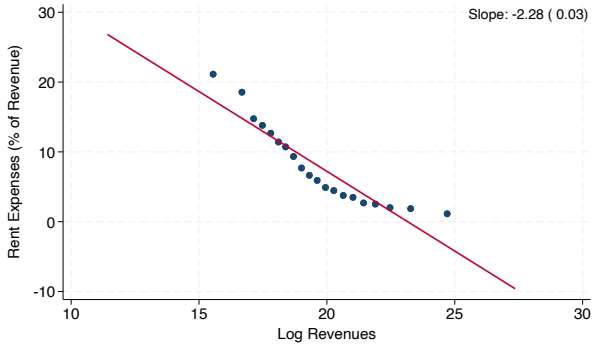



Figure A.14: Effects on Firm Revenues: Expanded Sample

*Notes:* The figure presents the effects of the reform on firm revenues, scaled by pre-reform values. The sample includes all firms with positive revenues in the pre-reform period, expanded to all years of the sample, with zero revenues assigned in any year in which the firm is not active. the sample comprises 142,677 firm-year observations and the regression is weighted by the log of pre-reform revenues. The horizontal axis denotes the number of years since the reform implementation. Each dot represents the point estimate of the treatment effect for a given year, with year −1 as the reference category. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with never-treated units serving as the comparison group. Standard errors are clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

Panel A: Rental Expenses Against Revenues




Panel B: Land and Buildings Against Revenues

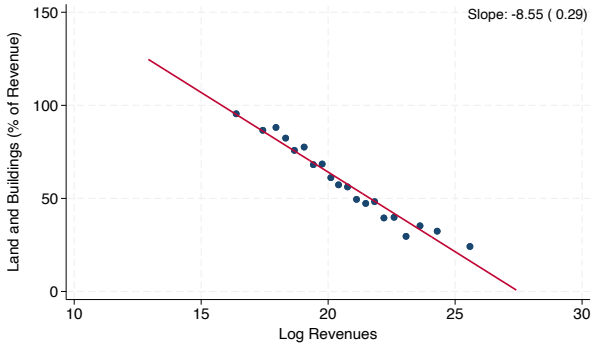



Figure A.15: Property Intensity and Firm Size

*Notes:* The figure presents binned scatterplots of the relationship between firm size and the intensity with which firms use commercial property in production. Property intensity is measured in two ways: the ratio of rental expenses to revenues (Panel A) and the ratio of the value of land and buildings to revenues (Panel B). Firm size is measured as log revenues in both panels. Panel A is restricted to firms that report positive rental expenses; Panel B is restricted to property owners. Both intensity measures are winsorized at the 5th and 95th percentiles to limit the influence of extreme values. The solid line in each panel is the linear fit through the underlying firm-level observations, with the slope coefficient and its standard error reported in the upper corner. Observations are weighted by the logarithm of pre-reform revenues. The sample is restricted to the pre-reform period, and both the binned means and the linear fit are residualized on sector fixed effects. Panel A comprises 16,929 observations; Panel B comprises 6727 observations.

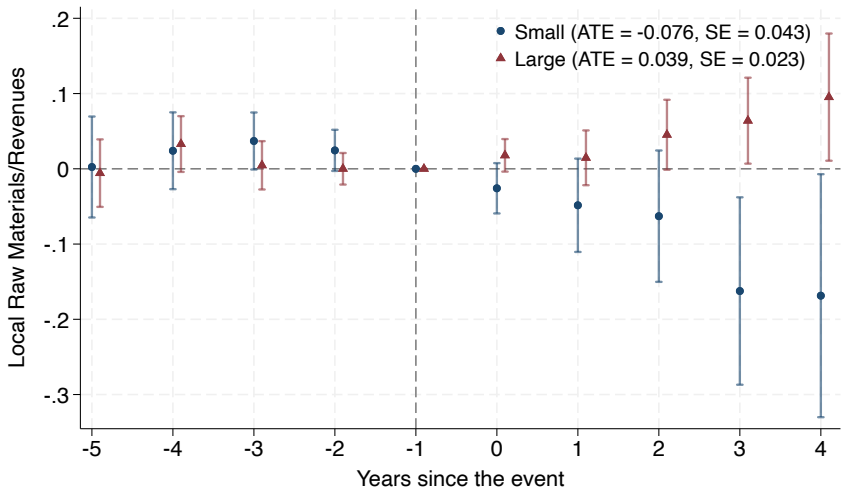



Figure A.16: Effects on Local Raw Material Purchases by Firm Size

*Notes:* The figure presents the effects of the reform on local raw material purchases for firms of different sizes. Firms are classified as small or large depending on whether their average pre-reform revenues fall below or above the sample median. The outcome is normalized by average revenue over the two pre-reform years. Regressions are estimated on an unbalanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. The estimation sample comprises 30,511 firm-year observations for small firms and 30,530 for large firms. Coefficients are estimated using the staggered difference-in-differences approach of Callaway and Sant'Anna (2021), with never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Average treatment effects for each firm category are reported in the legend, with standard errors in parentheses.

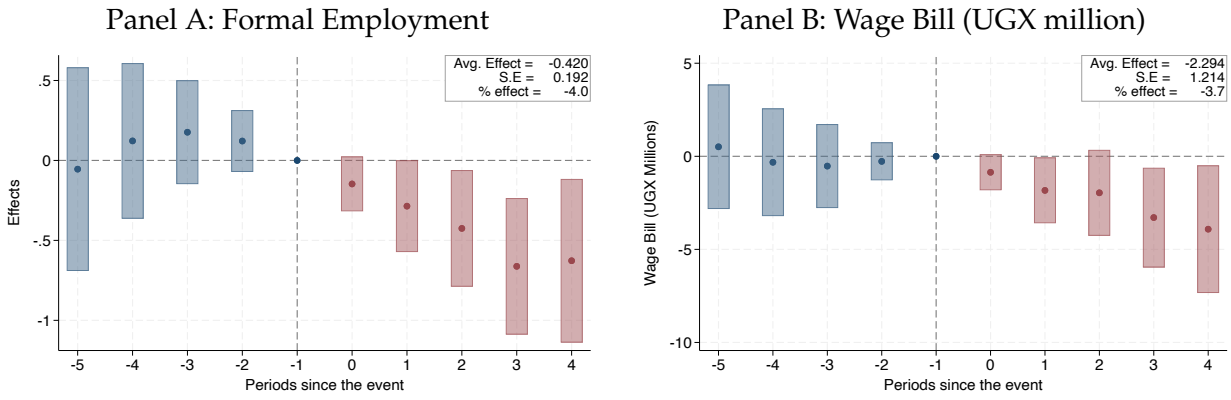



Figure A.17: Formal Employment and Wage Bill: *Unweighted Regressions*

*Notes:* The figure presents the effects of the reform on the number of formal workers and the total formal wage bill. All regressions are estimated on an unweighted sample of firms in a balanced panel. The percent effects are computed dividing the coefficient estimates by the pre-reform average of the outcome variable. The sample comprises 33,012 firm-year observations. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Vertical bars indicate 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

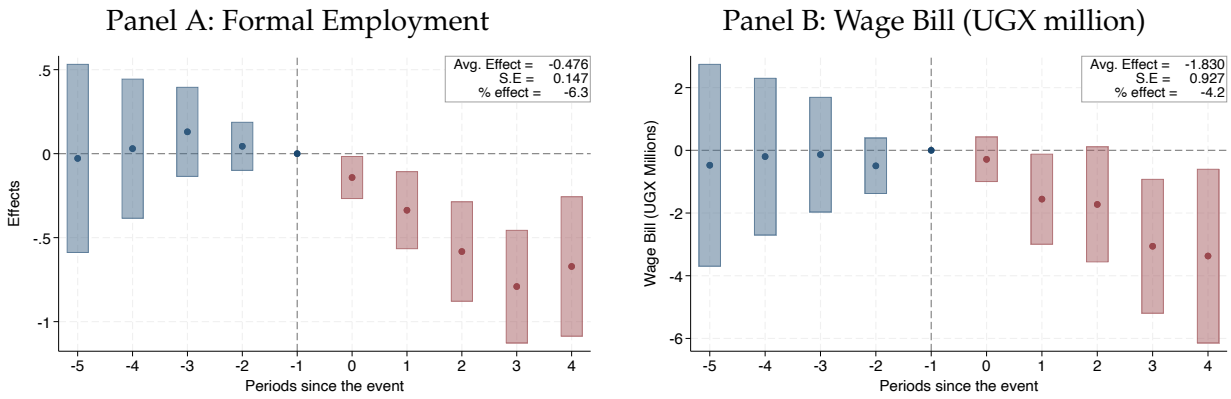



Figure A.18: Formal Employment and Wage Bill: *Unbalanced Panel*

*Notes:* The figure presents the effects of the reform on the number of formal workers and the total formal wage bill. Outcome variables are in level. All regressions are estimated using an unbalanced panel and are weighted by the logarithm of pre-reform revenues. The percent effects are computed dividing the coefficient estimates by the pre-reform average of the outcome variable. The sample comprises 61,041 firm-year observations. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Vertical bars indicate 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

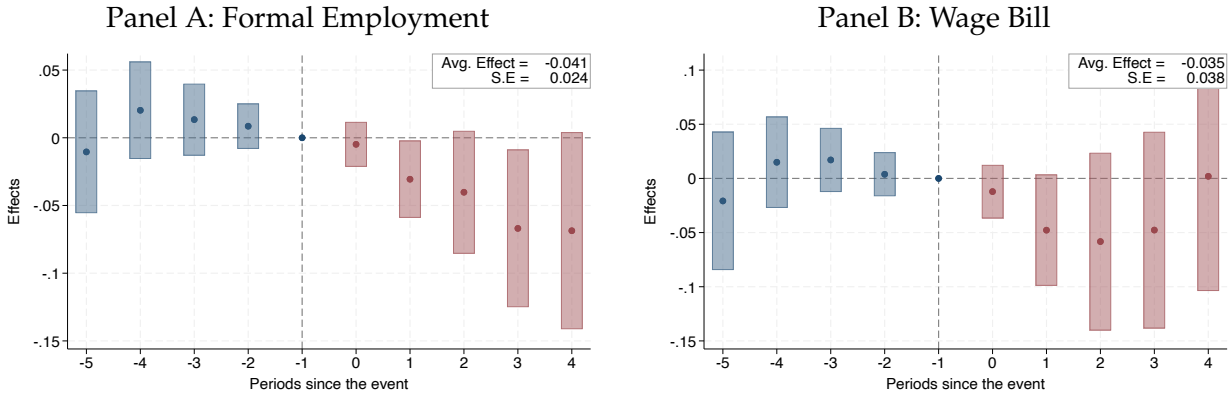



Figure A.19: Effects of the Reform on Formal Employment and Wage Bill: *Normalized by Pre-Reform Values*

*Notes:* The figure presents the effects of the reform on the number of formal workers and the total formal wage bill. The outcome variables are scaled by their pre-reform values so that coefficient can be interpreted as percent changes. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. The sample comprises 23,985 firm-year observations. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Vertical bars indicate 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

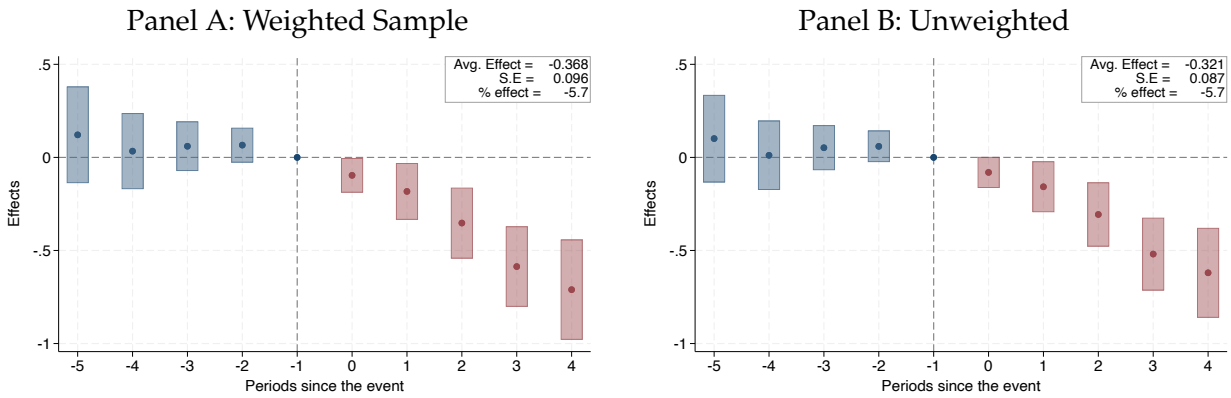



Figure A.20: Effects on Formal Employment: Expanded Sample

*Notes:* The figure presents the effects of the reform on formal employment. The sample includes all firms with positive revenues in the pre-reform period, expanded to all years of the sample, with zero formal employment assigned in any year in which the firm is not active. The outcome variable is formal employment in levels. In Panel A, the regression is weighted by the log of pre-reform revenues. Panel B presents results from unweighted regression. The sample comprises 142,677 firm-year observations. The horizontal axis denotes the number of years since the policy implementation. Each dot represents the point estimate of the treatment effect for a given year, with year −1 as the reference category. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach following Callaway and Sant'Anna (2021), with never-treated units serving as the comparison group. Standard errors are clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

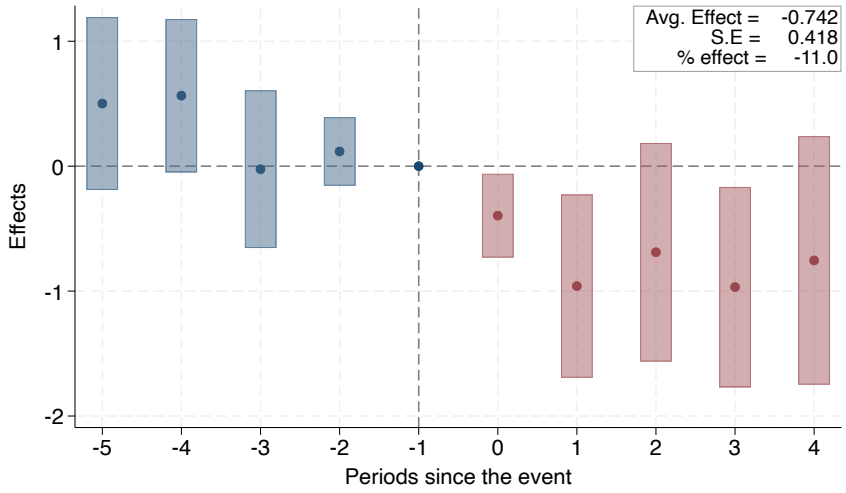



Figure A.21: Employment Effects Using Firms Outside GKMA as the Never-Treated Units

*Notes:* This figure presents employment effects of the reform using a comparison group restricted to firms located outside the Greater Kampala Metropolitan Area (GKMA), thereby excluding nearby firms that are most likely to absorb workers displaced from treated firms in the city. Specifically, this comparison group comprises firms in the 15 districts closest to the GKMA that are not themselves within the GKMA. The outcome variable the number of formal workers is in levels. The regressions are estimated on an unbalanced panel comprised of 62,886 firm-year observations. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

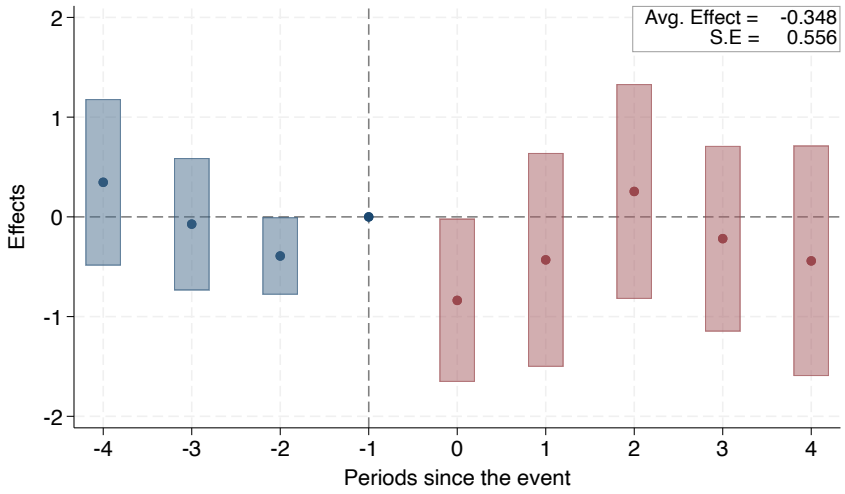



Figure A.22: Placebo Test: Employment Effects Among Firms Outside Kampala

*Notes:* The figure presents placebo estimates of the reform's employment effects. The sample is restricted to firms located outside Kampala City, comparing those within the greater Kampala metropolitan area (GKMA), which are geographically proximate to treated firms, to those located entirely outside the GKMA. The comparison group comprises firms in the 15 districts closest to the GKMA that are not themselves within the GKMA. If workers displaced by treated firms were absorbed by nearby firms rather than experiencing genuine job losses, employment among GKMA firms should increase differentially after the reform. The absence of significant positive post-reform effects is inconsistent with such relocation, supporting the interpretation that the estimated employment declines reflect real job losses rather than spatial reallocation of workers. The regressions are estimated on an unbalanced panel of 11,341 firm-year observations. The specification includes industry-size-year and pre-reform accounting month fixed effects. By construction, this comparison eliminates the staggered adoption and yields fewer pre-reform periods. The staggered difference-in-differences estimator therefore collapses to a traditional difference-in-differences. Standard errors are clustered at the firm level.

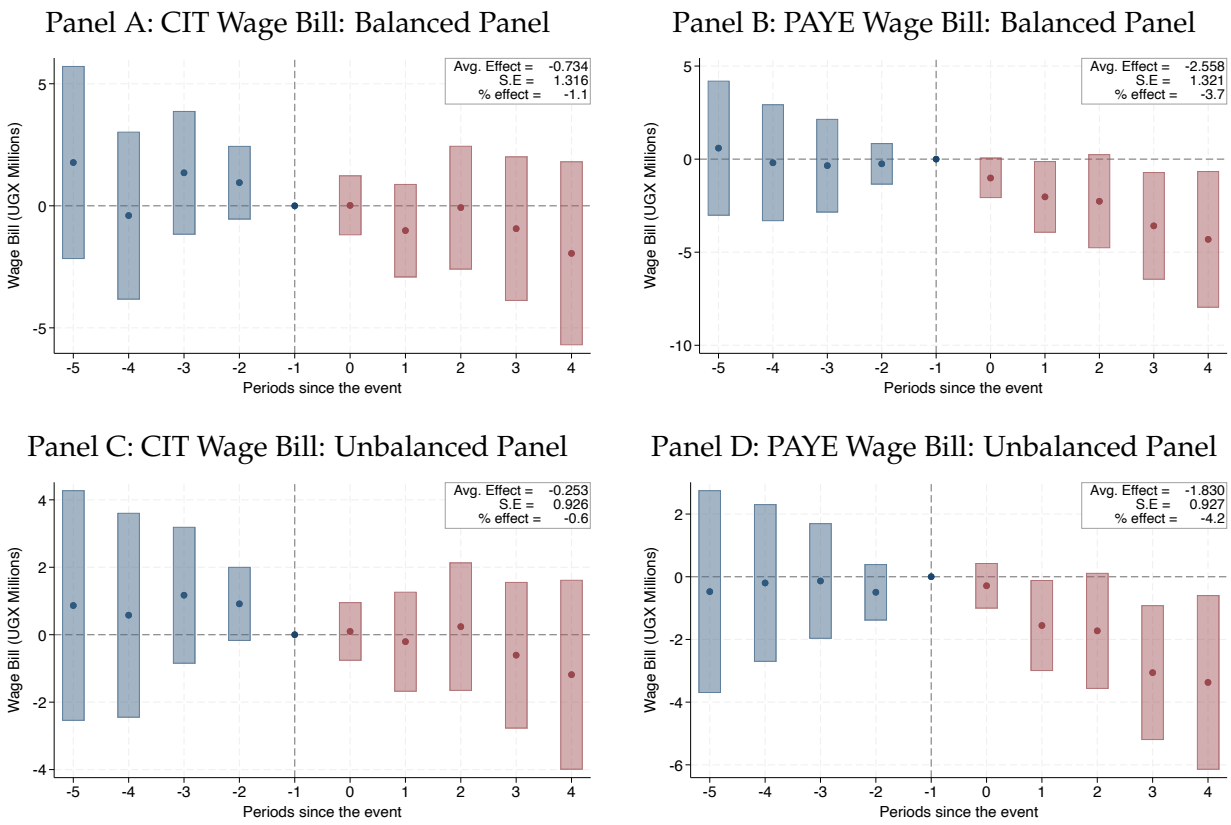



Figure A.23: CIT and PAYE Wage Bills

*Notes:* The figure presents the effects of the reform on firms' wage bills. Panels A and B present results from the balanced panel with 33,012 firm-year observations while Panels C and D present results from unbalanced panel with 61,041 firm-year observations. All regressions are weighted by the logarithm of pre-reform revenues. The wage bills are expressed in millions of Uganda shillings, constant 2016/17. Each dot represents the estimated treatment effect for a given event year, with event time −1 omitted as the reference period. Vertical bars indicate 95 percent confidence intervals. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

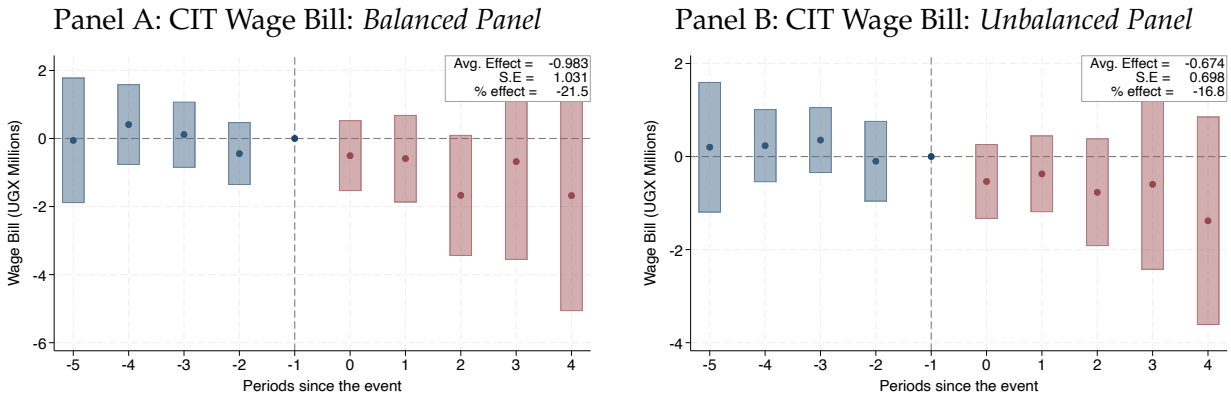



Figure A.24: Effects on the Reported CIT Wage Bill Among Firms with no PAYE Workers

*Notes:* The figure presents the effects of the reform on firms' CIT wage bills, restricting the sample to firms with no formal workers in the pre-reform period. Panels A presents results from the balanced panel with 33,012 firm-year observations while Panel B presents results from unbalanced panel with 61,041 firm-year observations. All regressions are weighted by the logarithm of pre-reform revenues. The wage bills are expressed in millions of Uganda shillings, constant 2016/17. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

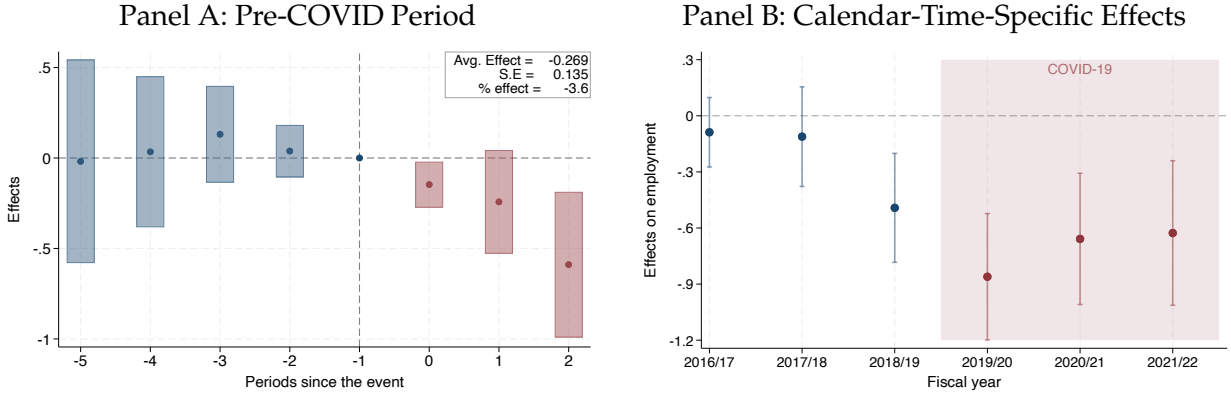



Figure A.25: Effects on Employment: *Addressing COVID-19*

*Notes:* Panel A presents event-study estimates of the effects on employment, restricting the sample to the pre-COVID period (ending in 2018/19), with 42,263 firm-year observations. Panel B presents calendar-time-specific treatment effects on employment, estimated on an unbalanced panel of 61,041 firm-year observations. All regressions are weighted by the logarithm of pre-reform revenues. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with never-treated units as the comparison group. Standard errors are clustered at the firm level. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

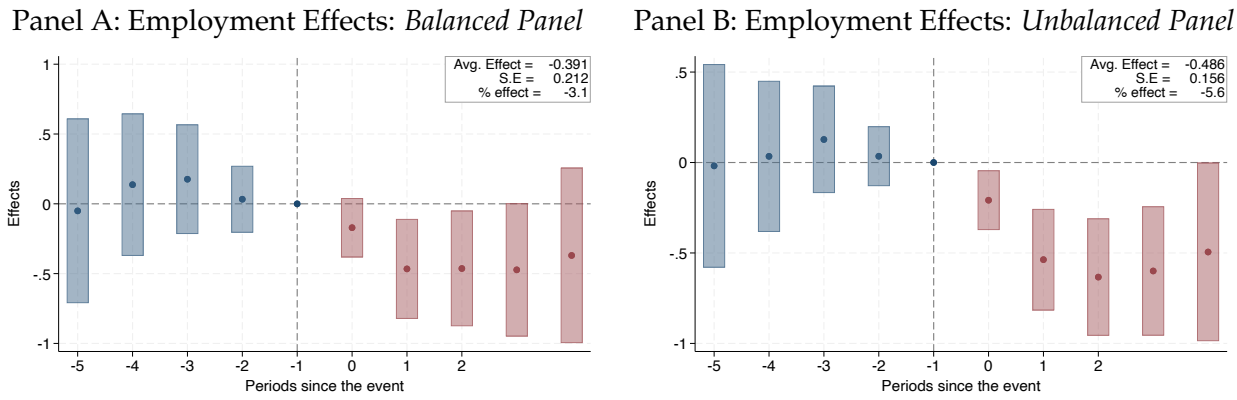



Figure A.26: Effects of the Reform on Employment, Excluding Central Division

*Notes:* The figure presents the effects of the reform on the number of formal workers, while excluding the central division from the sample. Panels A presents results from the balanced panel with 18,630 firm-year observations while Panel B presents results from unbalanced panel with 34,675 firm-year observations. All regressions are weighted by the logarithm of pre-reform revenues. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

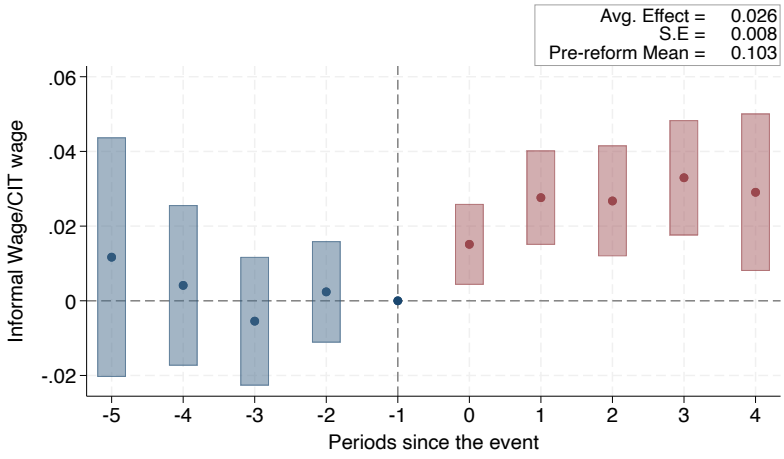



Figure A.27: Effects on Wage Informality: *Unweighted Regression*

*Notes:* The figure presents the effects of the reform on wage informality, measured as the fraction of the wage bill reported in corporate income tax returns that does not appear in individual employee income tax filings. This means coefficients can be interpreted as a percentage point changes. All regressions are estimated on a balanced panel of active firms. The sample comprises 24,579 firm-year observations. The horizontal axis reports the number of years relative to the reform implementation. Each dot represents the estimated treatment effect for a given year, with year −1 as the reference category. The vertical bars denote the corresponding 95 percent confidence intervals. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

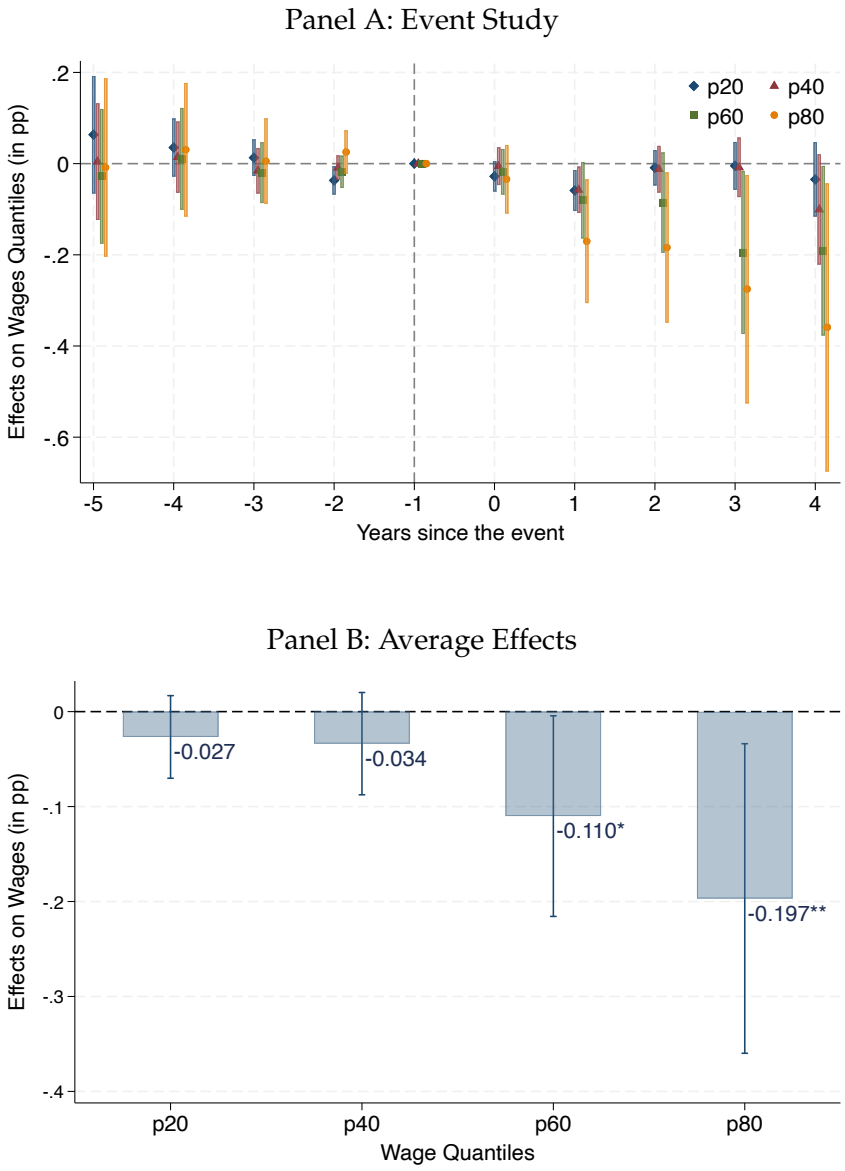



Figure A.28: Effects on Within-Firm Wage Quantiles, Scaled by Pre-reform Wage Bill

*Notes:* The figure presents the effects of the reform on within-firm wage quantiles across the wage distribution. The dependent variables are the $20^{th}$, $40^{th}$, $60^{th}$, and $80^{th}$ percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm $i$'s $q^{th}$ wage quantile at time $t$ and $W_{i,0}$ denotes firm $i$'s pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B reports the average treatment effects. The coefficients are estimated using the staggered difference-in-differences design following Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

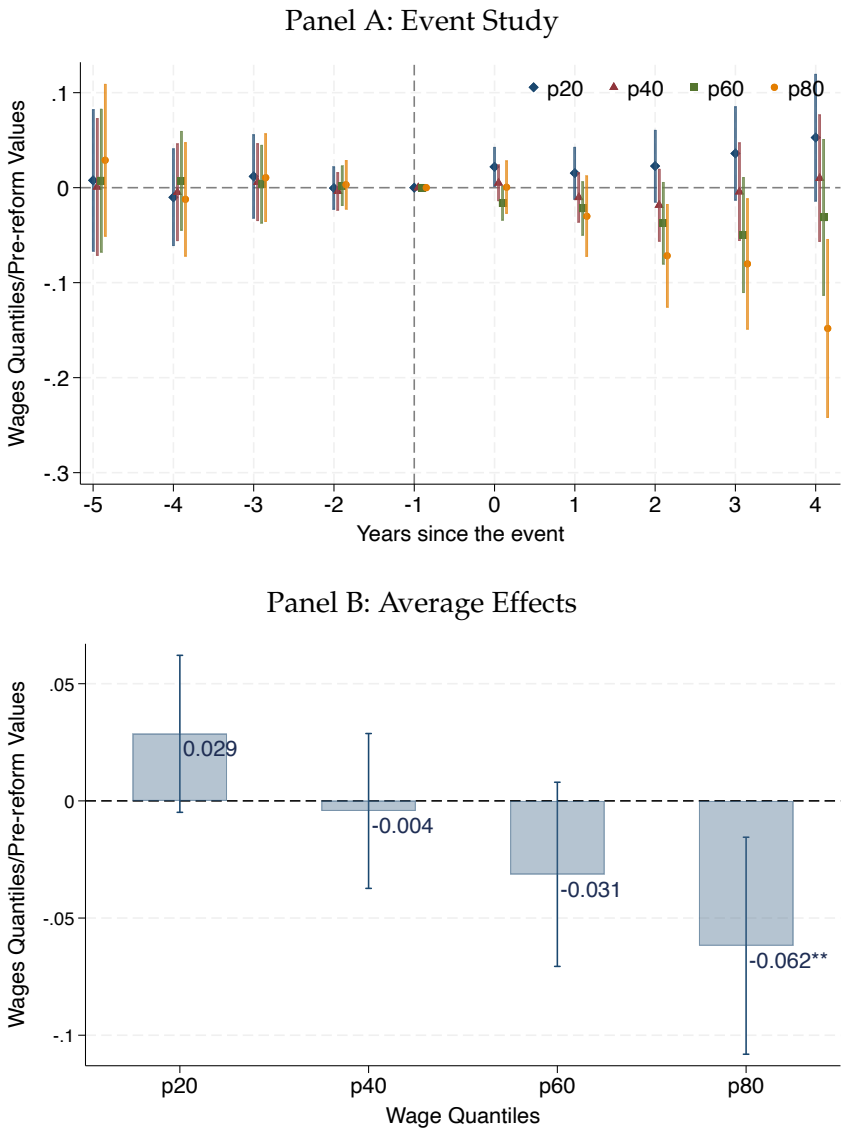



Figure A.29: Effects on Within-Firm Wage Quantiles, Scaled by Pre-reform Values

*Notes:* The figure presents the effects of the reform on within-firm wage quantiles across the wage distribution. The dependent variables are the $20^{th}$, $40^{th}$, $60^{th}$, and $80^{th}$ percentiles of the within-firm wage distribution, scaled by their pre-reform values. Accordingly, the coefficient estimates should be interpreted as percent changes. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B reports the average treatment effects. The sample comprises 19,656 firm-year observations. All regressions are weighted by the logarithm of pre-reform formal wage bill. The coefficients are estimated using the staggered difference-in-differences design following Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

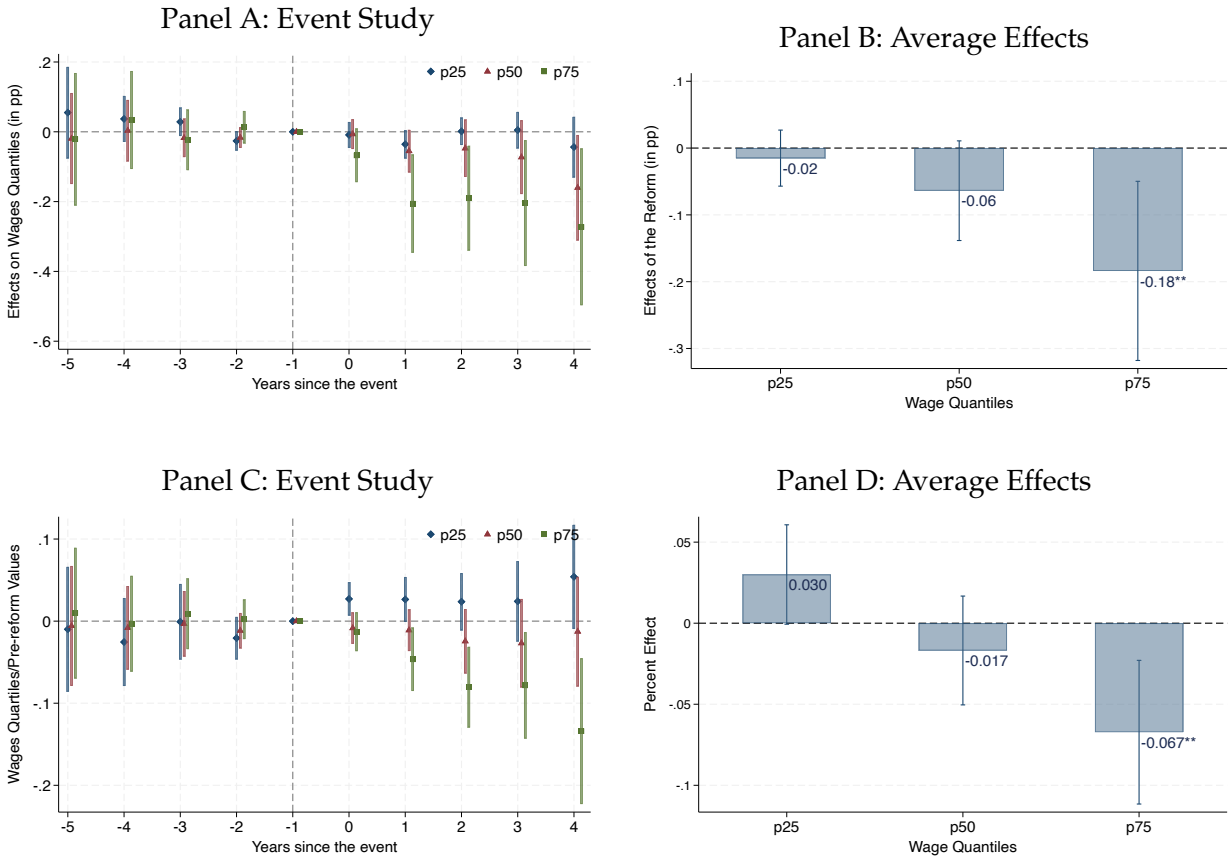



Figure A.30: Effects on Wage Quartiles

*Notes:* The figure presents the effects of the reform on within-firm wage quartiles. The dependent variables are the $25^{th}$, $50^{th}$, and $75^{th}$ percentiles of the within-firm wage distribution. Panels A and B present the results with the outcome variable scaled by the pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm $i$'s $q^{th}$ wage quantile at time $t$ and $W_{i,0}$ denotes firm $i$'s pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. Panels C and D present the results with outcome variable scaled by their pre-reform values allowing coefficients to be interpreted as percent changes. The sample comprises 19,656 firm-year observations. All regressions are weighted by the logarithm of pre-reform formal wage bill. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

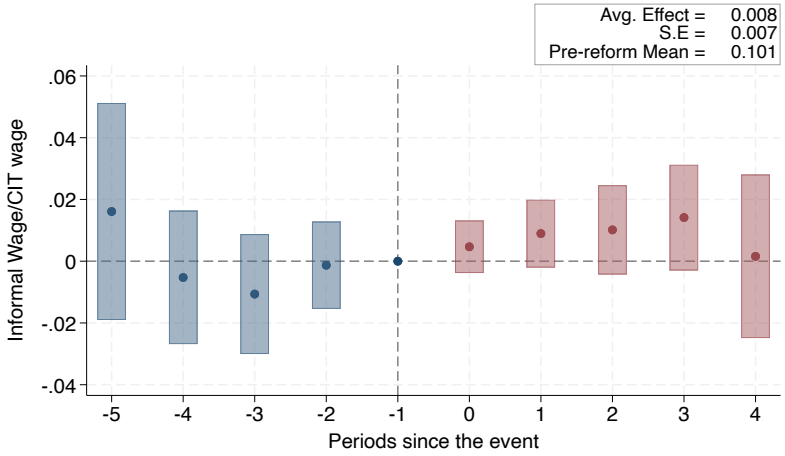



Figure A.31: Effects on Wage Informality Among Firms with at Least Five Pre-Reform Formal Workers

*Notes:* The figure presents the effects of the reform on wage informality among firms with at least five pre-reform workers. wage informality is measured as the fraction of the wage bill reported in corporate income tax returns that does not appear in individual employee income tax filings. The coefficients are estimated using the staggered difference-in-differences design following Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

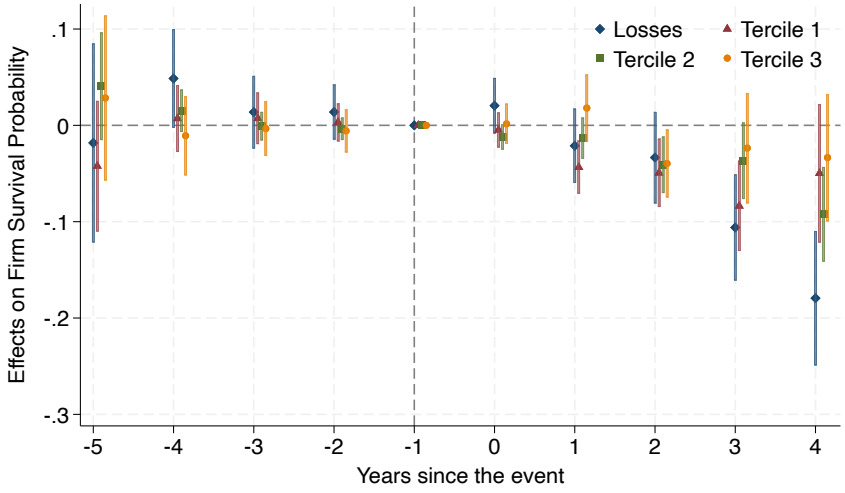



Figure A.32: Survival Effects by Profitability

*Notes:* The figure presents the effects of the reform on firm survival across different points of the pre-reform profit margin distribution.. The outcome variable is an indicator equal to one if the firm is operating at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

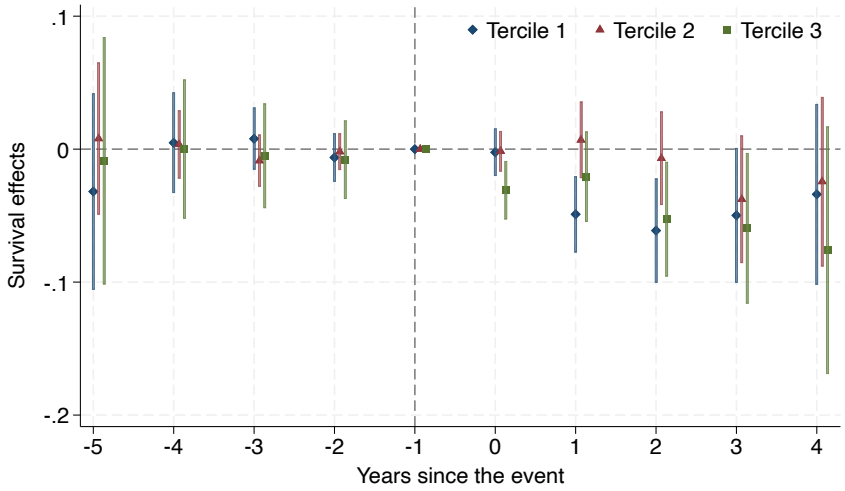



Figure A.33: Survival Effects by Leverage

*Notes:* The figure presents the effects of the reform on firm survival across firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

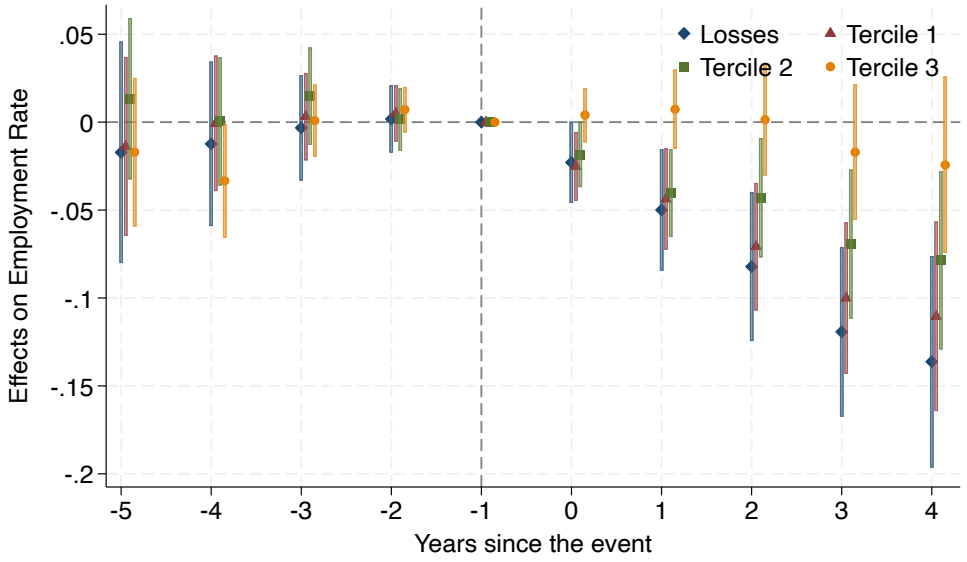



Figure A.34: Employment Effects by Profitability

*Notes:* The figure presents the dynamic effects of the reform on formal employment across different points of the pre-reform profit margin distribution. The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

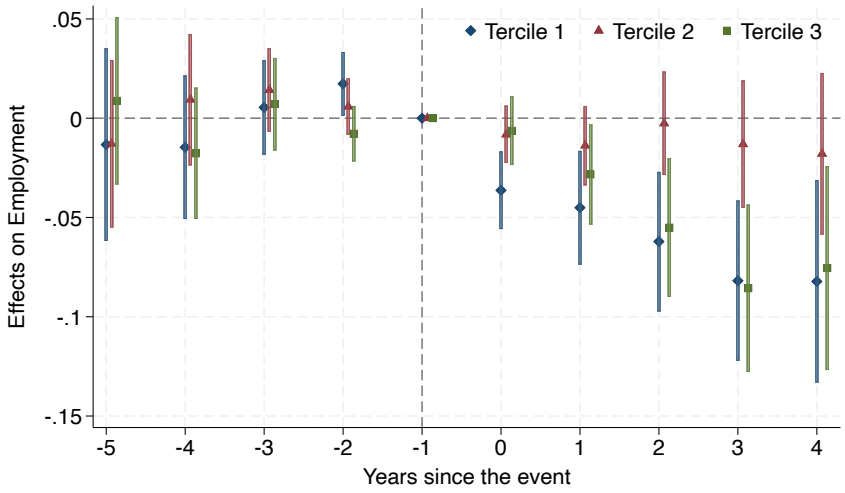



Figure A.35: Employment Effects by Leverage

*Notes:* The figure presents the dynamic effects of the reform on formal employment across firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at −1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

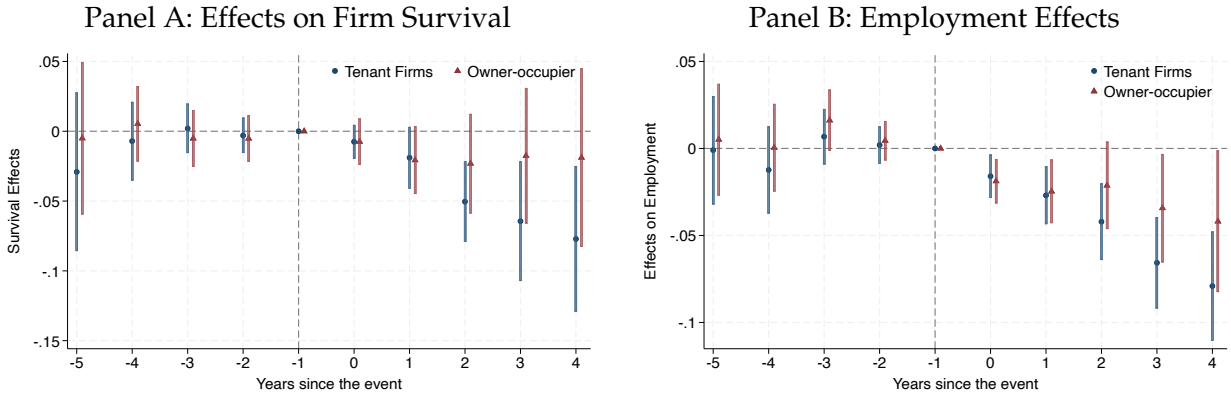



Figure A.36: Employment and Survival Effects by Occupancy Status

*Notes:* The figure presents the dynamic effects of the reform on firm survival and formal employment, separately for tenant and owner-occupier firms. In Panel A, the outcome variable is an indicator equal to one if the firm operated at time $t$, zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. In Panel B, the outcome variable is the number of workers scaled by its pre-reform value. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned occupancy status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at −1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using the staggered difference-in-differences approach, following Callaway and Sant'Anna (2021). The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

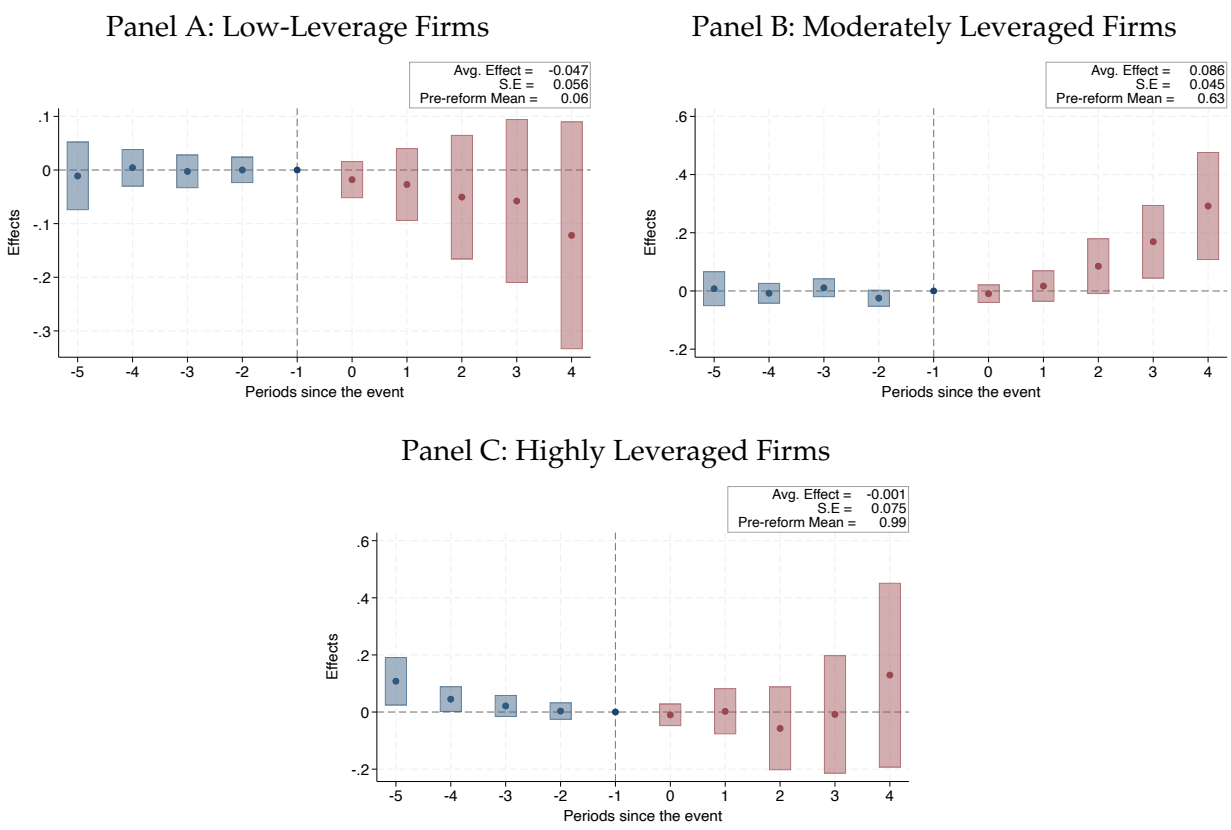



Figure A.37: Effects of the Reform on Firms' Debt

*Notes:* In all panels, the outcome variable is total debt scaled by pre-reform total assets. In Panel A, I restrict the sample to firms in the first tercile of the pre-reform leverage distribution. Panel B restricts the sample to firms in the second tercile, while Panel C restricts the sample to firms in the third tercile (highly leveraged firms). The coefficients are estimated using a staggered difference-in-differences design following Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, accounting month, and accupancy status (owner-occupier vs tenant). Standard errors are clustered at the firm level.

## *I.E. Business Rental Cost Channel*

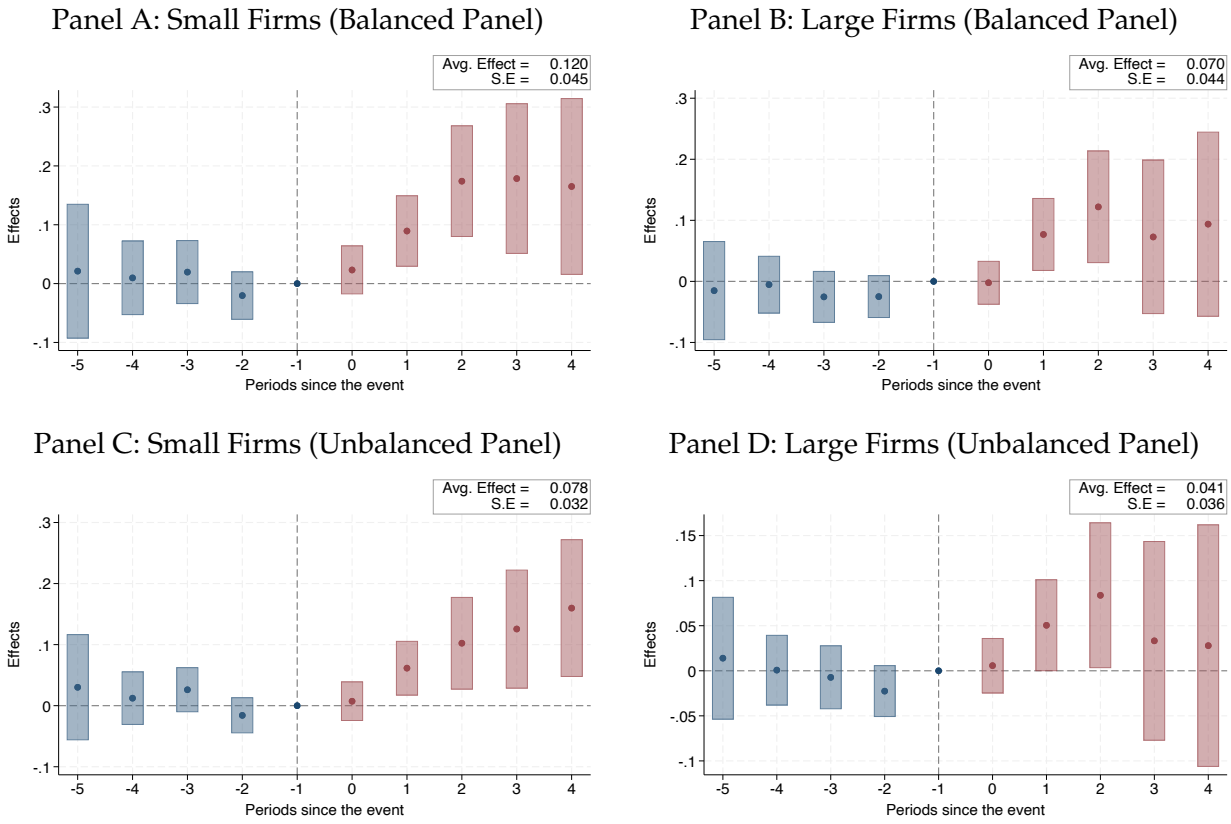



Figure A.38: Effects of the Reform on Business Rental Expenses by Firm Size

*Notes:* The figure shows the effects of the reform on business rental expenses by firm size. Firms are classified as small and large using the median level of pre-reform employment as the cutoff. Panels A and B present the results estimated on a balanced panel of firms that reported paying rent in the pre-reform period while Panels C and D present the results from an unbalanced panel. The dependent variable is rent costs scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals.

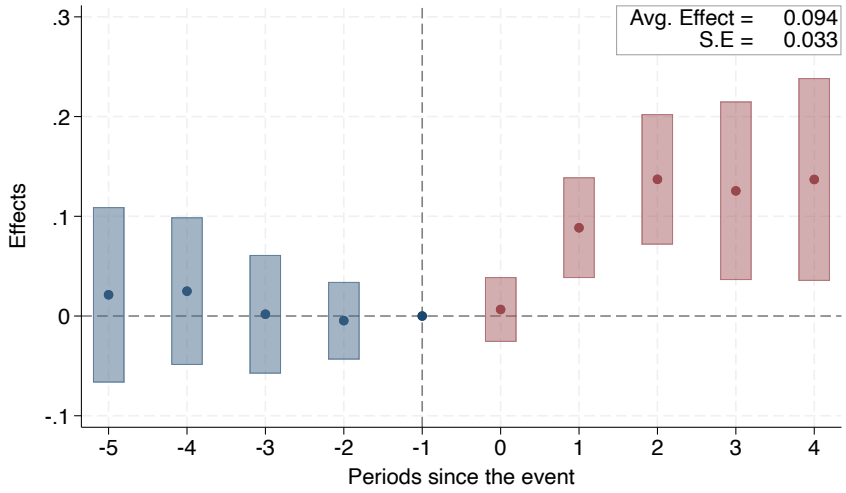



Figure A.39: Effects of Property Taxes on Business Rental Expenses: Never-Treated Firms as the Comparison Group

*Notes:* The figure presents the dynamic effects of property taxes on firms' rental expenses. The dependent variable is rental costs scaled by their pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), using never-treated units as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

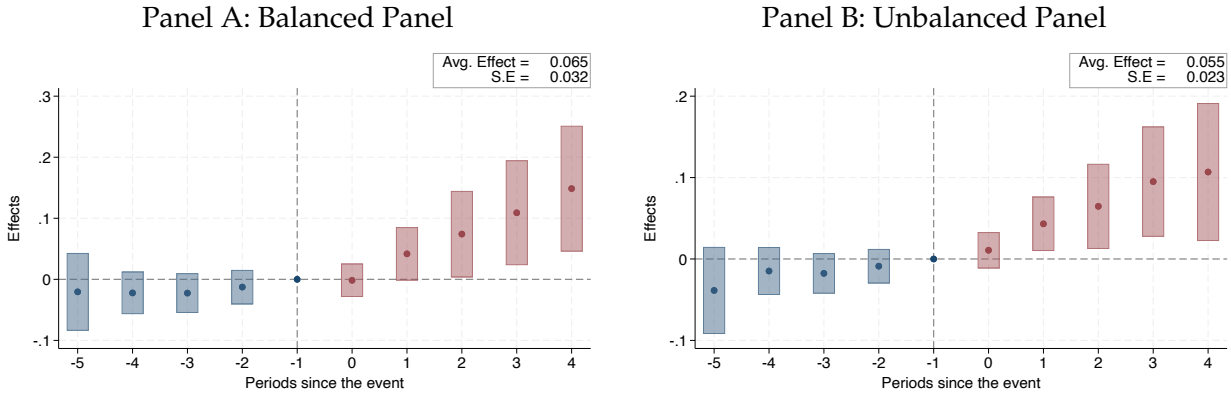



Figure A.40: Effects of the Reform on Business Rental Expenses, Restricting Comparison Group to Firms Outside GKMA

*Notes:* This figure presents the effects of the reform on business rental expenses with the never-treated group restricted to firms located outside the Greater Kampala Metropolitan Area (GKMA), thereby excluding nearby firms that are most likely to experience spillovers arising from the general equilibrium adjustments. Specifically, the never-treated group comprises firms in the 15 districts closest to the GKMA that are not themselves within the GKMA. This group is exactly similar to that used in SUTVA validation in Section VI.A and V.A. Panel A presents estimates based on a balanced panel of firms that reported paying rent in the pre-reform period, while Panel B presents estimates based on an unbalanced panel. The sample in Panel A comprises 29,466 firm-year observations, whereas the sample in Panel B comprises 54,476 firm-year observations. The dependent variable is rent costs scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month.

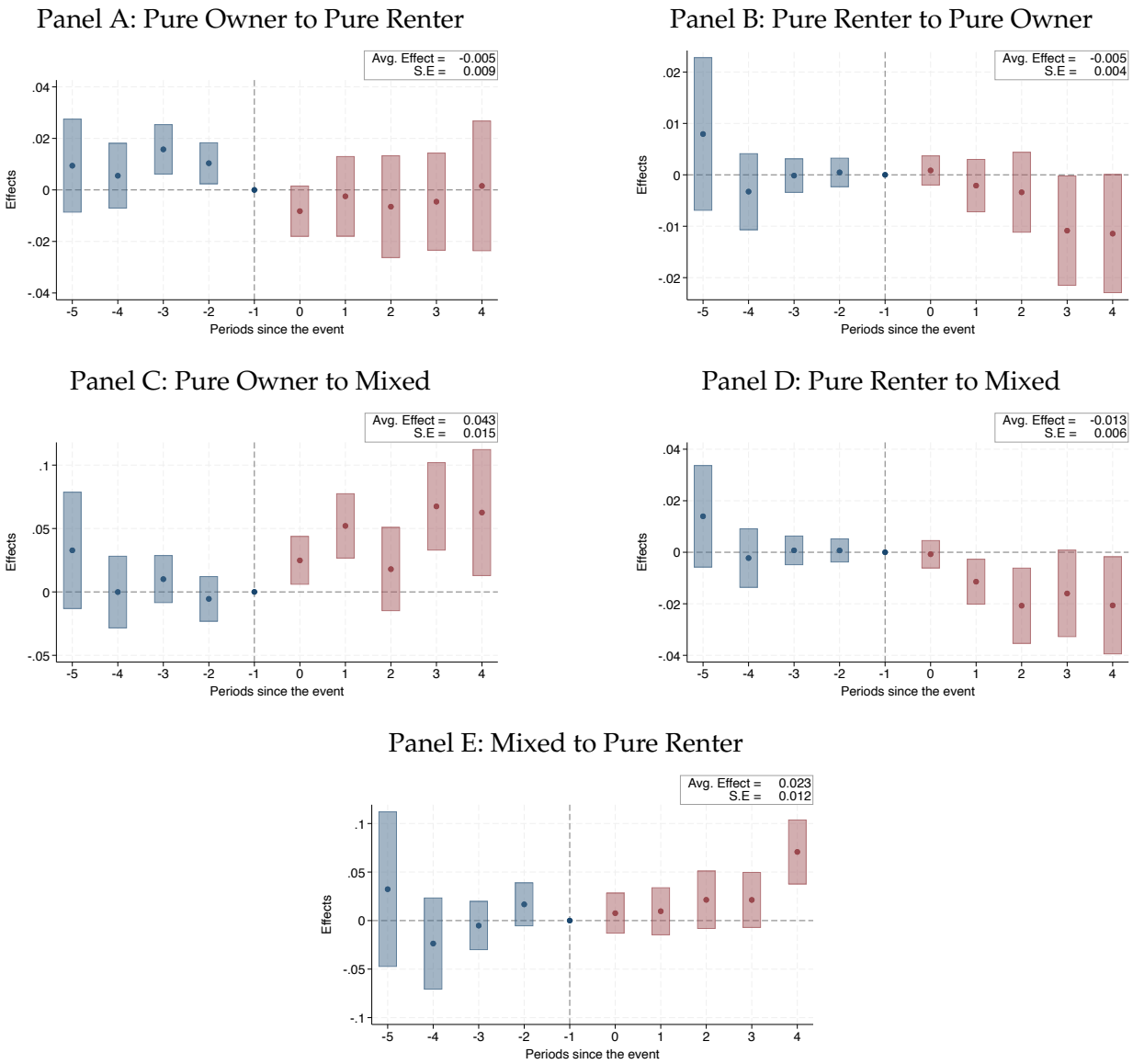



Figure A.41: Effects of the Reform on Occupancy Decisions: *Own or Rent?*

*Notes:* The figure presents the dynamic effects of the reform on firms' occupancy decisions, examining transitions across occupancy categories at the extensive margin. All outcome variables are binary indicators. The categories are defined as follows: a pure owner is defined as a firm that reports positive land and building values but no rental expenses; a pure renter is a firm that reports positive rental expenses but no land and building values; and a mixed firm reports positive values for both categories. Panel A examines the probability that a pure owner transitions to pure renter following the reform. Panel B examines the probability that a pure renter transitions to pure owner. Panel C examines the probability that a pure owner transitions to the mixed category. Panel D examines the probability that a pure renter transitions to the mixed category. Together, Panels C and D assess whether the reform induced firms to move toward partial occupancy arrangements—simultaneously owning and renting space— from previously exclusive ownership or tenancy arrangements. Panel E examines the probability of a mixed-category firm transitioning to pure renter status. All regressions are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

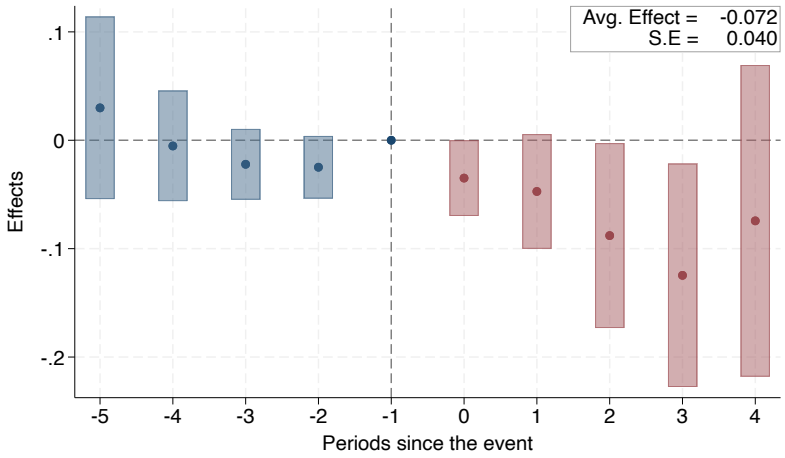



Figure A.42: Reported Land and Building Values: Sample of Pure Owners

*Notes:* The figure presents the dynamic effects of the reform on reported land and building values among pure owners—defined as firms reporting positive land and building values but no rental expenses in the pre-reform period. The outcome variable is the reported value of land and buildings, scaled by their pre-reform values, such that the coefficients can be interpreted as percentage changes relative to the pre-reform period. The estimation conditions on firms' baseline sector of operation, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

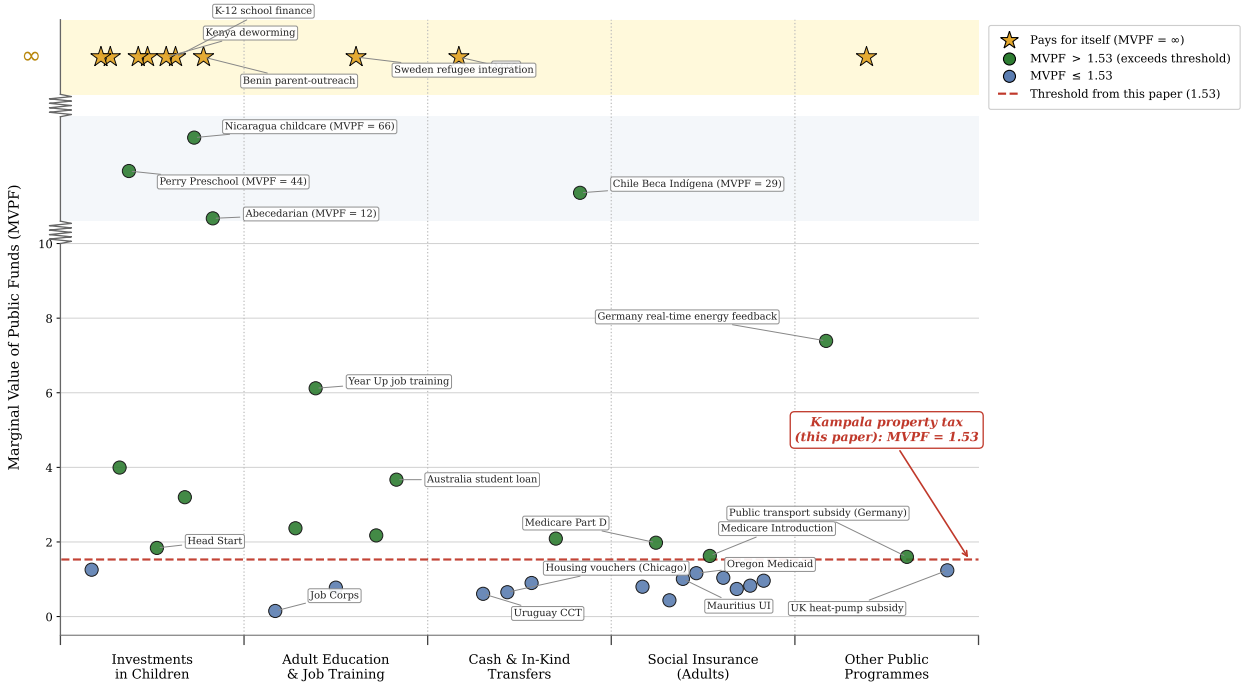



Figure A.43: Property-Tax MVPF in Context: Comparison with Government Spending Programs

*Notes:* presents estimates of the marginal value of public funds (MVPF) for selected public spending programs, comparing them to the estimated MVPF for the property tax reform. MVPFs compiled from Hendren and Sprung-Keyser (2020) supplemented with more recent estimates from the Policy Impacts Library (policyimpacts.org). Sample: 40 government spending programs. A complete list of the programs is presented in Appendix Table A.13.

# II. ADDITIONAL TABLES

## *II.A. Business Activity*

Table A.1: Lee-Style Bounds on Revenues and Profits

| | Log revenues | | | Profits/Revenues | | |
|---|---|---|---|---|---|---|
| | (1) Main | (2) LB | (3) UB | (4) Main | (5) LB | (6) UB |
| Average ATT | -0.005 | -0.070*** | 0.024 | 0.009 | 0.002 | 0.042*** |
| | (0.024) | (0.024) | (0.024) | (0.006) | (0.006) | (0.005) |
| | [-0.052, 0.042] | [-0.117, -0.023] | [-0.023, 0.070] | [-0.003, 0.020] | [-0.010, 0.014] | [0.032, 0.052] |
| Observations | 96,744 | 96,338 | 96,330 | 96,744 | 96,599 | 96,355 |
| Sector-Size | Yes | Yes | Yes | Yes | Yes | Yes |
| Accounting Month | Yes | Yes | Yes | Yes | Yes | Yes |

Note: Notes: This table reports estimates of the average treatment effect on firm revenues and profits (scaled by pre-reform revenues), together with Lee-style bounds that adjust for differential post-reform exit. The sample is restricted to firms with positive revenue in the pre-reform year, which ensures that all firms in the analysis were active at baseline. Columns 1 and 4 report main estimates on the full baseline sample. Columns 2–3 and 5–6 report lower and upper bounds constructed in the spirit of Lee (2009), adapted for the staggered DiD setting. The bounds are built in three steps: (i) I compute the differential survival rate between never-treated and treated firms at each post-reform year $t$, defined as $q(t) = [P(\text{survive} \mid \text{never-treated}, t) - P(\text{survive} \mid \text{treated}, t)]/P(\text{survive} \mid \text{never-treated}, t)$, where survival corresponds to having positive revenue in year t; (ii) at each year $t$ with $q(t) > 0$, I compute the $q(t)^{th}$ and $(1-q(t))^{th}$ percentiles of the never-treated distribution of each outcome, using outcome-specific cutoffs (log revenue for columns 1 – 3 and profits for columns 4 – 6); and (iii) I re-estimate the average treatment effects after dropping, in each year, the bottom $q(t)$ share of control observations (lower bound) or the top $q(t)$ share (upper bound). The lower bound thus compares treated firms to the higher-performing surviving controls, and the upper bound compares them to the lower-performing surviving controls. All specifications condition on sector-by-size and accounting-month. Standard errors are clustered at the firm level and reported in parentheses. Numbers in brackets are 95% confidence intervals on each bound estimate. The adaptation of Lee (2009) to a staggered DiD setting differs from the canonical cross-sectional procedure in that trimming operates at the firm-year level rather than at the unit level. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

Table A.2: Effects on Formal Employment and Wage Bill

| | Formal Workers | | Wage Bill (UGX Million) | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| ATE | -0.471** | -0.434** | -2.832** | -2.558* |
| | (0.201) | (0.200) | (1.320) | (1.321) |
| % effect | -4.1 | -3.8 | -4.1 | -3.7 |
| Sector | Y | Y | Y | Y |
| Accounting Month | N | Y | N | Y |
| Pre-Reform Mean | 11.39 | 11.39 | 68.79 | 68.79 |
| N | 33012 | 33012 | 33012 | 33012 |

Note: The table presents the average effects of the reform on the number of formal workers and the total formal wage bill. Outcome variables are in level. The percent effects are computed dividing the coefficient estimates by the pre-reform average of the outcome variable. All regressions are estimated on a balanced panel of active firms, with observations weighted by the logarithm of pre-reform revenues. Coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021), with standard errors clustered at the firm level.

Table A.3: Effects on Formal Employment: Expanded Sample

| | Weighted | | | Unweighted | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| ATE | -0.394*** | -0.354*** | -0.368*** | -0.343*** | -0.309*** | -0.321*** |
| | (0.103) | (0.104) | (0.096) | (0.092) | (0.093) | (0.087) |
| % Effect | -6.1 | -5.5 | -5.7 | -6.1 | -5.5 | -5.7 |
| Sector-Size-Year | Y | Y | Y | Y | Y | Y |
| Accounting Month | N | Y | Y | N | Y | Y |
| Occupancy Status | N | N | Y | N | N | Y |
| Pre-reform Mean | 6.4 | 6.4 | 6.4 | 5.6 | 5.6 | 5.6 |
| Observations | 162729 | 155799 | 155799 | 162729 | 155799 | 155799 |
| Jobs Lost per $1 Million | 343 | 308 | 320 | 298 | 269 | 279 |

Note: The table presents the average effects of the reform on formal employment. The sample includes all firms with positive revenues in the pre-reform period, expanded to all years of the sample, with zero formal employment assigned in any year in which the firm is not active. The outcome variable is formal employment in levels. Each column reports estimates from a different specification with an alternative set of controls. Columns 1 - 3 present results from regressions weighted by the log of pre-reform revenues while columns 4 - 6 present results from unweighted regressions. The bottom row reports the citywide reform-induced job loss per additional $1 million in aggregate commercial property tax liability, calculated as $\frac{\hat{\beta} \cdot N}{\Delta T^{\text{City}}}$, where $\hat{\beta}$ is the reduced-form coefficient, $N$ is the number of firms with positive revenues in the pre-reform period, and $\Delta T^{\text{City}}$ is the reform-induced change in citywide commercial property tax liability.

Table A.4: Effects on Wage Informality

| | Balanced Panel | | Unbalanaced Panel | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| ATE | 0.025*** | 0.026*** | 0.028*** | 0.027*** |
| | (0.008) | (0.008) | (0.007) | (0.007) |
| Sector-Year FE | Y | Y | Y | Y |
| Accounting Month | N | Y | N | Y |
| Pre-Reform Mean | 0.10 | 0.10 | 0.11 | 0.11 |
| N | 24579 | 24579 | 38200 | 38200 |

Note: The table presents the effects of the reform on wage informality, defined as $\frac{\text{CIT wage bill - Total wage bill from employee returns}}{\text{CIT wage bill}}$. This means coefficients can be interpreted as a percentage point changes. Columns 1 and 2 present results from the balanced panel of active firms while columns 3 and 4 present results from the unbalanced panel. Standard errors are clustered at the firm level.

Table A.5: Effects on Wage Bills (UGX Millions)

| | Balanced Panel | | | Unbalanaced Panel | | |
|---|---|---|---|---|---|---|
| | (1) $\hat{\beta}^{PAYE}$ | (2) $\hat{\beta}^{Informal}$ | (3) $\frac{\lvert\hat{\beta}^{Informal}\rvert}{\lvert\hat{\beta}^{PAYE}\rvert}$ | (4) $\hat{\beta}^{PAYE}$ | (5) $\hat{\beta}^{Informal}$ | (6) $\frac{\lvert\hat{\beta}^{Informal}\rvert}{\lvert\hat{\beta}^{PAYE}\rvert}$ |
| Estimates | -2.558* | 1.745** | 0.682 | -1.830** | 1.133*** | 0.619 |
| | (1.321) | (0.714) | | (0.927) | (0.430) | |
| Pre-reform Mean | 59.879 | 5.045 | | 38.126 | 4.765 | |
| Observations | 33012 | 33012 | | 61041 | 61041 | |

Note: The table reports the effects of the reform on firms' wage bill. Columns 1 to 3 are estimated on a balanced panel while columns 4 to 6 are estimated on an unbalanced panel. All regressions are weighted by the logarithm of pre-reform revenues and include controls for sector-year and accounting month fixed effects. The wage bills are expressed in millions of Uganda shillings, constant 2016/17. Columns 1 and 4 report the estimated treatment effects on the formal wage bill constructed from employee tax returns (PAYE) while columns 2 and 5 present effects on informal wage bill computed as the difference between the firm's wage bill on CIT returns and the payroll returns. Column 3 and 4 report the share of the decline in the formal wage bill attributable to increased informal wage payments, computed as $\frac{\lvert\hat{\beta}^{Informal}\rvert}{\lvert\hat{\beta}^{PAYE}\rvert}$. Standard errors are clustered at the firm level.

Table A.6: Effects on Selected Categories of Operating Expenses

| | (1) Training | (2) Travel | (3) Telephone | (4) Computer | (5) Machine Repair | (6) Expenses |
|---|---|---|---|---|---|---|
| ATE | -0.000 | -0.006 | -0.004 | -0.001 | -0.022*** | -0.151 |
| | (0.001) | (0.006) | (0.004) | (0.001) | (0.006) | (0.113) |
| Sector of Operation | Yes | Yes | Yes | Yes | Yes | Yes |
| Other Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| Pre-reform Mean | 0.073 | 0.073 | 0.066 | 0.007 | 0.040 | 2.057 |
| Observations | 33012 | 33012 | 33012 | 33012 | 33012 | 33012 |

Note: The table reports the effects of the reform on selected categories of operating expenditures. Each outcome variable is scaled by pre-reform total operating expenses. All regressions are weighted by the logarithm of pre-reform revenues and include controls for sector-year and accounting month fixed effects. Standard errors are clustered at the firm level.

Table A.7: Effect of the Reform on Workers' Social Security Coverage

| | (1) | (2) | (3) |
|---|---|---|---|
| Treated x Post | -0.035* | -0.041** | -0.036** |
| | (0.017) | (0.015) | (0.015) |
| District FE | Y | Y | Y |
| Year FE | Y | Y | Y |
| Sector of Activity | N | Y | Y |
| Worker Characteristics | N | N | Y |
| Pre-Reform Mean | 0.148 | 0.148 | 0.148 |
| $R^2$ | 0.019 | 0.227 | 0.239 |
| Observations | 2930 | 2930 | 2930 |

*Notes:* The table reports difference-in-differences estimates of the effect of the reform on workers' social security coverage. Estimates come from two repeated cross-sections of the Uganda National Household Survey (2016/17 and 2019/20). The outcome is an indicator equal to one if the worker reports receiving employer-provided social security contributions in their current job and zero otherwise. The treatment group is workers residing in Kampala while the comparison group is workers outside the city but within the Greater Kampala Metropolitan Area. All specifications include county and year fixed effects. Worker controls include age, sex, and marital status of the workers. Regressions are estimated on the pooled sample and weighted by individual survey weights supplied by the Uganda Bureau of Statistics. Standard errors, clustered at the county level, are reported in parentheses. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Effects on Wage Distribution: $25^{th}$, $50^{th}$, and $75^{th}$ Percentiles

| | (1) p25 | (2) p50 | (3) p75 |
|---|---|---|---|
| ATE | -0.004 | -0.050 | -0.147** |
| | (0.022) | (0.040) | (0.071) |
| Sector of Operation | Yes | Yes | Yes |
| Pre-reform Size | Yes | Yes | Yes |
| Other Controls | Yes | Yes | Yes |
| Pre-reform Mean | 0.356 | 0.476 | 0.780 |
| Observations | 19656 | 19656 | 19656 |

Note: The table reports the average effects of the reform on within-firm wage quantiles. The dependent variables are the $25^{th}$, $50^{th}$, and $75^{th}$ percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm $i$'s $q^{th}$ wage quantile at time $t$ and $W_{i,0}$ denotes firm $i$'s pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. The estimation conditions on firms' baseline sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.9: Effects on Wage Distribution: $20^{th}$, $40^{th}$, $60^{th}$, and $80^{th}$ Percentiles

| | (1) p20 | (2) p40 | (3) P60 | (4) p80 |
|---|---|---|---|---|
| ATE | -0.018 | -0.028 | -0.098* | -0.171* |
| | (0.025) | (0.031) | (0.059) | (0.093) |
| Sector of Operation | Yes | Yes | Yes | Yes |
| Pre-reform Size | Yes | Yes | Yes | Yes |
| Other Controls | Yes | Yes | Yes | Yes |
| Pre-reform Mean | 0.363 | 0.454 | 0.599 | 0.966 |
| Observations | 19656 | 19656 | 19656 | 19656 |

Note: The table reports the average effects of the reform across the within-firm wage distribution. The dependent variables are the $20^{th}$, $40^{th}$, $60^{th}$, and $80^{th}$ percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm $i$'s $q^{th}$ wage quantile at time $t$ and $W_{i,0}$ denotes firm $i$'s pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. The estimation conditions on firms' baseline sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.10: Property Taxes and Worker Composition

| | (1) $< p25$ | (2) p25-50 | (3) p50-75 | (4) p75-100 |
|---|---|---|---|---|
| Average Effect (in pp) | 0.404 | 0.698 | 2.212** | -2.838** |
| | (1.213) | (1.046) | (0.879) | (1.394) |
| Sector of Operation | Y | Y | Y | Y |
| Pre-reform Size | Y | Y | Y | Y |
| Other Controls | Y | Y | Y | Y |
| Pre-reform Mean | 21.411 | 22.885 | 24.831 | 29.341 |
| Observations | 19125 | 19125 | 19125 | 19125 |

Note: The table reports the average effects of the reform on within-firm employment composition. The dependent variables are the shares of employment in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,q,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm $i$'s number of workers in quartile $q$ at time $t$ and $E_{i,0}$ denotes firm $i$'s total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to firms' pre-reform total employment. The sample is restricted to active firms with at least five workers in the pre-reform period. The estimation conditions on firms' baseline sector of operation and pre-reform size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.11: Implications for Central Government Tax Revenue

| | CIT | | Employee Tax | | Combined | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| ATE | 0.003 | -0.119 | -0.477*** | -0.494*** | -0.474** | -0.613** |
| | (0.135) | (0.150) | (0.155) | (0.172) | (0.231) | (0.252) |
| Controls | No | Yes | No | Yes | No | Yes |
| Pre-reform Mean | 1.664 | 1.664 | 7.391 | 7.391 | 9.056 | 9.056 |
| Observations | 142677 | 142677 | 142677 | 142677 | 142677 | 142677 |
| Central Revenue Loss per UGX 1 | | | | | 0.231 | 0.299 |
| 95% Confidence Interval | | | | | [0.011, 0.452] | [0.058, 0.540] |

Note: The table reports the average effects of the reform on central government tax revenue. All analyses are aggregated at the firm level. The outcome variables are measured in millions of constant 2016/17 Uganda shillings (UGX). Columns 1 and 2 report the effects on corporate income tax revenue, Columns 3 and 4 report the effects on employee tax revenue (PAYE), while Columns 5 and 6 report the effects on combined tax revenue. The sample is a panel of all firms that were active in the pre-reform year, with tax revenue set to zero following firm exit. The estimation conditions on firms' baseline sector of operation, pre-reform accounting month, and occupancy status (tenant versus owner-occupier. Standard errors are clustered at the firm level.

Table A.12: Comparison of the MVPFs Across Tax Instruments

| **Reform & Authors** | **MVPF** | **95% CI** |
|---|---|---|
| *Property Taxation* | | |
| **Commercial Property Tax — This Paper** | **1.53** | **[0.97, 2.09]** |
| *Corporate Income Taxation* | | |
| CIT Tax Cut: Uganda — Global Tax Research Initiative (2026) | 1.68 | |
| CIT Tax Cut: Kenya — Global Tax Research Initiative (2026) | 1.39 | |
| CIT Tax Cut: Tanzania — Global Tax Research Initiative (2026) | 1.71 | |
| CIT Tax Cut: Rwanda — Global Tax Research Initiative (2026) | 1.73 | |
| Corporate Tax Cut — Kennedy et al. (2026) | 1.47 | [1.21, 1.91] |
| *Personal and Business Income Taxation* | | |
| Taxing Business Income Directly — Risch (2024) | 1.46 | |
| Taxing Business Owners through PIT — Risch (2024) | 1.38 | |
| Top Marginal Tax Rate — Risch (2024) | 1.36 | |
| US Top Tax Rate Increase, 1993 — Hendren and Sprung-Keyser (2020) | 1.85 | [1.19, 4.07] |
| US Top MTR Reduction, 1986 — Hendren and Sprung-Keyser (2020) | 44.27 | [1.04, $\infty$] |
| US Top MTR Increase, 2013 — Hendren and Sprung-Keyser (2020) | 1.16 | [0.87, 1.93] |
| US Top Tax Rate Reduction, 2001 — Hendren and Sprung-Keyser (2020) | 1.37 | [0.92, 2.86] |
| Payroll Tax Cuts — Lobel (2024) | 1.66 | |
| US Capital Gains Tax Rate — Agersnap and Zidar (2021) | 2.65 | |
| *Tax Enforcement* | | |
| Tax Audits: High-Income Taxpayers — Boning et al. (2025) | 1.15 | |
| Tax Audits: Low-Income Taxpayers — Boning et al. (2025) | 1.30 | |

Note: The table compares estimates of the marginal value of public funds (MVPF) across different tax instruments and tax enforcement policies. Confidence intervals are reported where available in the original studies. For tax-cut policies, the MVPF measures the welfare gain per dollar of tax reduction. For tax increases and tax enforcement policies, the MVPF is interpreted as the welfare cost per dollar of revenue raised. All comparison estimates are taken from the cited sources without recalculation; readers should note that MVPFs across studies may differ in the assumed incidence, discount rate, fiscal externality margins, and time horizon, so direct numerical comparisons should be interpreted with these differences in mind.

## Table A.13: Marginal Value of Public Funds Across Public Programs

| Broad Group | Policy | Country | Policy Year | Page | MVPF | Original Authors |
|---|---|---|---|---|---|---|
| Investments in Children | Argentina Pre-Primary Classroom Construction | Argentina | 1993–1999 | | Pays for itself | Berlinski, Cruces, Galiani, Gertler, Gonzalez (2026) |
| Investments in Children | Improving Student Outcomes by Calling Parents | Benin | 2022 | | Pays for itself | Yedomiffi (2025) |
| Investments in Children | Early-Childhood Education in India | India | 2016 | | Pays for itself | Ganimian, Muralidharan, and Walters (2021) |
| Investments in Children | K–12 School Finance Reform | United States | 1991 | | Pays for itself | Jackson, Persico, and Johnson (2016) |
| Investments in Children | Medicaid Expansions to Pregnant Women & Infants | United States | 1986 | | Pays for itself | Currie and Gruber (1996); Dave, Decker, Kaestner, Simon (2015); Miller and Wherry (2018) |
| Investments in Children | School-Based De-Worming Treatments in Kenya | Kenya | 1998 | | Pays for itself | Baird, Hicks, Kremer, and Miguel (2016) |
| Investments in Children | Medicaid Expansions to Young Children | United States | 1986 | | Pays for itself | Brown, Kowalski, and Lurie (2015; 2020) |
| Investments in Children | Public Childcare | Nicaragua | 2014 | | 66.00 | Hojman and Boo (2022) |
| Investments in Children | Perry Preschool Program | United States | 1962 | | 43.61 | Heckman, Moon, Pinto, Savelyev, and Yavitz (2010) |
| Investments in Children | Georgia HOPE Scholarship | United States | 1995 | | 4.00 | Cornwell, Mustard, and Sridhar (2006) |
| Investments in Children | College Financial Aid in Colombia (Test Score Cut-Off) | Colombia | 2014 | | 3.20 | Londoño-Vélez, Rodriguez, Sanchez, and Álvarez-Arango (2025) |
| Investments in Children | Carolina Abecedarian Study | United States | 1975 | | 11.89 | Barnett and Masse (2007); Campbell et al. (2012) |
| Investments in Children | Head Start Impact Study | United States | 2002 | | 1.84 | Kline and Walters (2016) |
| Investments in Children | Universal Childcare Provision in Quebec | Canada | 1997–2003 | | 1.26 | Montpetit, Beauregard, and Carrer (2026) |
| Adult Education & Job Training | Refugee Integration Intervention in Sweden | Sweden | 2017 | | Pays for itself | Dahlberg, Egebark, and Vikman (2023) |
| Adult Education & Job Training | Sectoral Job Training: Year Up | United States | 2000 | | 6.12 | Narain and Noray (2023) |
| Adult Education & Job Training | Income-Contingent Student Loan Repayment in Australia | Australia | 2005 | | 3.67 | Tim de Silva (2025) |
| Adult Education & Job Training | Hope Tax Credit, Joint Filers at Phase End | United States | 2007 | | 2.37 | Bulman and Hoxby (2015) |
| Adult Education & Job Training | Pell Grants Introduction to Adults | United States | 1973 | | 2.18 | Seftor and Turner (2002) |
| Adult Education & Job Training | Work Advance | United States | 2012 | | 0.78 | Hendra, Greenberg, Hamilton, Pennington, Schaberg, and Tessler (2016); Schaberg (2017) |
| Adult Education & Job Training | Job Corps | United States | 1995 | | 0.15 | Schochet (2018); Schochet, Burghardt, and McConnell (2006) |
| Cash & In-Kind Transfers | Moving to Opportunity Experiment (MTO) | United States | 1996 | | Pays for itself | Goering et al. (1999); Sanbonmatsu et al. (2011); Chetty, Hendren, and Katz (2016) |
| Cash & In-Kind Transfers | Chile's Indigenous Grants Program (Beca Indígena) | Chile | 1991 | | 29.00 | Lucas, McEwan, and Irribarra (2025) |
| Cash & In-Kind Transfers | Removing SNAP Reporting Requirements | United States | | | 2.09 | Unrath (2021) |
| Cash & In-Kind Transfers | Work Requirements for SNAP | United States | 2013 | | 0.90 | Gray, Leive, Prager, Pukelis, and Zaki (2023) |
| Cash & In-Kind Transfers | Housing Vouchers in Chicago | United States | 1997 | | 0.65 | Jacob and Ludwig (2012); Jacob, Ludwig and Kapustin (2014) |
| Cash & In-Kind Transfers | Conditional Cash Transfers in Uruguay | Uruguay | 2009 | | 0.61 | Bergolo and Cruces (2021) |
| Social Insurance (Adults) | Introduction of Medicare Part D | United States | 2006 | | 1.98 | Wettstein (2020) |
| Social Insurance (Adults) | Medicare Introduction in 1965 | United States | 1965 | | 1.63 | Finkelstein and McKnight (2008) |
| Social Insurance (Adults) | Oregon Health Insurance Experiment (Provided to Single Adults) | United States | 2008 | | 1.16 | Finkelstein et al. (2012); Finkelstein, Hendren, and Luttmer (2019) |
| Social Insurance (Adults) | Supplemental Security Income (SSI) for Adults | United States | 2000 | | 1.04 | Deshpande, Gross and Su (2021) |
| Social Insurance (Adults) | Unemployment Insurance in a Large Informal Economy | Mauritius | | | 1.01 | Liepmann and Pignatti (2024) |
| Social Insurance (Adults) | Disability Insurance (DI) Changes in Benefit Generosity | United States | 2004 | | 0.96 | Gelber, Moore, and Strand (2017) |
| Social Insurance (Adults) | Unemployment Insurance Duration Extensions in Missouri | United States | 2011 | | 0.83 | Johnston and Mas (2018) |
| Social Insurance (Adults) | Introduction of Old-Age Pensions in the UK | United Kingdom | 1909 | | 0.80 | Giesecke and Jaeger (2021) |
| Social Insurance (Adults) | Maternity Leave in Denmark | Denmark | | | 0.74 | Brenøe, Canaan, Harmon, and Royer (2020) |
| Social Insurance (Adults) | Unemployment Insurance Benefit Changes in Missouri (Recession Estimates) | United States | 2010 | | 0.44 | Card, Johnston, Leung, Mas, and Pei (2015) |
| Other Public Programmes | Electronic Monitoring in Sweden | Sweden | 1997 | | Pays for itself | Grenet, Grönqvist, and Niknami (2024) |
| Other Public Programmes | Real-Time Feedback Intervention to Reduce Resource Use | Germany | 2020–2021 | | 7.39 | Andor, Goette, Price, Tilling, and Tomberg (2025) |
| Other Public Programmes | German Public Transport Subsidy (9-Euro Ticket) | Germany | 2022 | | 1.60 | Liebensteiner, Losert, Necker, Neumeier, Paetzold, Wichert (2026) |
| Other Public Programmes | Boiler Upgrade Scheme (Heat Pump Subsidy) | United Kingdom | 2022 | | 1.24 | Bernard, Hacket, Metcalfe, and Schein (2024) |

Note: The table presents estimates of the marginal value of public funds (MVPF) for selected public spending programs. The estimates are compiled from Hendren and Sprung-Keyser (2020) and supplemented with more recent estimates from the Policy Impacts Library (policyimpacts.org).

## III. NEGATIVE-WEIGHT DIAGNOSTICS AND ROBUSTNESS TO ALTERNATIVE ESTIMATORS

A conventional approach to estimating the average and dynamic effects of the reform would be to use two-way fixed-effects (TWFE) difference-in-differences regressions. However, recent work has shown that, under staggered treatment adoption and heterogeneous treatment effects, TWFE estimators may place negative weights on some treatment effects arising from using already-treated units as controls for later-treated units, potentially yielding estimates that are difficult to interpret (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Borusyak et al. 2024). In this Appendix section, I first assess the extent of the negative-weighting problem for the main outcomes of interest. I then examine the robustness of the dynamic and average treatment effects by comparing the preferred Callaway and Sant'Anna (2021) estimates with those from conventional TWFE event-study regressions and a stacked difference-in-differences design following Cengiz et al. (2019).

Table A.14 reports the number of treatment effects receiving positive and negative weights and the aggregate weights assigned to them under TWFE estimation. The the decomposition is based on De Chaisemartin and d'Haultfoeuille (2020). Across the four outcomes, between 27 and 35 percent of the treatment effects receive negative weights, although the aggregate mass of these negative weights is relatively small under each outcome. The prevalence of negatively weighted treatment effects nevertheless implies that, if the effects receiving negative weights differ systematically in magnitudes from those receiving positive weights, the average effects from the conventional TWFE may be may substantially biased.

Table A.14: TWFE Negative-Weight Diagnostics

| | Firm Survival | | Firm Revenues | | Formal Employment | | Total Tax Revenue | |
|---|---|---|---|---|---|---|---|---|
| | # ATTs | $\sum$ weights | # ATTs | $\sum$ weights | # ATTs | $\sum$ weights | # ATTs | $\sum$ weights |
| Positive weights | 37,770 | 1.0825 | 9,803 | 1.0598 | 10,735 | 1.0983 | 40,953 | 1.0380 |
| Negative weights | 17,556 | -0.0825 | 3,622 | -0.0598 | 5,834 | -0.0983 | 15,628 | -0.0380 |
| Total | 55,326 | 1.0000 | 13,425 | 1.0000 | 16,569 | 1.0000 | 56,581 | 1.0000 |

Notes: The table reports the number of treatment effects receiving positive and negative weights and the sum of these weights in conventional two-way fixed-effects specifications, following De Chaisemartin and d'Haultfoeuille (2020). For simplicity, the decomposition focuses on four main outcomes: survival, firm revenues, employment, and tax revenue. All specifications include firm and year fixed effects and allow the relevant baseline covariates to have year-specific effects. All regressions use the respective samples and weighting procedures described in the main text.

Figure A.44 compares event-study estimates from the preferred Callaway and Sant'Anna

(2021) estimator with those from stacked difference-in-differences and conventional TWFE specifications for five main outcomes of interest: firm survival, firm revenues, profitability, formal employment, and total tax revenue. There are notable differences between the TWFE and Callaway and Sant'Anna (CSDID) estimates for the tax revenue outcome (Panel E), with the estimates diverging in both the pre- and post-treatment periods. In contrast, the pre- and post-treatment estimates are relatively similar across the other outcomes. Reassuringly, the stacked difference-in-differences estimates closely track the Callaway and Sant'Anna estimates in both the pre- and post-treatment periods across all outcomes.

Figure A.45 presents the average treatment effects across the five outcomes using the three alternative estimators. The stacked DiD estimates are broadly consistent with the Callaway and Sant'Anna estimates across all five outcomes, although the estimated effect on total tax revenue is somewhat smaller in absolute magnitude. In contrast, conventional TWFE produces attenuated estimates for firm survival, formal employment, and total tax revenue, with the tax revenue effect becoming statistically insignificant. This pattern is consistent with the well-known weighting problems that can arise in TWFE specifications under staggered treatment adoption and heterogeneous treatment effects. More importantly, the close correspondence between the Callaway and Sant'Anna and stacked DiD estimates indicates that the baseline results are robust across alternative estimators that avoid the problematic treatment comparisons underlying conventional TWFE estimation.

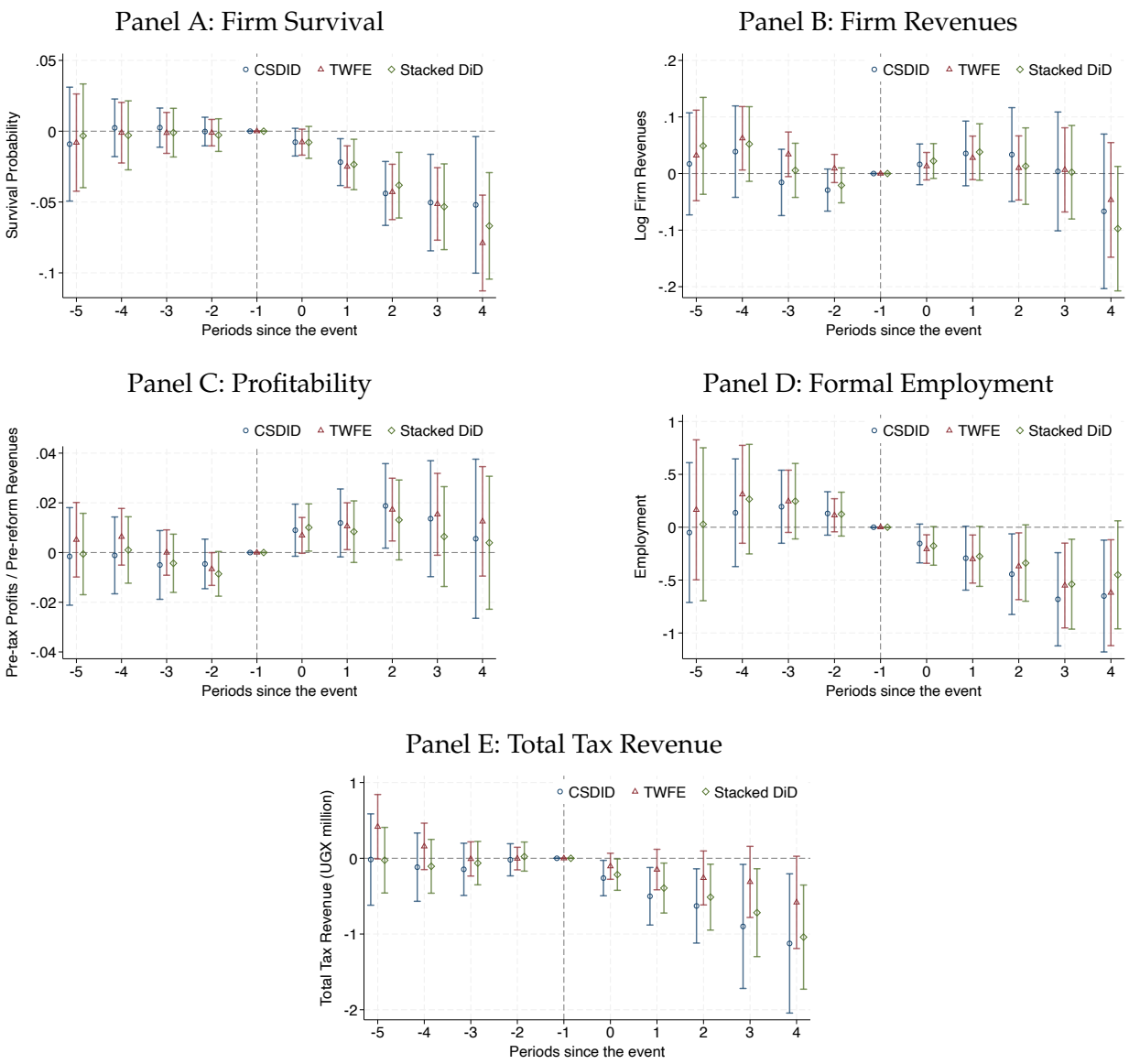



Figure A.44: Robustness to Alternative Estimators

*Notes:* The figure compares event-study estimates from the preferred Callaway and Sant'Anna difference-in-differences estimator (CSDID), a conventional two-way fixed-effects (TWFE) estimator, and a stacked difference-in-differences estimator. The decomposition focuses on five main outcomes: survival (Panel A), firm revenues (Panel B), profitability (Panel C), employment (Panel D), and tax revenue (Panel E). Event time $-1$ is the omitted reference period. The TWFE specifications include firm and year fixed effects and allow the relevant baseline covariates to have year-specific effects. The stacked specifications construct separate samples for each treatment cohort, comparing firms in that cohort with never-treated firms over the event window, and include stack-specific firm and year fixed effects together with the corresponding covariate-by-stack-year effects. All specifications use the respective samples, covariates, and weighting procedures described in the main analysis. Standard errors are clustered at the firm level, and vertical bars report 95 percent confidence intervals.

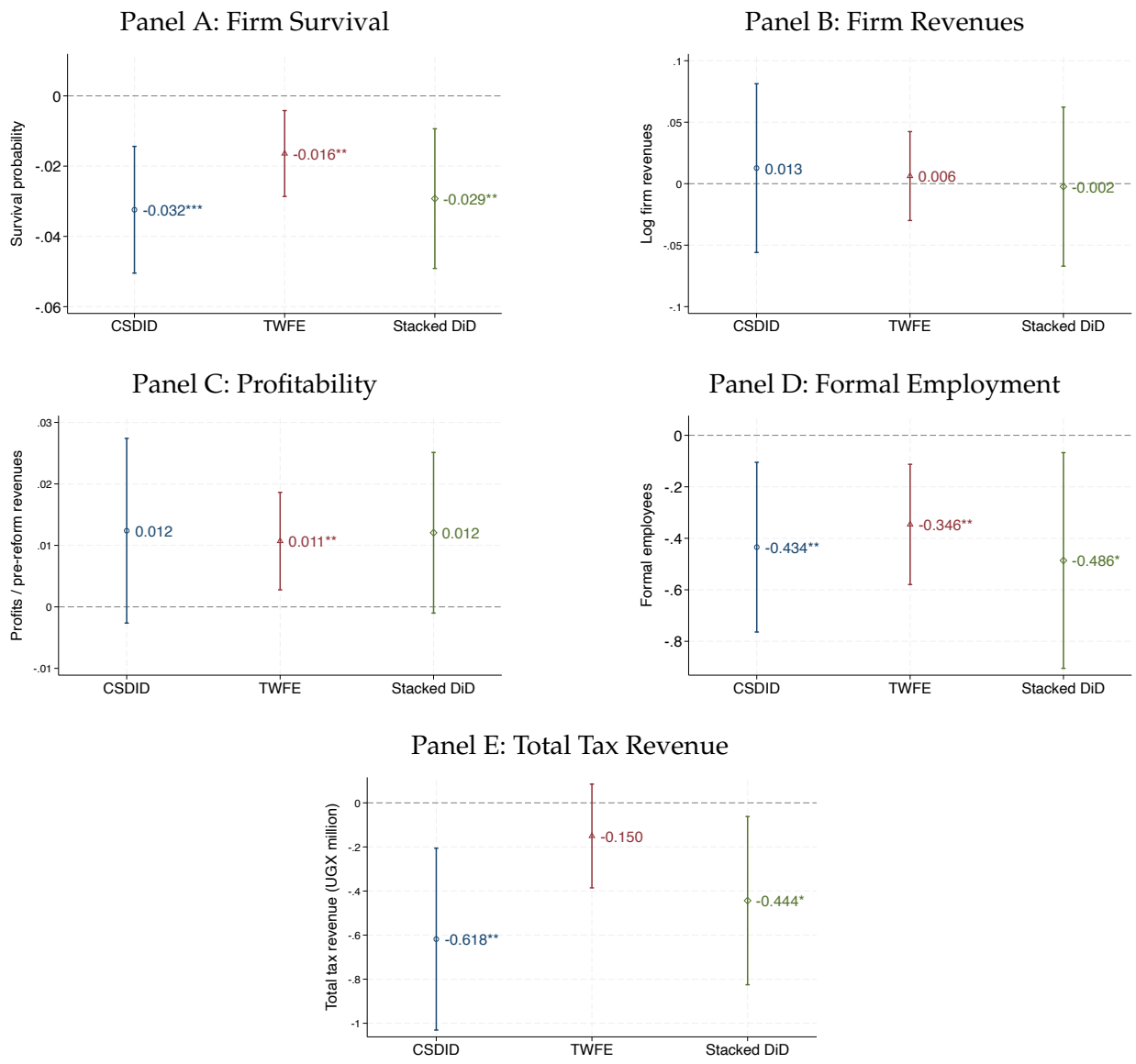



Figure A.45: Robustness to Alternative Estimators: *Average Effects*

*Notes:* The figure compares the average effects from the preferred Callaway and Sant'Anna difference-in-differences estimator (CSDID), a conventional two-way fixed-effects (TWFE) estimator, and a stacked difference-in-differences estimator. The decomposition focuses on five main outcomes: survival (Panel A), firm revenues (Panel B), profitability (Panel C), employment (Panel D), and tax revenue (Panel E). The TWFE specifications include firm and year fixed effects and allow the relevant baseline covariates to have year-specific effects. The stacked specifications construct separate samples for each treatment cohort, comparing firms in that cohort with never-treated firms over the event window, and include stack-specific firm and year fixed effects together with the corresponding covariate-by-stack-year effects. All specifications use the respective samples, covariates, and weighting procedures described in the main analysis. Standard errors are clustered at the firm level, and vertical bars report 95 percent confidence intervals.

## IV. A MODEL OF HETEROGENEOUS FIRM RESPONSES TO PROPERTY TAXATION

The empirical evidence in Section V documents an asymmetric response to the property tax reform: large firms experience gains in both revenues and profit margins, while small firms lose revenue with no significant change in their profits. This section develops a simple neoclassical general-equilibrium model that rationalizes the pattern. The model builds on the local-tax incidence framework of Suárez Serrato and Zidar (2016), adapted to the property-tax setting. Two features drive the results: firms differ in the intensity with which they use commercial space, and presence of a locally supplied input $X$ that clears in general equilibrium so that a shock to one input propagates through the equilibrium price of $X$.[36]

*Environment.*— Consider a local economy hosting a mass of firms that use two priced inputs to produce a homogeneous output. The inputs are commercial space $R$, supplied by property owners at rental price $r$, and a locally supplied input $X$, supplied at price $q$.[37] The setting comprises both owner-occupier and tenant firms. The reform raises the marginal cost of using commercial space for both groups: tenant firms face higher rents through the pass-through channel, while owner-occupier firms face a higher direct tax liability on the property they own and occupy. In either case, the shadow price of space rises with the reform, and the firm's optimization responds through the same first-order channel.[38] I therefore use $r$ to denote the shadow price of space applicable to all firms, with the understanding that $dr$ reflects both a rental-cost shock for tenants and a direct tax-cost shock for owner-occupiers. An equivalent interpretation is that owner-occupier firms lease the property to themselves at rate $r$, which responds to the tax in the same way as the market rent. What ultimately matters for the model's predictions is how demand for $X$ responds to the local cost shock and its implications for firms of different size, rather than the incidence chain of the tax.

The output market is competitive and the output price is normalized to one, capturing the assumption that firms operate in a small local economy whose output faces demand at prices determined outside the model. The price of $X$ is determined endogenously by local market clearing and is the key general-equilibrium object of the model.

[36] A model that uses equilibrium adjustments in the output market produces a similar conclusion.

[37] Candidates include local labor, particularly for low-to-medium-skill workers with limited mobility across jurisdictions; local intermediate inputs supplied by other firms in the same geographic market; or local services such as logistics whose prices reflect local supply and demand conditions.

[38] The magnitude of the shock differs across the two ownership arrangements—owner-occupiers face the full tax increase while tenants absorb only the share passed through by landlords—but the direction of the response and the mechanism of the general-equilibrium reallocation through input $X$ are identical.

*Firm Heterogeneity and Production Technology.*— There are two types of firms; small ($S$) and large ($L$), with masses $m_S$ and $m_L$. The distinction between them lies in their production technology, described below. Firms of type $j \in \{S, L\}$ produce output according to

$$Y_j \;=\; z_j \left( R_j^{\alpha_j} X_j^{1-\alpha_j} \right)^{\theta}, \tag{6}$$

where $z_j$ is total factor productivity, $\alpha_j \in (0,1)$ is the property share in production, and $\theta \in (0,1)$ captures decreasing returns to the priced inputs at the firm level. Following Lucas (1978), I refer to $\theta$ as the span-of-control parameter: it summarizes the technological limit on how much a firm can produce from a doubling of its priced inputs, reflecting the fact that as the firm grows, the entrepreneur's fixed managerial capacity is stretched across a larger operation.

The technology above embeds two ingredients that jointly ensure the coexistence of small and large firms in equilibrium. First, decreasing returns to the priced inputs, $\theta < 1$, guarantee that the firm's optimal scale is finite: marginal cost rises with output, and the firm chooses the scale at which the value of the marginal product equals the input price. Second, heterogeneity in the productivity–intensity pair $(z_j, \alpha_j)$ generates different optimal scales across firm types, with more productive and less space-intensive entrepreneurs operating at larger scales. Together, these features deliver an interior firm-size distribution in which the two types coexist.

The key dimension of heterogeneity across firm types is

$$\alpha_S \;>\; \alpha_L, \tag{7}$$

that is, small firms use commercial space more intensively than large firms. This assumption is supported by the descriptive evidence in Appendix Figure A.15, which shows that both rental expenses and the value of land and buildings as a share of firms' revenues fall monotonically with firm size across the cross-section. Small firms operate with a cost structure disproportionately weighted toward space. Because the technology is Cobb–Douglas in $R$ and $X$ with factor shares summing to one, the assumption $\alpha_S > \alpha_L$ mechanically implies $1 - \alpha_S < 1 - \alpha_L$: large firms use the locally supplied factor $X$ more intensively than small firms.

*Firm Optimization.*— Given prices $r$ and $q$, a firm of type $j$ chooses $R_j$ and $X_j$ to maximize profits,

$$\pi_j \;=\; \max_{R_j, X_j} \left\{ Y_j - r R_j - q X_j \right\}. \tag{8}$$

The first-order conditions yield the cost-minimizing input ratio $rR_j/(qX_j) = \alpha_j/(1-\alpha_j)$.

Substituting into the production function and solving gives closed-form expressions for output, input demands, and profits:

$$\ln Y_j = \frac{1}{1-\theta}\left[\ln z_j + \theta \ln \theta + \theta\alpha_j \ln \frac{\alpha_j}{r} + \theta(1-\alpha_j)\ln \frac{1-\alpha_j}{q}\right], \tag{9}$$

$$X_j = \frac{(1-\alpha_j)\,\theta}{q}\, Y_j, \tag{10}$$

$$\pi_j = (1-\theta)\, Y_j. \tag{11}$$

Profits are a constant fraction $(1-\theta)$ of revenues, a standard result under Cobb–Douglas technology with span-of-control frictions.

*Partial-equilibrium effects of the reform.*— The reform raises the rental price by $dr > 0$. Holding the price of $X$ fixed at $q$, the direct output response of a firm of type $j$ is

$$\left.\frac{d\ln Y_j}{d\ln r}\right|_{q \text{ fixed}} = -\frac{\alpha_j \theta}{1-\theta} < 0. \tag{12}$$

All firms lose in partial equilibrium, but small firms lose proportionally more because $\alpha_S > \alpha_L$. Partial equilibrium, however, is not the end of the story. Because firms are linked through the local market for $X$, the aggregate contraction in output feeds back into the equilibrium price of $X$.

*Local market clearing for $X$.*— Aggregate demand for $X$ in the local economy is $X^d(r,q) = m_S X_S(r,q) + m_L X_L(r,q)$, where $m_j$ is the mass of type $j$ and each $X_j$ is given by equation (10). The local supply of $X$ is upward-sloping, $X^s(q) = Bq^{\varepsilon}$ with $\varepsilon > 0$ the supply elasticity of $X$. The local market for $X$ clears at a price $q^*$ satisfying

$$X^d(r, q^*) = X^s(q^*). \tag{13}$$

Totally differentiating equation (13) and simplifying yields the equilibrium price response of $X$ to a change in the property rent:

$$\frac{d\ln q^*}{d\ln r} = -\frac{\bar{\alpha}\,\theta}{\varepsilon\,(1-\theta) + (1-\bar{\alpha}\theta)}, \tag{14}$$

where $\bar{\alpha} \equiv s_S\alpha_S + s_L\alpha_L$ is the $X$-share-weighted average property intensity across firms, with $s_j = m_j X_j / X^d$ the share of aggregate $X$ demand accounted for by firms of type $j$.

The equilibrium price of $X$ falls in response to a rise in the property rent. Increase in property tax liabilities, leading to rise in $r$, causes all firms to contract, aggregate demand

for $X$ falls, and because the supply of $X$ is upward-sloping and local, its price falls to restore market clearing. The magnitude of the price decline is governed by three parameters: the average property intensity $\bar{\alpha}$ (higher average intensity means a larger aggregate contraction), the supply elasticity $\varepsilon$ (more elastic supply means a smaller price response), and the span-of-control parameter $\theta$ (which controls how sensitive firm scale is to input prices).

*Total effects and the source of heterogeneity.*— Given the equilibrium price response, the total effect on output for a firm of type $j$ combines the direct rent effect with the indirect price effect from $X$:

$$\frac{d\ln Y_j}{d\ln r} = \underbrace{-\frac{\alpha_j\theta}{1-\theta}}_{\text{direct rent effect } (<0)} + \underbrace{\left[-\frac{(1-\alpha_j)\theta}{1-\theta}\right]\cdot\frac{d\ln q^*}{d\ln r}}_{\text{indirect price effect } (>0)}. \tag{15}$$

The direct rent effect is unambiguously negative and is larger in magnitude for small firms. The indirect effect is positive, since $d\ln q^*/d\ln r < 0$, and is larger for large firms in absolute value because they use $X$ more intensively. Whether a firm gains or loses on net depends on which effect dominates.

Substituting equation (14) into equation (15) and rearranging, a firm of type $j$ gains if and only if

$$\frac{\alpha_j}{1-\alpha_j} < \Lambda \equiv \frac{\bar{\alpha}\,\theta}{\varepsilon\,(1-\theta)+(1-\bar{\alpha}\theta)}. \tag{16}$$

This defines a threshold on property intensity: firms whose property share is below the threshold ratio $\Lambda$ gain from the reform, while firms above the threshold lose. Equivalently, there exists a critical property share $\alpha^*$ satisfying $\alpha^*/(1-\alpha^*) = \Lambda$ such that firms with $\alpha_j < \alpha^*$ benefit and firms with $\alpha_j > \alpha^*$ are harmed. The threshold rises with the average property intensity $\bar{\alpha}$ and falls with the supply elasticity of $X$: reallocation is stronger when the aggregate demand contraction is large and when the local factor $X$ is in relatively inelastic supply.

The model therefore generates the observed empirical pattern: small firms losing revenue and large firms gaining revenue whenever

$$\frac{\alpha_L}{1-\alpha_L} < \Lambda < \frac{\alpha_S}{1-\alpha_S}. \tag{17}$$

This is the sufficient condition for the pattern, which holds for a range of parameter values, and becomes easier to satisfy as the gap between $\alpha_S$ and $\alpha_L$ widens or as the local supply of $X$ becomes less elastic.

Appendix Figure A.16 provides direct evidence consistent with the reallocation predicted by the model. Following the reform, small firms reduce their local input purchases, while large firms modestly expand theirs, consistent with the shift in local input demand from space-intensive small firms toward less space-intensive large firms that the model predicts.

## V. A FRAMEWORK FOR PROPERTY TAX PASS-THROUGH

The finding in Section VIII showing that higher property tax liabilities increase business rental expenses contrasts with the predictions of neoclassical theory, which suggests little to no pass-through of property taxes (Mieszkowski 1972; Zodrow 2001). Underlying the neoclassical prediction is the assumption that demand for commercial space within a jurisdiction is elastic, since other jurisdictions offer comparable commercial units, while supply is inelastic. In that setting, the property tax falls on the immobile factor and landlords absorb the entire burden because attempts to raise rents would induce tenants to relocate to competing jurisdictions.

A necessary condition for this result, however, is that the property tax is a cost of *owning* the property, not a cost of *renting it out*. If the tax is enforced universally, the landlord owes the same liability whether the property is occupied or vacant, and the tax is effectively a lump sum that cannot influence the landlord's pricing decision. But if the tax is enforced selectively—collected only when the property is occupied and generating observable economic activity—then renting out the property triggers the tax liability. In this case, the tax functions as a marginal cost of supplying space to the rental market, and standard incidence logic predicts positive pass-through to tenants.

To rationalize my findings, I develop a simple framework that predicts non-zero pass-through and that does not depend on the elasticities of demand and supply in the standard sense. The framework embeds a key institutional feature of property tax enforcement in developing countries: *targeted enforcement*, where enforcement and collection probabilities are lower for vacant properties than for occupied properties. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use and generating income (Franzsen and McCluskey 2017). Under limited administrative capacity, targeting occupied properties becomes a cost-effective enforcement strategy: occupied properties are easier to identify, their owners are easier to locate, and the revenue yield per enforcement action is higher because an active tenant signals ability to pay. Consistent with the pattern of targeted enforcement documented above, Ahabwe et al. (2020) notes that demand letters in Kampala are disproportionately directed toward high-value property owners, reflecting the administrative reality that limited enforcement resources incentivize tax authorities to prioritize collections where the potential yield is greatest.

*Setup.*— Consider a landlord who owns a unit of commercial space and must choose the rent $R$ to charge. If the unit is occupied at rent $R$, the landlord earns rental income $R$

and pays the property tax liability $T = \tau V$ with certainty, yielding a payoff of $R - T$.[39] If the unit remains vacant, the landlord earns no rental income but may still face the tax. The tax is enforced on the vacant unit with probability $p_e \leq 1$, yielding an expected payoff of $-p_e T$. [40]

The probability that the unit is occupied depends on the rental price charged. Let $q(R)$ denote the probability that the unit is filled at rent $R$, with $q'(R) < 0$ (higher rents reduce the likelihood of finding a tenant) and $q(0) \leq 1$. Assuming a unit mass of potential tenant firms, $q(R)$ also represents the aggregate demand for commercial space, i.e., the share of businesses willing to pay rent $R$ rather than take their outside option.[41]

*Landlord's Maximization Problem.*— The landlord sets rent to maximize his expected utility:

$$\max_R \; q(R)\,(R - T) + (1 - q(R))\,(-p_e T) \tag{18}$$

The first term is the expected payoff if the unit is filled: the landlord earns $R - T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1 - q(R)$.

Expanding (18) and dropping the terms that do not depend on $R$, the landlord effectively maximizes:

$$\max_R \; q(R)\,[R - (1 - p_e)T] \tag{19}$$

This formulation has a straightforward economic interpretation: the landlord behaves as if facing a per-unit cost of $(1 - p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero, i.e., the landlord pays $T$ regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax $T$, i.e., each occupied unit triggers the entire tax bill.

The first-order condition of (19) is:

$$q'(R)\,[R - (1 - p_e)T] + q(R) = 0 \tag{20}$$

[39] $\tau$ is the property tax rate and $V$ is the assessed value.

[40] In a setting with imperfect compliance, even when the unit is occupied, as is common in many developing countries, $p_e$ can be interpreted as the relative probability of compliance when the unit is vacant compared to when it is occupied. If we assume landlords are more likely to comply when the unit is occupied than when it is vacant, then $p_e < 1$.

[41] This equivalence holds when each potential tenant independently decides whether to rent the unit. If tenants have heterogeneous reservation values for the location, $q(R)$ is the fraction whose reservation value exceeds $R$.

Solving for $R$:

$$\boxed{R^* = (1 - p_e)T - \frac{q(R^*)}{q'(R^*)}} \tag{21}$$

The optimal rent has two components. The first, $(1 - p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

*Characterizing the Pass-through Rate.*— Defining $c \equiv (1 - p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent $R^*$ as a function of the effective cost $c$, and hence of $T$. Define the function:

$$G(R, c) \equiv q'(R)(R - c) + q(R) = 0 \tag{22}$$

By the implicit function theorem:

$$\frac{dR^*}{dc} = -\frac{\partial G/\partial c}{\partial G/\partial R} \tag{23}$$

Computing the partial derivatives:

$$\frac{\partial G}{\partial c} = -q'(R)$$

$$\frac{\partial G}{\partial R} = q''(R)(R - c) + q'(R) + q'(R) = q''(R)(R - c) + 2q'(R)$$

Therefore:

$$\frac{dR^*}{dc} = -\frac{-q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} = \frac{q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} \tag{24}$$

From the first-order condition (20), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$\frac{dR^*}{dc} = \frac{q'}{q''\left(-\frac{q}{q'}\right) + 2q'} = \frac{q'}{-\frac{q\,q''}{q'} + 2q'} = \frac{(q')^2}{-(q)(q'') + 2(q')^2} \tag{25}$$

Dividing numerator and denominator by $(q')^2$:

$$\frac{dR^*}{dc} = \frac{1}{2 - \dfrac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}} \tag{26}$$

Since $c = (1 - p_e)T$ and hence $dc/dT = (1 - p_e)$, the pass-through rate with respect to the tax liability is:

$$\boxed{\rho = \frac{dR^*}{dT} = \frac{dR^*}{dc} \cdot \frac{dc}{dT} = \frac{1 - p_e}{2 - \theta}}, \tag{27}$$

where $\theta \equiv \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by Weyl and Fabinger (2013).

The parameter $\theta$ governs the distribution of the tax burden for a given $p_e \in [0, 1)$. The pass-through rate is increasing in $\theta$ for any given $p_e$. Equation 27 also permits over-shifting if $\theta > 1$ (strongly log-convex demand) and $p_e$ is sufficiently small. If we assume a simple, linear demand, $q(R) = q_0 - q_1 R$, we have $q'' = 0$ and therefore $\theta = 0$. The pass-through rate is $\rho = (1 - p_e)/2$. Under this special case, the burden of property taxes is equally shared between the landlord and the tenant if $p_e = 0$, the case in which there is no enforcement on vacant units. Under linear demand and strictly no enforcement on vacant properties, the landlord bears more than half the burden when $\theta < 0$ (log-concave demand), while the tenant bears more than half when $\theta > 0$ (log-convex demand).

Note that the pass-through formula (27) requires $\theta < 2$ for $\rho \geq 0$. This restriction follows from the second-order condition for the landlord's optimization problem, which ensures that the optimal rent $R^*$ is a well-defined interior maximum. To see this, note that the second-order condition requires:

$$\frac{d^2}{dR^2}\left[q(R)(R - c)\right] = q''(R)(R - c) + 2q'(R) < 0 \tag{28}$$

From the first-order condition (20), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$q''(R^*)\left(-\frac{q(R^*)}{q'(R^*)}\right) + 2q'(R^*) = -\frac{q(R^*)\, q''(R^*)}{q'(R^*)} + 2q'(R^*) \tag{29}$$

Factoring out $q'(R^*)$:

$$= q'(R^*)\left(2 - \frac{q(R^*)\, q''(R^*)}{[q'(R^*)]^2}\right) = q'(R^*)\,(2 - \theta) \tag{30}$$

Since $q'(R^*) < 0$, the expression in (30) is negative if and only if $2 - \theta > 0$, i.e., $\theta < 2$. When $\theta \geq 2$, the landlord's objective function is locally convex at the candidate solution, and no interior optimum exists.